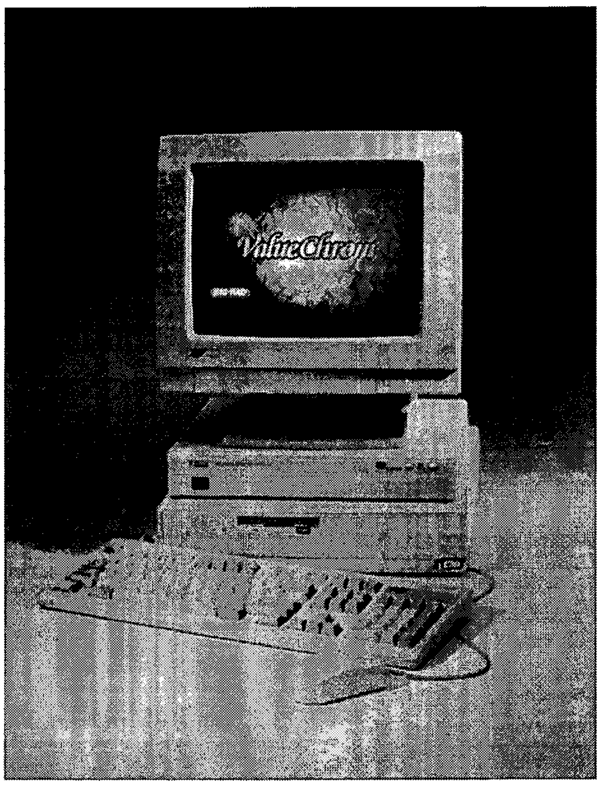
ValueChrom™ Chromatography Software



User's Manual

For Technical Service
Call your Local Bio-Rad Office or
in the U.S. Call **1-800-4BIORAD**
(1-800-424-6723)

400-5004 Rev B

ValueChrom Contents

Introduction

Product Description

ValueChrom, based on MicroSoft Windows is simple to use and includes the following powerful features :

- Data collection from up to two detectors from each of two systems
- Displays and re-analyzes stored data while new data is collected
- Accurate quantitation with single- and Multi-level calibrations (up to five levels)
- Integration parameters can be determined automatically or edited manually for customized data handling of even the most difficult separations
- Manual baseline editing and automatic re-analysis of stored data
- Total system automation with pump, fraction collector, and relay control
- Easy access to stored data files with the Chromatogram Database.

About this Product

Information in this document pertains to ValueChrom version 4.x. This information is subject to change without notice and does not represent a commitment on the part of Bio-Rad Laboratories. The software described in this document is furnished under a license agreement or nondisclosure agreement and may be used or copied only in accordance with the terms of that agreement. It is against the law to copy the software except as specifically allowed in the license or nondisclosure agreement.

Warranty agreement

This software is supplied under a software license agreement. Bio-Rad Labs Inc. specifically disclaims all other warranties, expressed or implied, including but not limited to implied warranties of merchant ability and fitness for a particular purpose. In no event shall Bio-Rad Labs Inc. be liable for any loss of profit or any commercial damage, including but not limited to special, incidental, consequential or other damages.

Trademarks

Soft-Start is a registered trademark of Bio-Rad Laboratories.

MicroSoft is a registered trademark and Windows is a trademark of MicroSoft Corporation.

Computer and Software Requirements

The following specifications are minimum requirements. For optimal performance a fast 80486DX processor with 8 Mbytes RAM is recommended.

Minimum System Requirements *

Machine Type:	IBM PC (or 100% Compatible)
Machine Class:	Pentium, 80486DX, 80486SX, 80386DX, or 80386SX (80486DX is recommended)
Minimum RAM:	4Mb
Disk Drives:	40 megabyte (100 megabyte or greater is recommended) internal hard disk and 1.44 megabyte, 3.5", internal diskette
Video System:	VGA (or 100% compatible) color graphics -or- SVGA
Serial Ports:	One (1) RS-232 (9 or 25 pin) for asynchronous communications, configured as COM 1 (IRQ4, I/OPort Address 03F8)** Two ports are required for Autosampler or Bio-Dimension control.
Mouse:	MicroSoft (or 100% compatible) Serial mouse, PS-2 mouse or IRQ12 preferred
Hardware Options:	Parallel Port: One (1) Centronix (or 100% compatible) for a printer (IRQ7) configured as LPT1 Printer: Centronix (or 100% compatible) parallel interface with memory capable of processing full page graphics at 300 dpi
Operating System Requirement:	MS (or PC) DOS: Version 5.x, or Version 6.x***
Environment Requirement:	MicroSoft Windows 3.1

* All hardware documentation, including computer and adapter card settings, must be available.

** When using an autosampler, two (2) RS-232 (9 or 25 pin) for asynchronous communications, configured as COM 1 (IRQ4, I/O Port Address 03F8) and COM 2 (IRQ3, I/O Port Address 02F8) are needed.

*** Disk compression programs such as Double Space are incompatible with this system

Installation

Location

The HPLC system should be placed on a level bench top in typical laboratory ambient conditions. Allow adequate space for HPLC modules, computer, and any additional output device and printer. Allow at least 5 inches of clear space between the rear panel of the system components and any wall or obstruction. This provides access to the rear panel connections and allows free flow of cooling air. The system must not be operated in direct sunlight. Ambient temperature during operation must not exceed 40°C. An isolated or filtered power line is highly desirable to minimize noise which might interfere with computer function or UV detection.

Unpacking

When you receive your **ValueChrom** system, carefully inspect the containers for any damage which may have occurred in shipping. Severe damage to a container may indicate damage to its contents. If you suspect damage to the contents may have occurred, immediately file a claim with the carrier in accordance with their instructions before contacting **Bio-Rad Laboratories**.

Save the packing materials to facilitate shipping in case repair should be required at one of our depot service centers.

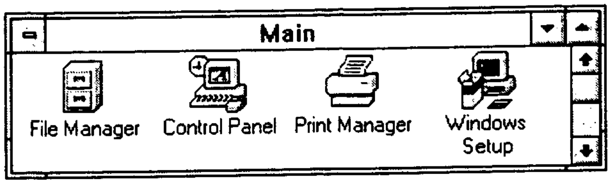
Software Installation

Computers supplied by **Bio-Rad** are shipped with both **Windows** and **ValueChrom** installed. If you are installing **ValueChrom** on a computer not supplied by **Bio-Rad**, you must first install MicroSoft Windows (v3.1 or higher). Refer to your Windows reference manual for instructions.

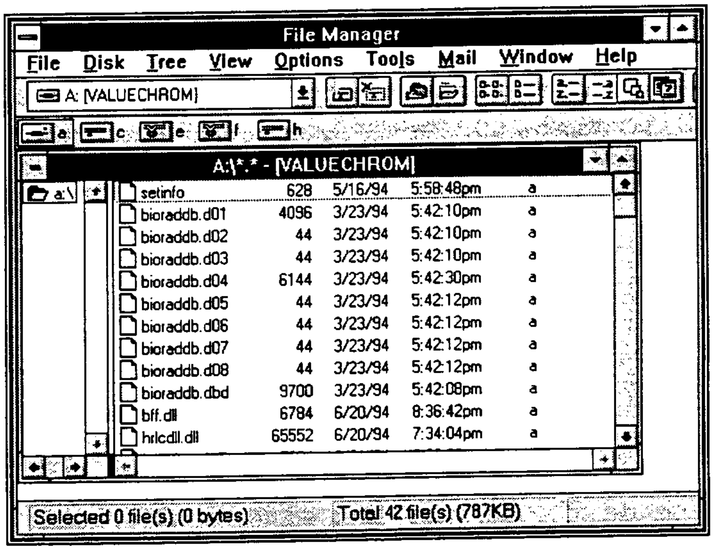
If you are installing **ValueChrom** for the first time, use the **SETUP.EXE** program on the SETUP disk. This program loads the **ValueChrom** software onto your computer and makes necessary changes to the **WIN.INI** file and the **AUTOEXEC.BAT** file. The hard disk retains copies of the old versions of these files for your convenience. For more information about these files see the end of this section.

To install the ValueChrom software:

Double click on the **File Manager** icon in the Main window of the Windows software.

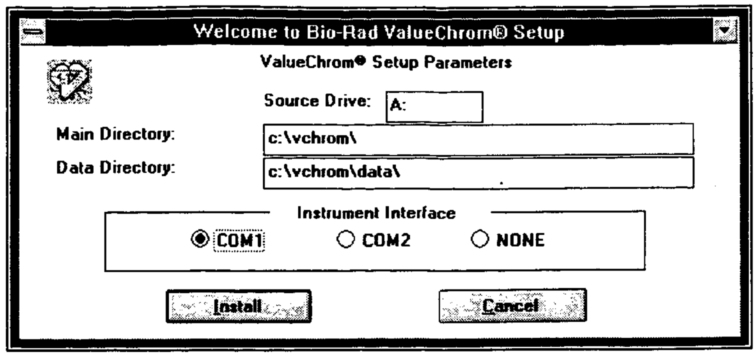


Insert the **Setup** disk into the **A** drive of your computer. Then click on the A disk icon in the upper left of the File Manager screen.



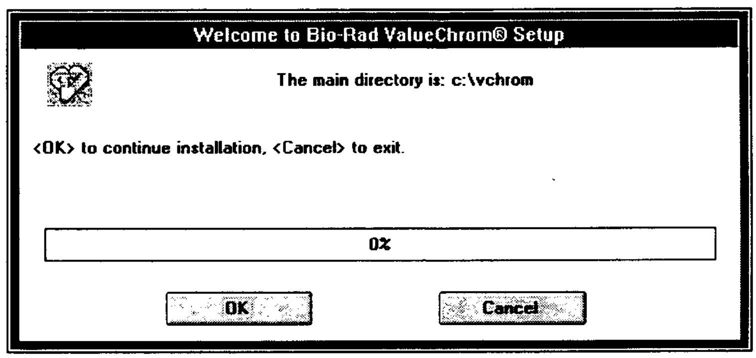
Double click the SETUP.EXE file.

The window below will appear asking you to specify a **Main Directory** (default is C:\vchrom\), **Data Directory** (default is C:\vchrom\data), and **Instrument Interface** communications port (default is **COM1**) for the **ValueChrom** software. The examples and tutorials in this manual are based on installation in the default drive and directories.



Click on **Install** to proceed.

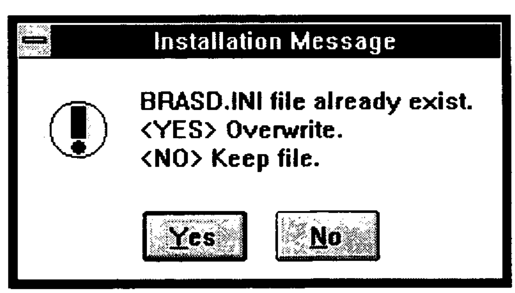
The window below will confirm your choice for the main directory.



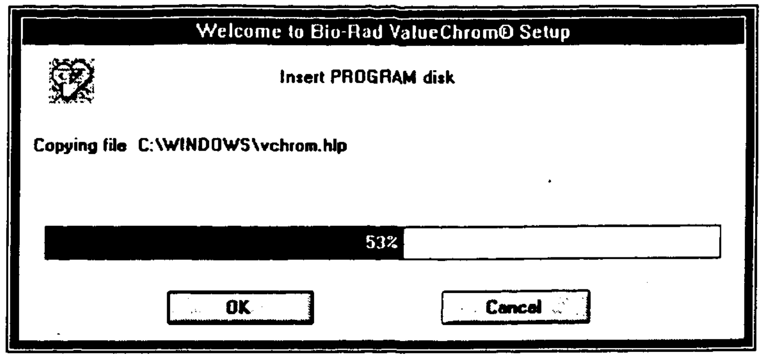
If the directory is correct, Click on **OK**.

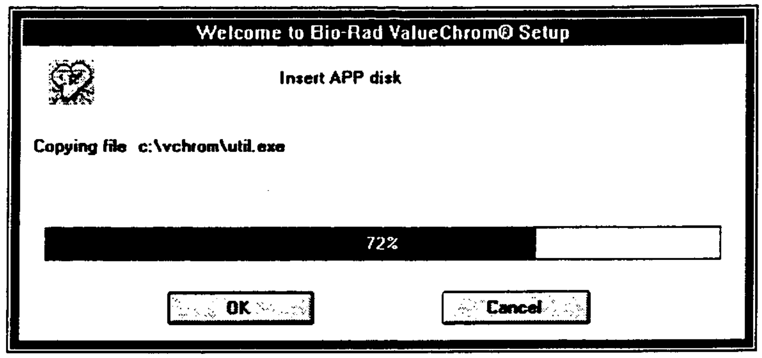
The window above will list **ValueChrom** files as they are copied to your hard disk.

You may also see the following message. Generally you should click on **no** at the following message.

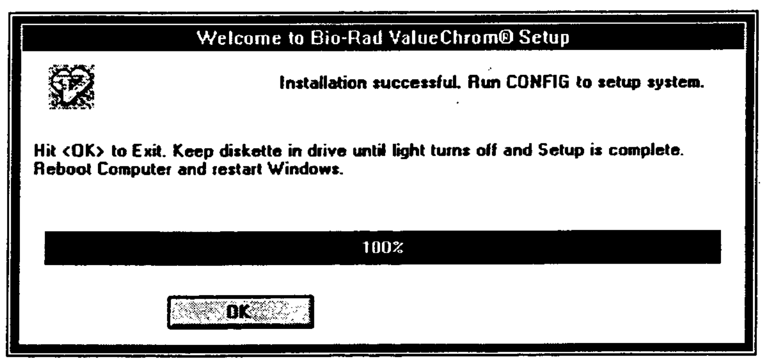


The installation program will ask you to insert the Program and App disks.



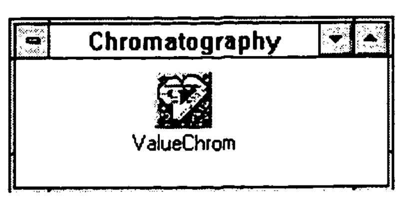


When the software installation is completed the window below will appear. Follow the instructions to run **ValueChrom.**



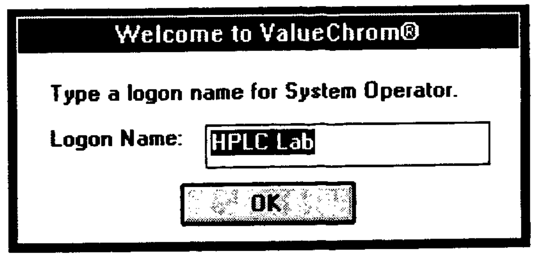
If problems occur during installation contact your local Bio-Rad support office. In the U.S. call **1- (800) 4 - BIORAD.**

The software installation program will open a Windows program group titled **Chromatography** and create a **ValueChrom** icon.



To run the **ValueChrom** software, double-click this icon

The window below will appear asking for a **Logon Name**. The name entered in this field will appear as the default for any field that asks for operator identification, throughout the software.



Click on **OK**.

If it becomes necessary to make a **ValueChrom** icon:

1. Select **New** from the **File** menu in **Program Manager**.
2. Select **Program Item**, then click on **OK**.
3. Type **ValueChrom** in the box labeled **Description**.
4. Type **C:\vchrom\vchrom.exe** in the **Command Line** box.
5. Type **C:\vchrom** in the box labeled **Working Directory**.
6. Click on **OK**. This will create an icon for the **ValueChrom** program.

Double click on the icon to start the program.

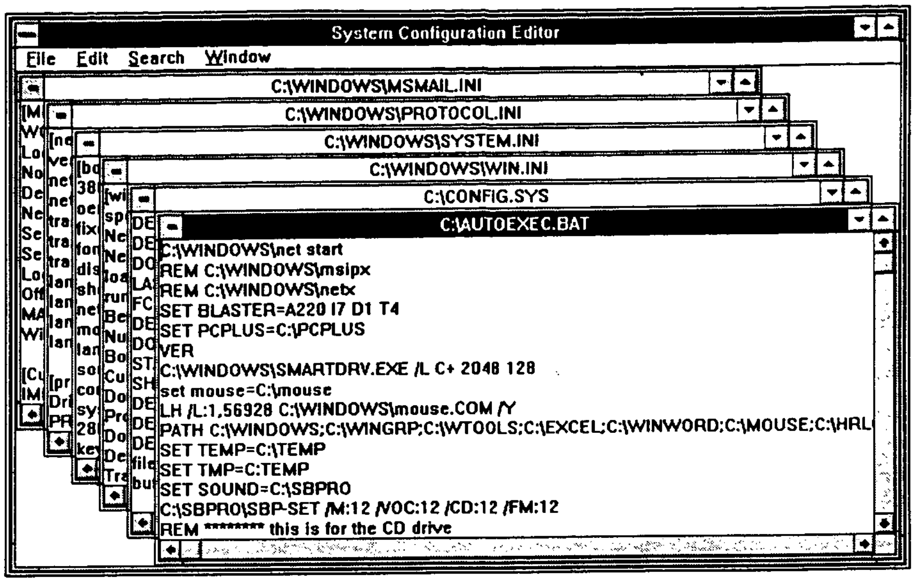
Editing System Files

Under normal conditions you will not need to edit system files to run ValueChrom, however, some programs will change these files causing ValueChrom not to operate. This information is provided for your convenience in case you need to edit these files.

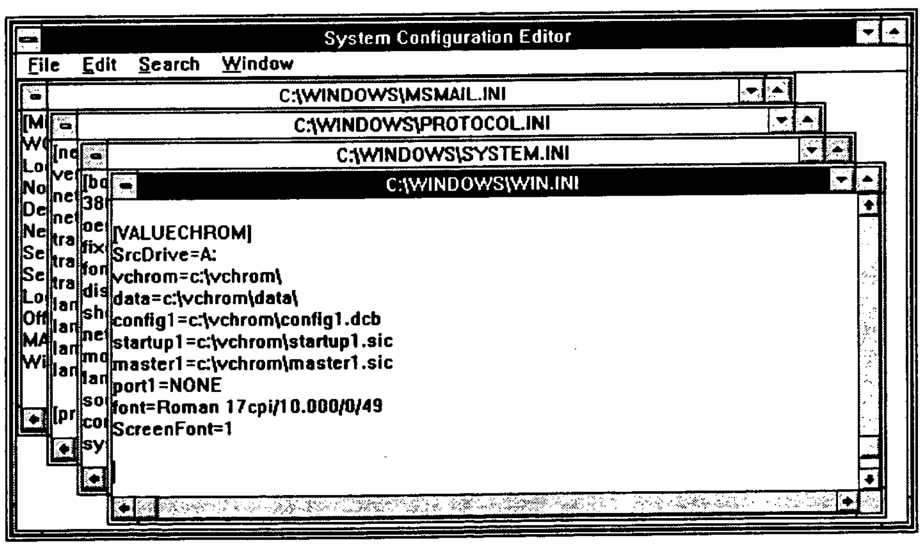
The ValueChrom installation program makes changes to the **AUTOEXEC.BAT** and **WIN.INI** files that are necessary to run the software. Copies of the original versions of these files are retained for your convenience. The Windows Accessory SYSEDIT can be used to edit these files at any time.

Below is an example of an **AUTOEXEC.BAT** file. The ValueChrom installation program adds **C:\vchrom** to the **PATH** statement and runs the **SHARE.EXE** program. If the statement **WIN** is included in the **AUTOEXEC.BAT** file (to start Windows automatically when the computer is turned on), the **SHARE.EXE** statement must come before the **WIN** statement. **SHARE.EXE** allows the system to share files. If it is not executed before running windows (i.e. if **WIN** appears in the **AUTOEXEC.BAT** before **SHARE.EXE**), windows programs such as

ValueChrom will not be able to share files. A symptom of this problem in ValueChrom is the Stripchart opening up without any pens assigned.



Below is an example of the **WIN.INI** file. This file contains instructions necessary for Windows to run program files. The ValueChrom installation program adds a section to this file titled **[VALUECHROM]**.



Software Operation

Windows Terminology

Working knowledge of MicroSoft Windows 3.x software is necessary to operate **ValueChrom**. If you are not already familiar with Windows, please refer to the Windows User's Guide for details.

*If you are familiar with Windows and HPLC you may want to proceed directly to "**Quick-Start Tutorial**" on page 23.*

MicroSoft Windows is an easy-to-use interface for MS-DOS personal computers. This interface provides standardized controls for each window available. **ValueChrom** was written to take advantage of the power of Windows, by allowing multiple windows to be open at once, and different programs to operate at the same time.

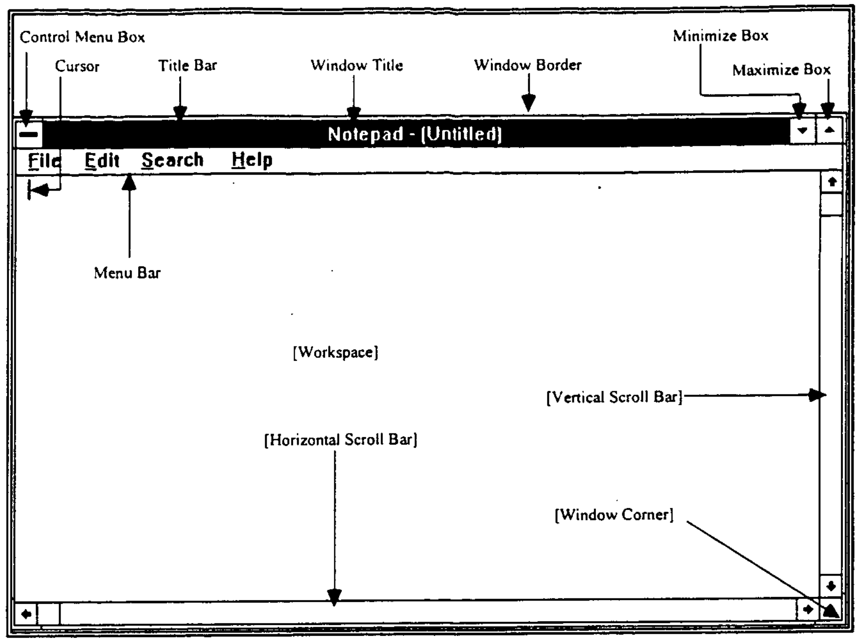
This section describes some of the Windows terminology contained in this manual. For more information, refer to the MicroSoft Windows User's Guide supplied with your Windows software. To benefit most from your HPLC system software, take some time to become familiar with Window's powerful features.

Windows Controls

Each window has several ways to perform most available commands. This is illustrated in the figure below.

The Control-menu box, in the upper left corner of the window, supplies a list of available commands for that window. Place the mouse pointer on the Control-menu box and press the left mouse button. A list of commands available for that window will appear.

The Menu bar contains a list of menu items, each of which contain commands that effect the open window. To access these commands, place the mouse pointer on the desired item on the menu bar and press the left mouse button. A list of commands will be displayed and you may choose different selections by moving the mouse pointer to them and clicking the left mouse button.



Mouse Interaction Technique

There are four basic operations for using the mouse. They are:

Click: A *click* consists of pressing and releasing the left mouse button without moving the mouse pointer. Click is the method of selecting objects and actions.

Double-click: A *double-click* consists of two rapid, consecutive clicks without moving the mouse pointer. **When you double-click on an object that has a default action, that default action is performed**. For example, the default action of the control menu box is closing the window. When users double-click on the control menu box for any window, that window is closed.

Drag Select: *Drag select* consists of pressing the left mouse button and holding it down while moving the mouse so that the pointer travels to a different location on the screen. Dragging ends when the mouse button is released. All items between the button down and button up points are selected.

Direct Manipulation: *Direct manipulation* is similar to *drag select*; the mouse is positioned over a particular item on the screen, the left mouse button is pressed and held down while moving the mouse to a new location, and the item beneath the mouse pointer is changed in size or position.

Direct manipulation allows users to control the appearance of the computer desk top using the mouse. For example, users can size a window by selecting the window's border segment with the mouse and moving the window border until the window is the desired size.

ValueChrom Terminology

The following is a list of terms that are important to know while going through this software manual and using the **ValueChrom** software.

Calibration: At a uniform flow, peak area is directly related to the concentration of the eluting compound. A **Calibration Curve** is a plot of the absorbencies of standards over a known concentration range. The absorbance of an unknown can be compared to this plot to determine its concentration. **Calibration curves** are generated through a **Sequence table**, and linked to individual methods. When the method is run, a **Calibration report** may be printed along with the integration report.

Configure: The term configure is used both to designate what type of modules you have in your HPLC system and how you want them to operate. Before using **ValueChrom** you must establish which peripheral equipment you have connected to each system (see "Configuring the Software for Your System" on page 20). Use the **Configure** screen in the method table to select sampler type or mode of fraction collection.

Data File: Data files are the DOS files that contain the data from your HPLC detector. Raw data files are created by the analog to digital converter and are stored on your hard disk during an HPLC run. While data files cannot be edited or changed once they are collected, the parameters for analyzing the data can. Data files end in the .ACQ extension.

Data Channel: Each detector is associated with one of the available data channels on the Instrument Interface. Data channels are assigned to strip chart pens in the method table using the **Assign Pens** screen. To integrate data from a detector, you must also assign it to an integration channel.

Gradient: A gradient involves a change in solvent composition over time. Gradients are programmed on the **Timed Events Screen** of the method.

Integration: Integration is the analysis of a raw data file. Integration parameters determine the boundaries that the computer will use to define chromatographic peaks.

Main Menu Bar: The main menu bar is the collection of pull-down menus that appears across the top of the **ValueChrom** window.

Method: A method contains all programmed commands necessary to run an HPLC sample, for example, flow rate, run length, and integration parameters

Sample: A sample is the substance to be analyzed.

Sequence Table: A sequence table is a series of methods and samples programmed to run, automatically. Sequence tables are used when running more than one sample.

Strip Chart: A strip chart is used for graphical representation of data files (both raw and analyzed). The strip chart allows you to set the display parameters, alter the integration parameters, generate new integration reports and print chromatograms.

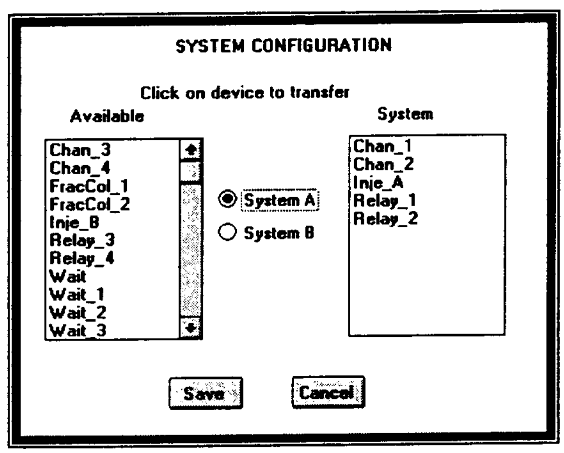
Configuring the Software for Your System

ValueChrom automatically configures the most common hardware setup at installation. For specialized systems you may need to make changes to the configuration. The default configuration is 2 data collection channels, 2 pumps, a manual injector, and 2 relay outputs. (Systems without pump control cannot have the pumps configured.)

To change your system configuration:

Select **Configure** from the **Control Panels** menu then select **Set System** from the sub menu to access the **System Configuration** screen. The box on the left lists all the modules the HPLC software can control. The box on the right lists all modules assigned to a system.

Note: Even if you do not plan to use them, installing two relays and two data channels is recommended.



To assign a module to a system, click on the device and it will be transferred to the selected system.

To un-assign a module click on it again.

Note: If the Instrument Interface is not turned on or is improperly connected, you will get an error message. Turn on or reconnect the Instrument Interface and repeat this sequence.

When the system is properly configured click on **Save**. The software will send the new configuration over the RS232 port to the Instrument Interface.

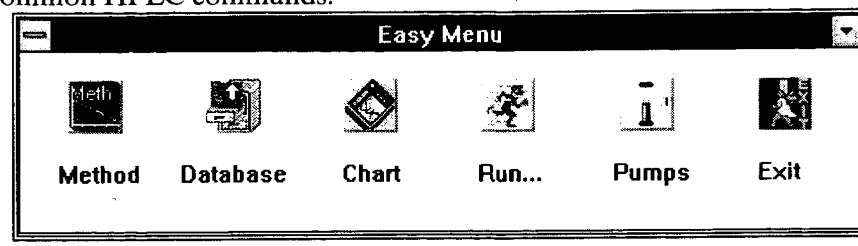
Some devices have different modes of operation. To select the mode of operation appropriate for your system you must configure the module. To configure a module:

Select **Configure** from the **Control Panels** menu then select the module you wish to configure; **Set Pumps**, **Set Inputs**, **Set Fraction Collectors**.

Select the appropriate information for each device, then click **Save**.

The Easy Menu

This menu is displayed when you first enter the **ValueChrom**. You can access the Easy Menu at any time by selecting **Easy Menu** from the main menu bar. The Easy Menu is a quick way to access the five most common HPLC commands.



Method:

Select this icon to open an existing HPLC method, Sequence Table, or Calibration Table. To create a new table, choose Create from the File menu, or open an existing table and Save As using a different name. See "Creating a Method" on page 33 for more information.

Run:

Select Run Method icon when you are ready to run either a method or Sequence Table. You will be asked for run information before starting the method.

Pumps:

The Pump control icon allows manual operation of the pumps. See " Pumps" on page 29 for more information.

Chart:

The Chart icon provides immediate access to the video strip chart.

Database:

The Database icon provides immediate access to the chromatogram database. From the database screen you can easily reanalyze or print reports from your data.

See "Chromatogram Database" on page 55 for details.

Exit:

This exits ValueChrom. When you exit from the Easy Menu you will be given the option to leave the pumps running when the software is no longer running.

Local / Remote Operation

From the **Solvent Delivery System** window you can switch to front panel control of the pumps by clicking the **Local** operation button. When pumps are in **Local** mode they will not respond to computer commands.

Flow Rate / Solvent Composition

Note: The Bio-Rad Model 1350, Model 2700 and Model 2800 pumps are equipped with the Soft-Start feature. When the pump is started, Soft-Start will gradually increase the flow rate over 1 minute, to the final programmed value to avoid sudden shock to the column, extending column life.

To set flow rate / solvent composition:

Enter the percentage of the total flow for each pump in the % box, near the bottom of the screen.

Enter the total flow rate in the box labeled **Total Flow**

Click the **SET** button.

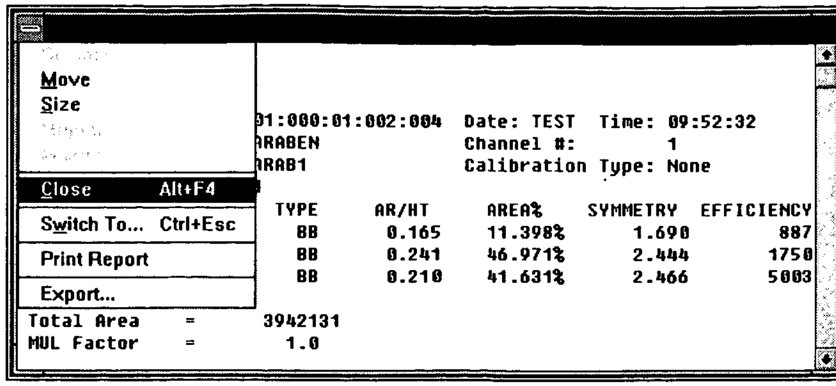
The actual flow rate of the individual pumps will appear in the bars below the **Stop** buttons.

To stop flow:

Click on the Stop button or enter 0 in the Total Flow box.

To resume flow:

Click on the Run button.



*Choose **CLOSE** or **ALT-F4** to close the report screen.*

Step 3. View the Method

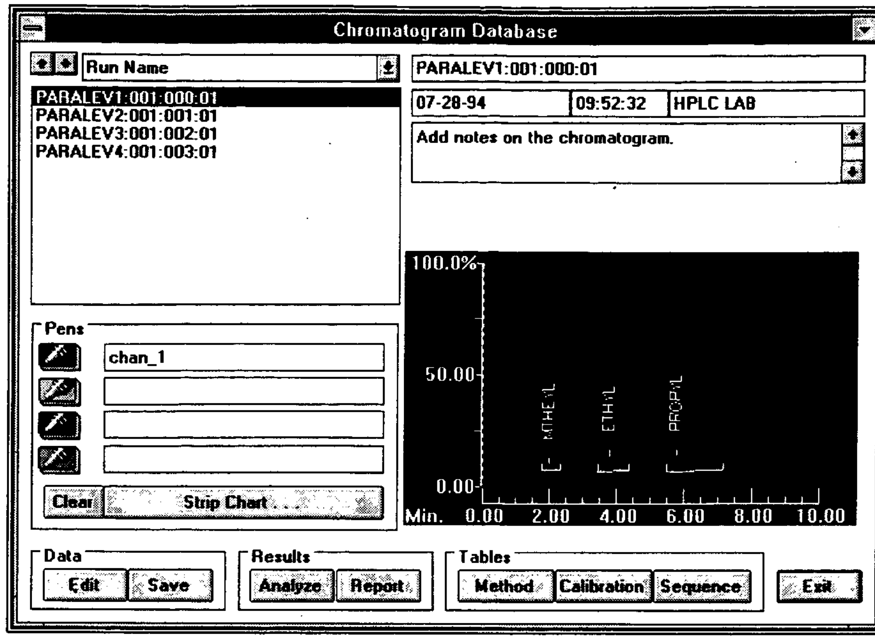
Hint: The easy way to create a method is to open a method, modify it, and save it as a different name.

Viewing a method is similar to viewing a report.

1. Select the chromatogram of interest.
2. Click on the **METHOD** button in the tables section.
3. The method will open showing the timed events screen.
4. Type ALT-F4 to close the method editing screen.

Step 4. Advanced Integration -- Baseline Editing

1. From the Database screen select a chromatogram to analyze.
2. Click on the strip chart button in the pen section of the database screen and the strip chart will enlarge.
3. Select Baseline edit from the strip chart menu bar.



The Database Screen

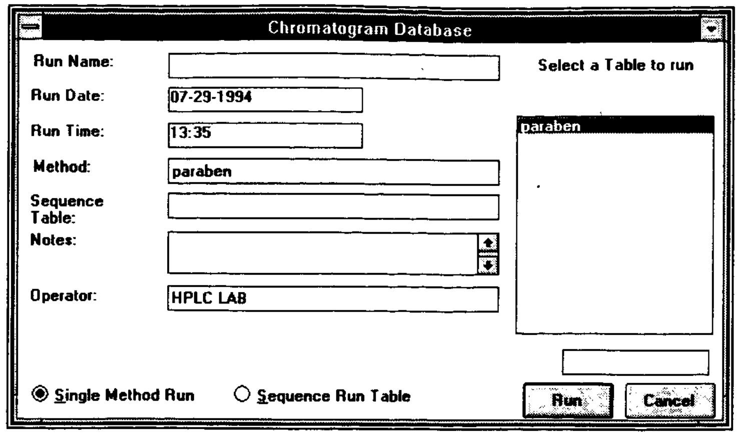
Note: If you did not select the default directory (C:\VCHROM), these files will not be present. You will need to edit the database, or reinstall the software using the default directories.

The database screen list all the runs stored in your system. If your system is new you will see four runs shipped with the software and the run you completed in "Running a Test Chromatogram" section. The run you completed will be named PARABEN and will have a group of numbers assigned to it. These number will be explained later in the manual.

To view each chromatogram click on the run in the list box in the left side of the screen. Each chromatogram will be displayed along with the corresponding pump trace.

To compare two chromatograms

1. Select the run PARALEV1 by clicking on it with the mouse.
2. Click the red pen icon in the screen section labeled pens
3. Click twice on the green pen icon in the pen section to clear the pen of any data.
4. Click on the green pen button again to activate it. (Active is pressed in.)
5. Select the second run PARALEV4 and both runs will be overlaid on the strip chart display.
7. Click the clear button in the pen section to clear the strip chart. (Later in the manual we'll see how to display different peak labels.)



The Run Chromatogram Screen

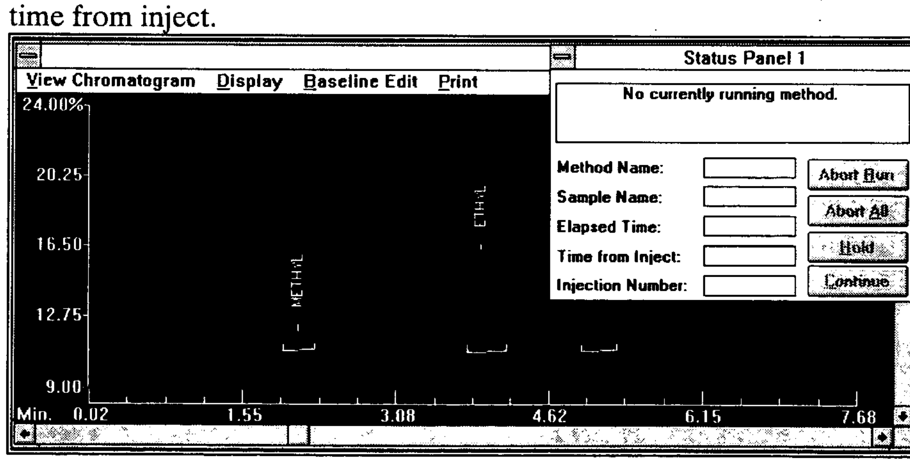
Click on **PARABEN** with the mouse and then click **RUN**. (We'll learn about the remaining information later in the manual.)

Step 3. Inject Your Sample

Note: The pumps will continue to flow at the initial programmed flow rate giving plenty of time for the system to stabilize, or if need be, for you to prepare the sample.

After about 1 minute the status box will say "Waiting for Inject". If your baseline looks stable inject the sample. If your baseline still needs to settle, wait a few minutes. (The red pen is from your detector, and the green pen is the pump information.)

Put the sample into the injector with the injector in the **load** position. When you move the handle from load to inject to strip chart will zero itself and data collection will start. The status box will start giving the time from inject.



Enlarged Strip Chart display showing Status Panel

Step 4. Strip Chart Controls

Once you sample is injected, you can manipulate screen controls to view the chromatogram differently. For example zooming.

To zoom in

Methods

Introduction

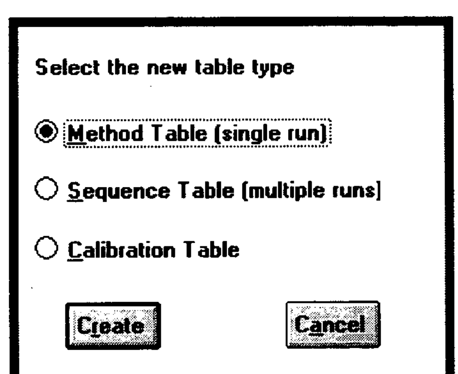
A Method Table contains all the information necessary to run an HPLC sample; pump flow, injection, gradient control, data collection and parameters for integrating data. A method may be calibrated to provide quantitative information. Fraction Collectors and relay operated devices may also be controlled.

Creating a Method

Hint: The fastest way to create a method is to open an existing method, make any changes and save it with a new name.

To create a new method:

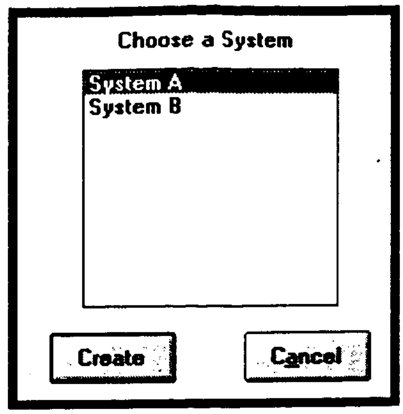
Select **Create** from the **File** menu.



In the box labeled **Select the new table type** select **Method Table [single run]**, then click **Create**.

__Remember:__ Each system, either A or B is configured with different hardware, pumps, detectors,...etc.

Before you can create a method you must configure your system.



Select the appropriate system, then click **Create**.

At a minimum, each method must contain a **Gradient Table**. Optional method information includes; **Assign Pens**, **Integration**, **Configure**, **Notes**, and **Calibration**.

See "Configuring the Software for Your System" on page 20 for more information.

Timed Events Screen

The **Timed Events Screen** is the first screen shown in the new method file. The layout of this screen will reflect the information you provided about your system configuration. For example; if you have configured pumps A and B to your system there will be two columns labeled **%Pump_A** and **%Pump_B**, if you have configured pumps C and D to your system there will be two columns labeled **%Pump_C** and **%Pump_D**.

Note: To delete a line from this screen...

1. Click on the line you want to delete

2. Click on the right mouse button

3. Press delete

Method: UNTITLED1

Configure Timed Events Assign Pens Integration Notes Next Previous Save

Time	% Pump_A	% Pump_B	Flow Rate	Inje_A	Relay_1	Relay_2	Fraction	Int #	Pk
0.00	100.0	0.0	1.500					1	
1.00			1.500	I				1	
6.00			1.500					1	

This screen is used to control all timed events; flow rates, gradients, injection times, relay closures and changes in integration parameters. To access this screen at any time while creating the method select **Timed Events** from the menu bar.

Pump / Injection Programming

The minimum information required to run a method includes flow rate, gradient, and injection information. You must have configured a pump and an injection device to your system. To create a new pump / injection program:

Note: Bio-Rad's HPLC pumps include the Soft-Start feature which, on startup, ramps the flow to the programmed rate to avoid pressure shock to the column. If using the Soft-Start feature in your method you should program at least 1 minute between time zero and inject.

At time zero, enter the initial pump settings (i.e., **%Pump_A**, **%Pump_B** and **Flow Rate**).

Enter a second line, at some time after zero, that has identical pump information (**%Pump_A**, **%Pump_B** and **Flow Rate**) and enter the letter "I" in the **Inje_A** column (**AutoInject_A** for an automatic sampler), to signal an injection. *You cannot enter an injection at time zero!*

To program a gradient into the method:

Enter a line that contains the starting solvent composition at the time you wish the gradient to start.

Enter a line that has the information reflecting the final solvent composition at a time that will signal the end of the gradient (**%Pump_A**, **%Pump_B**). See the example below.

Note: The appearance of this sample **Timed Events Screen** may be different than your own. The layout of this screen depends on the information provided in the system configuration when you first installed your system. For example ValueChrom Data Stations will not have pump control.

Method: UNTITLED1

Configure Timed Events Assign Pens Integration Notes Next Previous Save

Time	% Pump_A	% Pump_B	Flow Rate	Inje_A	Relay_1	Relay_2	Int #	Pk Width	Threshold	Skim	Integ
0.00	100.0	0.0	1.500				1				
1.00	100.0	0.0	1.500	I			1				
11.00	0.0	100.0	1.500				1				
11.50	100.0	0.0	1.500				1				

*Note: If you enter a time line that has no entries in the **%Pump_A** or **%Pump_B** columns there will be no effect on the solvent composition or gradient.*

In this example the initial conditions (time = 0.00) are 100% pump A, 0% pump B, and a flow rate of 1.5 ml/min. At time = 1.00 minute, the method will pause to wait for a signal from the injector. In the next line, time = 11.00 minutes, the pump settings are 0% pump A, 100% pump B. This means, beginning at the injection, the solvent composition will gradually change from 100% pump A to 100% pump B over 10 minutes. Between time 11.0 and 11.5 the pumps will gradually return to the initial conditions.

To delete a line in the Timed Events Screen:

1. Point to the time line to delete
2. Click the right mouse button
3. Press delete.

The last **Time** entered in the **Timed Events Screen** signals the end of the method.

Integration Using the Timed Events Screen

*Note: To edit integration parameters from the **Timed Events Screen** there must be an entry in the **Integ** column at time zero.*

Integration parameters control the boundaries the computer uses to define chromatographic peaks. Bio-Rad's **ValueChrom** software allows you to change integration parameters within a method using the **Timed Events Screen**. If the method uses the same integration parameters throughout, they may be set in the **Set Parameters** screen Any directives entered on the **Timed Events Screen** take priority over parameters entered on the **Set Parameters** Screen.

The column labeled **Int#** specifies the integration channel and is a non-editable field. Entries made to the right of this column apply to the integration channel shown.

The column labeled **PkWidth (Peak Width)**defines a time window over which the computer will average absorbance values. This value, together with the **Threshold** value are used to determine a rate of change. When the rate of change in the baseline (up or down) is greater than this number, the computer determines this the start of a chromatographic peak. When the rate of change returns to a value smaller than this rate of change, the computer considers this the end of the peak. Valid entries for this field are 0.01 to 10 minutes. If you are running an isocratic method, peak widths may increase over time. It may be useful to increase the **PkWidth** value to avoid integrating baseline noise.

*Note: **ValueChrom** can set threshold and peak width parameters automatically. For more information see "Slope Sensitivity Parameters" on page 43.*

Threshold works with **PkWidth** to determine a rate of change value. When the rate of change in the baseline (up or down) is greater than this number, the computer determines this as the start of a chromatographic peak. When the rate of change returns to a value smaller than the rate of change value, the computer considers this as the end of the peak. The lower the threshold setting the more sensitive the peak detection will be. Allowable entries for this field are 0.001 to 15.

The **Skim** setting determines whether the software will integrate overlapping peaks using a perpendicular drop line or a tangent skim line. If two peaks overlap, the integrator can drop a perpendicular line straight down from the valley point or it can skim the smallest peak. The skim setting is a ratio between peak heights. If the ratio of the peak heights is smaller than the skim setting, a drop line will be used for integration. The default value for this parameter is 10.

The integrator may be turned **ON** or **OFF** during a method in the column labeled **Integ**. *To use the **Timed Events Screen** to set integration parameters you must have an entry in this column at time zero.*

If there are areas of a chromatogram with no peaks of interest it may be useful to turn off integration to avoid integrating unnecessary information.

There are three allowable entries for the **Function** column; **ALLVV**, **HOREXT** and **NONE**. **ALLVV** instructs the software to integrate all overlapping peaks by drawing a baseline from the valley of one peak to the valley of the next peak. **HOREXT** will extend a baseline from the start of the first peak to the end of the last peak, integrating from the valley point down to the baseline. **NONE** will turn off the previous directive.

Turning **NegPks** on (**YES**) allows peaks that extend below the baseline to be integrated as regular chromatographic peaks.

To delete a line in the Timed Events Screen:

Point to the time line to delete, click the right mouse button, then press delete.

The last **Time** entered in the **Timed Events Screen** signals the end of the method.

Relay Control

If you have configured the Relay outputs on your system, you can control them with the **Timed Events Screen**.

To activate a relay, enter a "+" sign in the **Relay** column at the time you want the relay to close.

To open the relay, use a "-" sign.

To momentarily close a relay (1 second), use a "**P**" in the **Relay** column.

It may be useful to connect a relay contact closure to the zero input on the detector. You can zero the detector by pulsing the relay at the time of inject.

Fraction Collection

If you have configured a Fraction Collector to your system, you can control it from the **Timed Events Screen**. To turn the fraction collector on or off, enter **ON** or **OFF** in the **Fraction** column. Some functions of the Fraction Collector must be controlled in the **Configure Screen**.

Note: Some fraction collectors such as Bio-Rad's Model 2128, have advanced peak detection capability built in to the collector. To take full advantage of these capabilities refer to the instruction manual provided with the fraction collector.

Method: UNTITLED1

Configure Timed Events Assign Pens Integration Notes Next Previous Save

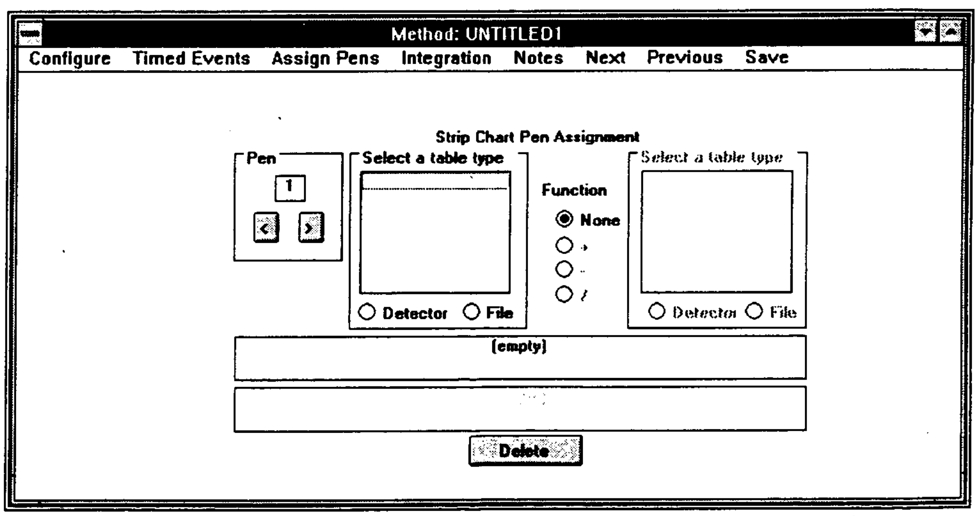
Time	% Pump_A	% Pump_B	Flow Rate	Inje_A	Relay_1	Relay_2	Fraction	Int #	Pk Width	Threshold	Skl
0.00	100.0	0.0	1.500					1			
1.00	100.0	0.0	1.500	I	P		ON	1			
11.00	0.0	100.0	1.500				OFF	1			
11.50	100.0	0.0	1.500					1			

Timed Event Screen showing Fraction Collection and Relay operation

Assign Pens

*Note: If you do not assign any pens, **ValueChrom** will assign pen 1 to the lowest number data channel configured to the system (i.e., **Chan_1**), and pen 2 to data channel 2, etc.*

Selecting **Assign Pens** from the menu bar brings up the **Strip Chart Pen Assignment** window. There are four strip chart pens available while running a method. You can display signal from a detector, pump information (gradient profile) or a stored data file (i.e., a standard chromatogram to compare to an unknown). Only two detector channels may be displayed during a run. These channels correspond to the 2 channels of data available to each system through the Instrument Interface.



Each pen is assigned individually by incrementing the pen selection box (< or >) in the upper left of the screen.

Viewing Real Time Data During A Run

Select the **Detector** option in the box labeled **Select a table type**. The header will change to **Select a device** and a list of the available real-time data collection options will appear.

Select the data you would like to see. When you have made a pen assignment it will appear in the box directly below **Select Table Type** (when the pen is not assigned, this box reads **[empty]**).

To delete a pen assignment, click on the **Delete** button at the bottom of the screen.

This data will be shown on the **Strip Chart** screen when the method is run.

View Previously Stored Data During a Run

Select the **File** option in the box labeled **Select a table type**.

Select **Method** for a single method run or **Sample** for a data file collected with a Sequence table.

Select the appropriate **Data File**. When you have selected a data file it will appear in the box directly below **Select Table Type** (when no channel is assigned, this box reads **[empty]**).

To delete a pen assignment click on the **Delete** button at the bottom of the screen.

When viewing stored data you can choose to see the data file added, subtracted, or divided by another data file. After selecting data file above;

Select the desired function (+, -, / buttons listed under **Function**).

Select a data file to add to, subtract from, or divide into the original data file.

This data will be shown on the **Strip Chart** screen when the method is run.

Integration

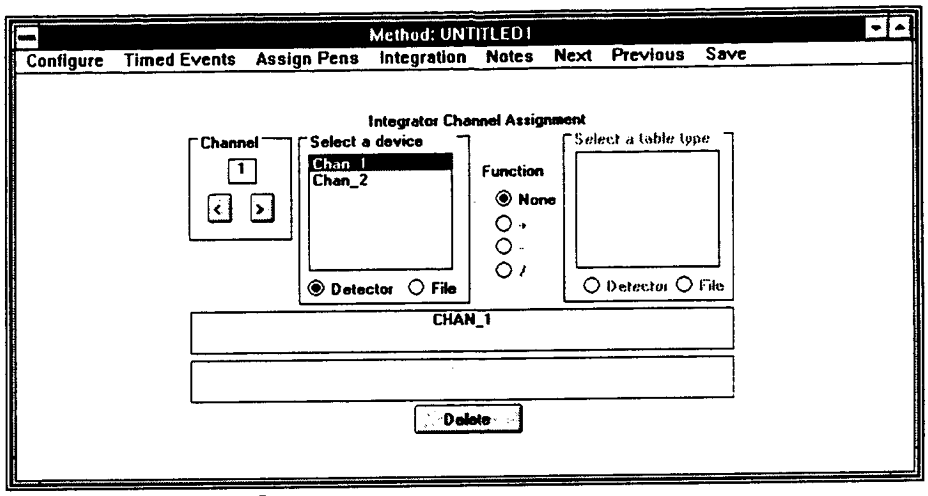
Within the menu item, **Integration**, there are three options. In the **Assign Integrator** window, integrator channels are assigned to incoming data. In the **Set Parameters** screen, parameters for data analysis are set and reports are programmed to print. **Calibration** allows you to link the method to an existing calibration table.

Assign Integrator

*Note: If you do not assign the data channels to integrator channels, **ValueChrom** will assign integrator channel 1 to the lowest number data channel configured to the system (i.e., **Chan_1**), and integrator channel 2 to data channel 2.*

Selecting **Assign Integrator** from within the **Integration** menu brings up the **Integrator Channel Assignment** window (see figure below). During a run the data will be automatically integrated, in real time, if integrator channels are assigned for the incoming data. Each system can have up to two channels of data integrated. Each channel can be integrated with separate integration parameters.

The box labeled **Channel** in the upper left corner allows you to toggle between the two available integration channels.



Integrator Assignment Screen

To assign a data channel to an integrator channel;

1. Select **Detector** in the box labeled **Select table type**. A list of available on-line devices will be displayed.
2. Select the detector you wish to assign to the current integration channel When a detector has been selected, it will appear in the box directly below **Select Table Type** (when no channel is assigned this box reads **[empty]**).
3. To delete an integrator channel assignment, click on the **Delete** button at the bottom of the screen.

Set Parameters

Selecting **Set Parameters** from the **Integration** menu brings up the **Integration Parameters and Reports** screen. From this screen, integration parameters are selected for data analysis and report types are selected to print at the end of the method.

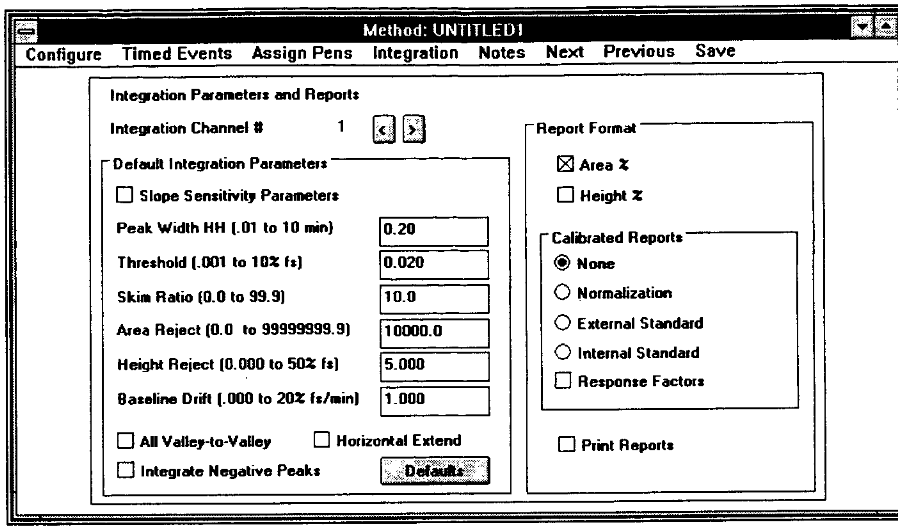
Integration Parameters

Integration parameters control the boundaries the computer uses to define chromatographic peaks. If the method uses the same integration parameters throughout, they may be set in the **Integration Parameters and Reports** screen. It is possible to set these parameters automatically or manually. See "Integrator 1 (2):" on page 106 for information on optimizing parameters for your chromatogram.

The **Integration Parameters and Reports** screen is also used to select the type of report that will be printed at the end of each run. Below is an example of the **Integration Parameters and Reports** screen.

*Note: If it is necessary to change the integration parameters within a method it must be done on the **Timed Events** screen.*

*Any directives entered on the **Timed Events Table** screen take priority over parameters entered in the **Integration Parameters and Reports** Screen.*



Hint: Click the default button to see what predefined parameters are available.

ValueChrom has two options for setting integration parameters from the Integration Parameters and Reports Screen. There are editable predefined Default integration parameters included with the software or, the Slope Sensitivity Parameters option which instructs the software to calculate appropriate integration parameters automatically.

Any of the integration parameters can be altered manually. An explanation of each parameter follows.

A method may contain separate integration information for each of two integration channels. Select the integration channel by clicking on **Integration Channel # [< , >]** in the upper left corner of the screen.

Peak Width at Half Height

PkWidthHH defines a time window over which the computer will average absorbance values. This value, together with the **Threshold** value are used to determine a rate of change value. When the rate of change in the baseline (up or down) is greater than this number, the computer determines this the start of a chromatographic peak. When the rate of change returns to a value smaller than the rate of change value, the computer considers this the end of the peak. Valid entries for this field are 0.01 to 10 minutes. If you are running an isocratic method, peak widths may increase over time. It may be useful to increase the **PkWidthHH** value to avoid integrating baseline noise.

Threshold

Threshold works with **PkWidthHH** to determine a rate of change value. When the rate of change in the baseline (up or down) is greater than this number, the computer determines this the start of a chromatographic peak. When the rate of change returns to a value smaller than the rate of change value, the computer considers this the end of the peak. The lower the threshold setting the more sensitive the peak detection will be. Allowable entries for this field are 0.001 to 15. Setting the appropriate threshold value can eliminate integration of background noise while allowing integration of peaks of interest.

Note: ValueChrom can set threshold and peak width parameters automatically. For more information see "Slope Sensitivity Parameters" on page 43.

Skim Setting

The **Skim** setting determines whether the integrator will analyze overlapping peaks using a perpendicular drop line or a tangent skim line. If two peaks overlap, the integrator can drop a perpendicular line straight down from the valley point or it can skim the smallest peak. The skim setting is a ratio between peak heights. If the ratio of the peak heights is smaller than the skim setting, a drop line will be used for integration. The default value for this parameter is 10.

Area and Height Reject

Area Reject is the minimum area the integrator will detect as a chromatographic peak. This parameter is useful for keeping the integrator from integrating background noise the area of which is less than the peaks of interest. Allowable entries for this field are 0.0 to 99999999.9.

Height Reject is the minimum peak height the integrator will integrate as a chromatographic peak. Allowable entries for this field are 0 to 50% of full scale.

Baseline Drift

Baseline Drift allows the entry of a percentage drift the integrator will interpret as drifting baseline. Allowable values are 2-20%. Valleys that extend below the drift setting are considered baseline resolved.

Baseline Marking

Selecting **All Valley-to-Valley** instructs the integrator to integrate all overlapping peaks by drawing a baseline from the valley of one peak to the valley of the next peak.

Horizontal Extend will extend a baseline from the start of the first peak to the end of the last peak, integrating from the valley point down to the baseline.

Turning **Integrate Negative Peaks** on allows peaks that extend below the baseline to be integrated as regular chromatographic peaks.

Predefined Default Integration Parameters

Predefined Defaults make integration easy. Select the peak types you have such as narrow, average, large, or Prep.

ValueChrom contains 4 sets of default integration parameters that are appropriate for many chromatograms. Each time you press the **Default** button on the **Integration Parameters and Reports** screen, the parameter windows will display the next set of values and the name of the preset defaults. You may change any of these values as necessary.

When the method is saved the displayed values will be saved with the method.

Slope Sensitivity Parameters

ValueChrom can automatically calculate integration parameters for you. To use this feature you must first run a method that has baseline monitoring (see "Baseline Monitoring"). When baseline monitoring is run, it measures the detector's noise and drift. This information is stored on the hard disk, and is used to determine the appropriate integration parameters.

Baseline Monitoring

Baseline monitoring is programmed using the **Timed Events Screen** of the method table.

Method: UNTITLED1

Configure   Timed Events   Assign Pens   Integration   Notes   Next   Previous   Save

Time	Flow Rate	Inje_A	Relay_1	Relay_2	Fraction	Int #	Pk Width	Threshold	Skim	Integ	Function	N
0.00	0.000					1				ON		
1.00		I				1				BMS		
10.00						1				BMF		

Timed Events Screen showing Baseline Monitoring

To monitor baseline :

1. Create a new method. The method should have your normal gradient, pens, and integration assigned.

2. On the **Timed Events Screen**, enter **ON** in the column labeled **Integ** at time zero.
3. Enter **BMS** (Baseline Monitoring Start) in the **Integ** column at the point which would normally signal the injection. Do not enter an injection in the method.
4. At the end of the run, enter **BMF** (Baseline Monitoring Finish) under **Integ**.
5. Save the method and run it.

After running the baseline monitoring method, **ValueChrom** will automatically calculate integration parameters and save them to the hard disk. They will be stored in a text file called **BRASD.INI**.

To use these integration parameters for your HPLC runs, select the box labeled **Slope Sensitivity Parameters** on the **Integration Parameters and Reports** screen.

How Baseline Monitoring Works

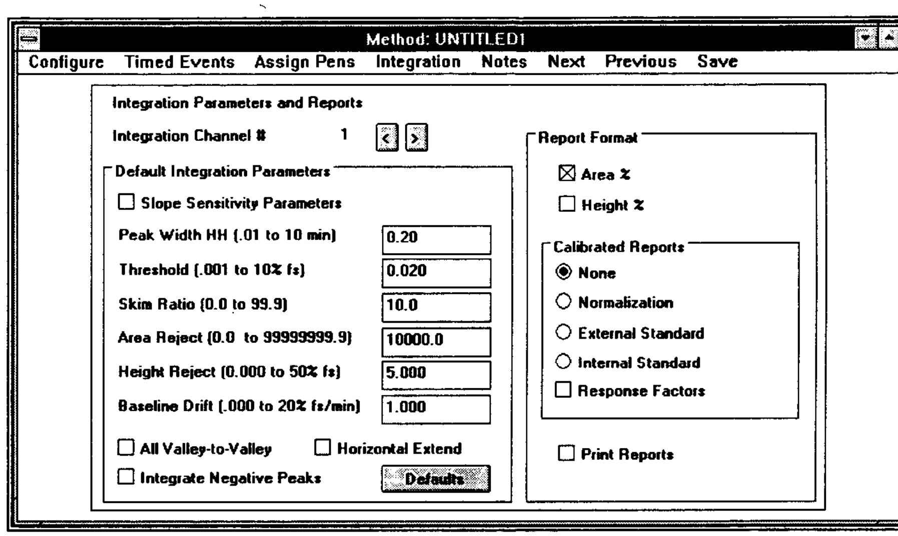
The text file, **BRASD.INI** contains a section labeled **[INTGAPP]**. The values in this section are calculated by the software when baseline monitoring is performed. The time and date are included so you can see when the baseline was last monitored. Do not alter this section of the **BRASD.INI** file. Below is an example of an **[INTGAPP]** section.

```
[INTGAPP]
threshold=0.156
slope=4.128
intercept=32704.647
drift=1.181
date=09-27-1991
time=15:51:43
sigma=107.3722
timefact=10
maxval=209715
param=3INTGAPP
```

Note: It may not be necessary to monitor the baseline for the length of an entire HPLC run. Often a 4 minute baseline monitoring method will provide enough data for automatic integration parameter calculation to work correctly. A possible alternative to a separate baseline monitoring method is to run baseline monitoring for the first few minutes of every run. To do this, monitor baseline for the first few minutes of every method, then do the injection. If you are running a gradient with high baseline drift, you should run baseline monitoring over the entire gradient.

Reports

From the **Integration Parameters and Reports** screen, several different report types can be selected to print at the end of the method. In the **Report Format** box on the right of the screen the report types are listed, including **Calibration Reports**. Each report type provides different information about the integrated data.



Area %

Selecting Area % will produce reports where peak size is reported in Area units. An area % will also be displayed for each peak relative to the total peak area of all of the peaks reported.

Height %

Selecting Height % will produce reports where peak size is reported in Height units. A Height% will also be displayed for each peak relative to the total peak height of all of the peaks reported.

Calibrated reports

Specifies how calibration reports will be printed out following the analysis of a calibration standard. The following options are available.

None

No calibration report will be printed.

Normalization

Produces report where the amount of each peak is given as a percentage of the area of all the peaks reported.

External Standards

Selecting External Standards produces a report calculated using the external standards specified in the calibration table.

Internal Standards

Selecting Internal Standards produces a report calculated using the internal standards specified in the calibration table

Response Factors

Includes the peak response factors for each identified peak in the report.

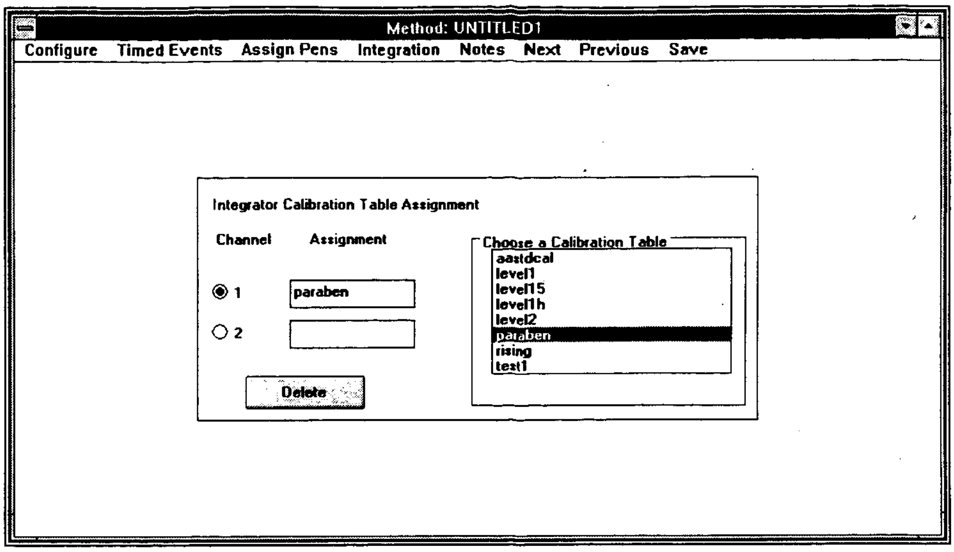
At the bottom of the **Report Format** section of the **Integration Parameters and Reports** screen is a box titled **Print Individual Reports**. If this box is checked, reports will also be printed on a separate page.

See also "Getting Your Results Ready to Publish" on page 124 and "Reports & Results" on page 64.

Calibration in Methods

Selecting **Calibration** from the **Integration** menu brings up the **Integrator Calibration Table Assignment** screen. From this screen you can link a method to a calibration table. Below is an example of a **Integrator Calibration Table Assignment** screen.

Response factors are determined by running samples in a sequence table; Each level is entered separately. See "Calibration" on page 83 for more information.

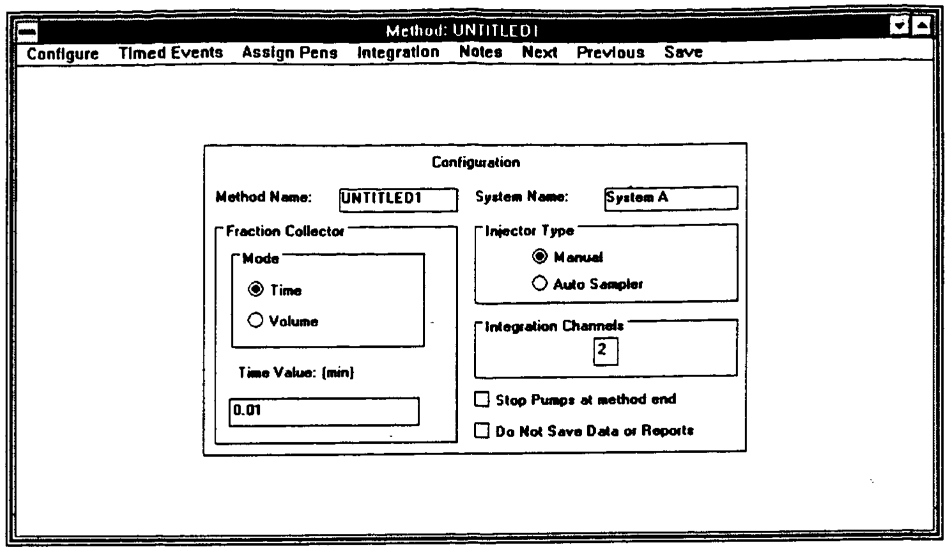


To assign a calibration table to a method;

1. Select an integrator channel (each integrator channel may be linked to a different calibration table).
2. Select a calibration table from the list of available calibration tables shown in the **Choose a Calibration Table** box on the right of the screen. The selected table will appear in the **Assignment** box.
3. To delete an assignment, click the **Delete** button at the bottom of the screen.

Configure Method

Selecting **Configure** from the menu bar brings up the **Configuration** screen. From this screen, adjustments can be made to the method based on the system configuration assigned in **Set System** in **Configure** in the **Control Panels** menu .



If there is a fraction collector in the system configuration, from this screen it is possible to:

Set the fraction collection mode; **Time** or **Volume.**

Set the collection volume; for **Time** (use **min**), for **Volume** (use **ml**).

If there are both an autosampler and a manual injector in the system configuration, you can select the device to be used for this method.

If the method requires only one of two configured integration channels the unused channel can be turned off.

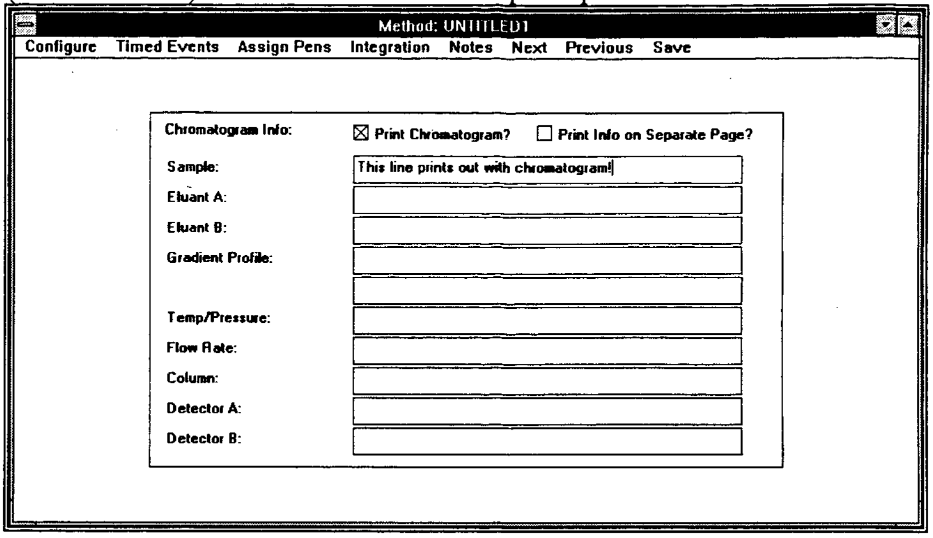
The **Stop Pumps at method end** option will turn off flow at the end of the method when selected. This is helpful when included in a method designed to flush the system at the end of the day.

Note: Do not perform reanalysis with this option selected or you will lose your data!

A method can be configured to run without saving any data to the hard disk. When the **Do Not Save Data or Reports** option is selected, the method will run, the report and chromatogram will be printed, but the data will not be stored on the hard disk when the run is finished. This is useful for equilibration methods, and any method where you do not nee to save the data.

Notes

The **Notes** section of the method allows **Chromatogram Info** and **Notes** (free form text) to be saved with the sample report.



This screen allows you to attach sample, mobile phase, gradient, flow, column and detection information to the method. If the **Print Chromatogram** box is checked the chromatogram will print at the end of the method. The information entered on this screen can be printed on the chromatogram or on a separate page.

The **Notes** screen allows free form text to be attached to the method.

Next and **Previous**

Selecting **Next** or **Previous** from the menu bar will move you from screen to screen in the method

Save

To save the method just created, select **Save** from the menu bar or double click on the control menu. You will be prompted to enter a name for the method.

Editing an Existing Method

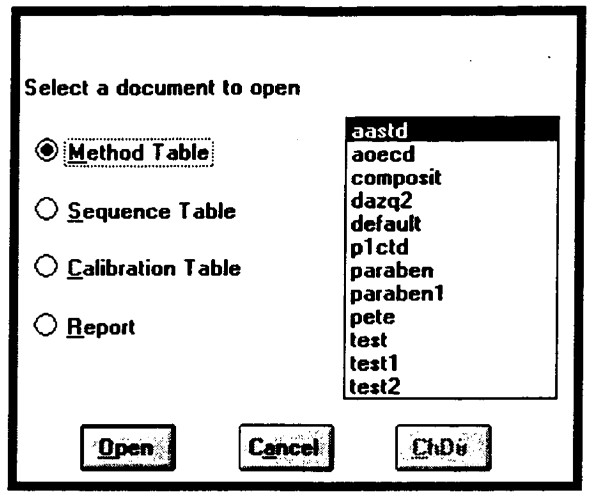
Once a method has been created it can be used as a template for other methods. Often large portions of the method will stay the same from run to run, for example, you may only change a gradient start point or a single integration parameter while developing a method. Editing an existing method rather than creating a new method for each run can save valuable time.

Method Icon

To edit an existing method:

1. Select the Method icon from the Easy Menu or Open from the File menu.

2. At the Select a document to open box select Method Table. A list of available methods will appear:



Select the method to edit

Click on **Open**.

Edit the portion of the method to be changed.

Saving an Edited Method

There are two options when saving an edited method. The new method can be saved under the same name as the old method, overwriting it, or the new method can be saved under a different name if you wish to keep both the new and old methods.

To overwrite the old method:

1. Select **Save** from the menu bar.
2. From the **Select A Table to save** screen select the name of the old method.
3. Click on **Save**. Only the new method will remain on disk.

To save the new method under a new name:

1. Double click on the control menu box.
2. In the box labeled **Save changes to** enter the new method name.
3. Click on **Save**. Both methods will remain on disk.

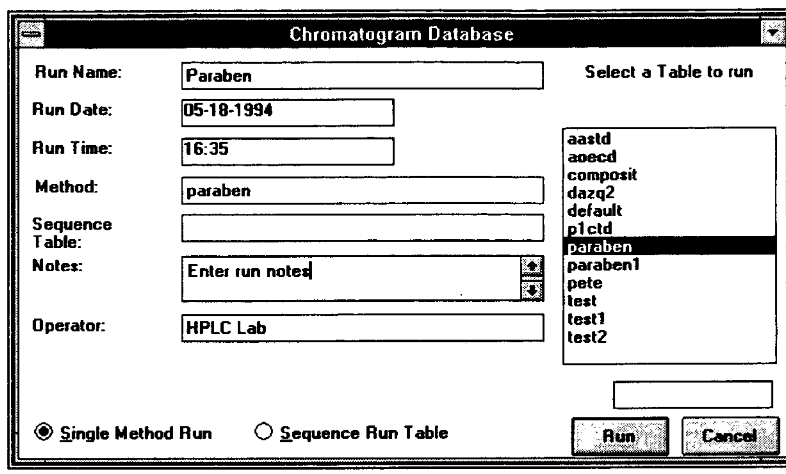
Running a Method

An HPLC method can be run individually or in a series as part of a Sequence table. To run an individual method:

The Run Icon

Select the **Run** icon from the **Easy Menu** or from the **Run** menu select **Method.**

This brings up the **Chromatogram Database** screen.



Each time a sample is run ValueChrom automatically enters the data into the Chromatogram Database. The Chromatogram Database allows you to attach various information to the sample (with the screen shown above) so that it can be organized for easy access at a later date. For a thorough discussion on using the Chromatogram Database to access data see "Chromatogram Database" on page 55.

The **Run Name** field allows you to assign each sample a meaningful name with up to 28 alphanumeric characters. If no name is specified by the user **ValueChrom** will generate one. A **ValueChrom** generated **Run Name** for a single method run consists of: the method name:NN:XX:Run ID

The **Run Date** and **Run Time** fields are filled by ValueChrom and are not editable. The data entered here corresponds to the date and time the sample is run.

The scroll box labeled **Select a Method to Run**, on the right of the screen, contains a list of all of the available methods. Click on a method name and it will appear in the **Method Name** field, or type the name of the method to be used.

The **Notes** field allows you to enter additional information relevant to the sample. This field will accept unlimited free form text.

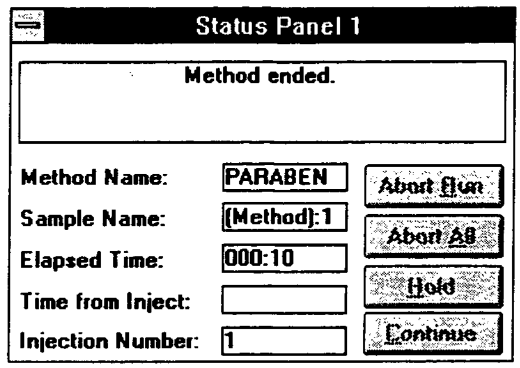
The **Operator** field will contain the **Log On** name provided when starting the software. This field may be edited if necessary.

Select the **Cancel** button to leave the **Chromatogram Database** window without starting a method.

To start the method click on **Run**. This will open the **Status Panel** and the **Strip Chart** screens, and begin the HPLC run.

The Status Panel

The **Status Panel** opens automatically with the start of every method or can be accessed at any time by selecting **Status Panel A** (or **B**) from the **Control Panels** menu.



The top field in the Status Panel screen contains up to three lines of text that relay current information about the status of the system, for example:

"**No method currently running,**" or "**Method in progress**" or "**Waiting for inject**"

The Status Panel also displays the name of the currently running method in the field titled **Method Name**, and the sample name from the Sequence table screen in the field titled **Sample ID**. In the case of a single method run, (**Method**) appears here.

Elapsed Time displays the time elapsed from "time 0.00" in the Timed Events Table screen in the method, this includes time during which the method is paused, for example, to wait for an injection.

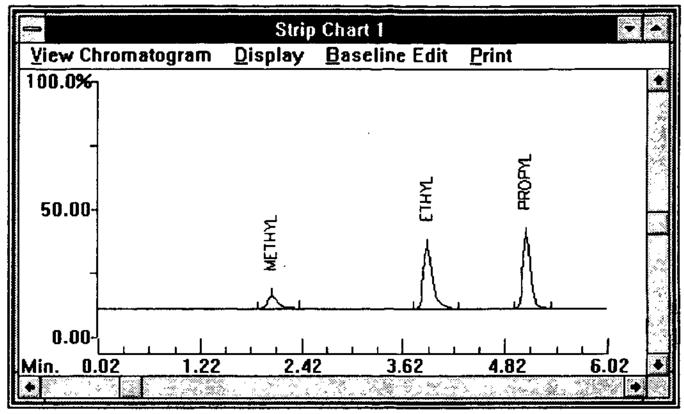
Time from Inject displays the time elapsed from the injection.

Injection Number displays the current injection number for samples running replicate injections.

From this screen you can abort the current run (**Abort Run**), abort all remaining runs in a sequence (**Abort All**), pause all timed events in a method (**Hold**), and resume all timed events in a method (**Continue**).

The Strip Chart

The Strip Chart opens automatically with the start of each method or can be accessed at any time by selecting **Strip Chart 1** (or **2**) from the **Chromatogram** menu.



When opened at the start of a method, the title of the window reflects the name of the current method, for example; **Strip Chart 1 Method: PARABEN**

During the method the Strip Chart shows a real time display of the data being collected by each of the devices assigned to the 4 available pens in the **Assign Pens** screen in the method. For more information see "The Strip Chart" on page 65.

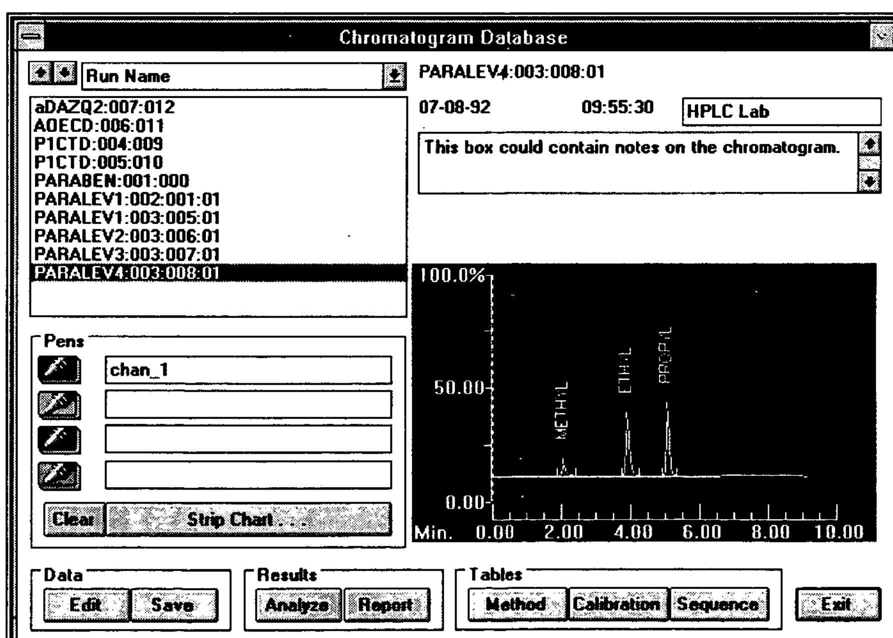
Chromatogram Database

Chromatogram Database Introduction

The Database Icon

ValueChrom contains a chromatogram database to make sorting and retrieving data files fast and easy. To open the database screen at any time select the Database icon from the Easy Menu.

Using the mouse, select an area of interest.



Entering Data into the Database

Real Time Data Collection

Each time a sample is run ValueChrom automatically enters the data into the Chromatogram Database. The Chromatogram Database allows you to attach various information to each sample at the start of each run so that the data files can be organized for easy access at a later date.

When an HPLC run is initiated (by clicking the Run icon in the Easy Menu, or selecting Run from the file menu) the Chromatogram Database window, above, opens. From this window it is possible to attach various information to each data file.

The **Run Name** field allows you to assign each sample a meaningful name with up to 28 alpha-numeric characters. If no name was specified by the user, **ValueChrom** will assign a name based on the method name and the time and date of injection. (When running a sequence table the **Run Name** will be assigned to each sample included in the sequence table.)

ValueChrom attaches a number to the **Run Name** to help identify the individual samples. These number correspond to the sequence table line number (1 for a single method run) and injection number.

The **Run Date** and **Run Time** fields are filled by ValueChrom and are not editable. The data entered here corresponds to the date and time the sample is run.

The list box labeled **Select a Table** (or **Method) to Run**, on the right of the screen, contains a list of all of the available sequence tables or methods. Click on a sequence table or method name and it will appear in the **Sequence Table** or **Method** field, or type the name of the sequence table or method to be used.

The **Notes** field allows you to enter additional information relevant to the samples being run. When running a sequence table all samples will have this information stored with the data file. This field will accept free form text. Choose the save button at the bottom of the screen to record any changes you make in the Notes field.

The **Operator** field will contain the **Log On** name provided when starting the software.

Beneath the list box on the right of the screen is a field titled **Run ID**. The **Run ID** is a number generated by **ValueChrom** and incremented by one with each injection. This provides a search parameter that is unique for each data file, all other parameters can have more than one sample associated with any given entry.

Select the **EXIT** button to leave the **Chromatogram Database** window without starting an HPLC run.

The **Table** section opens the method, sequence or calibration table for the run selected. For more information on editing an existing method see "Creating a Method" on page 33 or, sequence table see "Creating a Sequence Table" on page 77.

Adding HRLC Data files to the Chromatogram Database

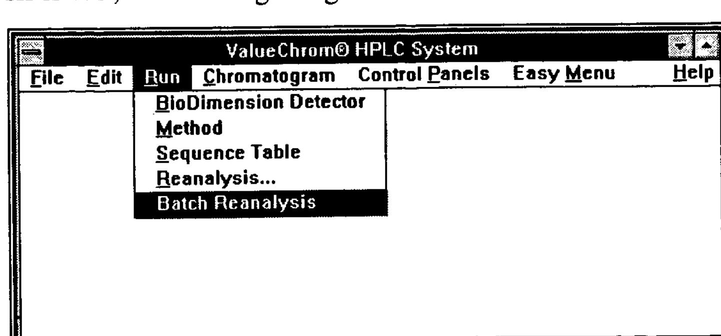
Data files that were collected with Bio-Rad's HRLC software can be added to the Chromatogram Database with the Batch Reanalysis feature. Bio-Rad's HRLC software stores raw data files, methods and sequence tables in the HRLC directory. To use this feature the data files, method, and sequence table must be moved into the VCHROM directory.

The Quick-Start Tutorial will not work on computers that have had HRLC loaded.

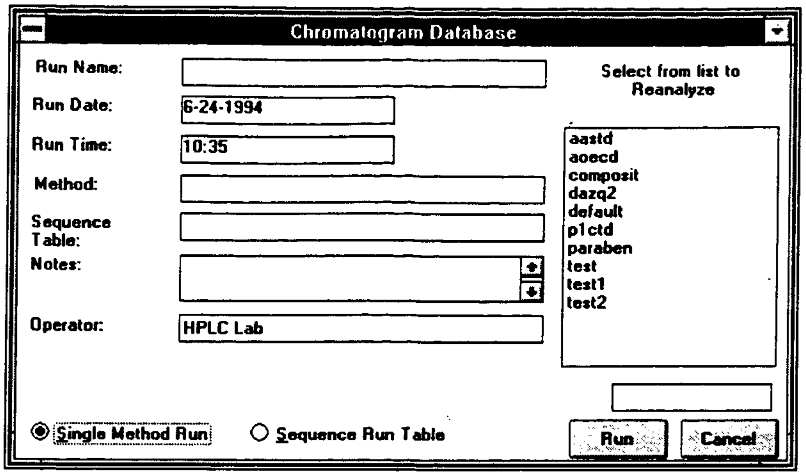
Open the Windows File Manager, located in the main window in the Main menu.

1. Copy the files in the HRLC\DATA\RAW\METHOD directory into the VCHROM\DATA\RAW\METHOD directory. Be sure to copy all sub-directories.
2. Copy the files in the HRLC\DATA\RAW\SAMPLE directory into the VCHROM\DATA\RAW\SEQUENCE directory. Be sure to copy all sub-directories.
3. Copy the files in the HRLC\DATA\TABLES\METHOD directory into the VCHROM\DATA\TABLES\METHOD directory.

Return to ValueChrom and select Batch Analysis from the Run menu. **Note: Before running Batch reanalysis you should open each method and turn off any printing.** It will also be faster to edit the method to not integrate. Open the method file, and choose the assign integrator screen. Delete the integrator channel assignment and save the method. When reanalysis is run, the existing integration information will be used.



This opens the Chromatogram Database screen.

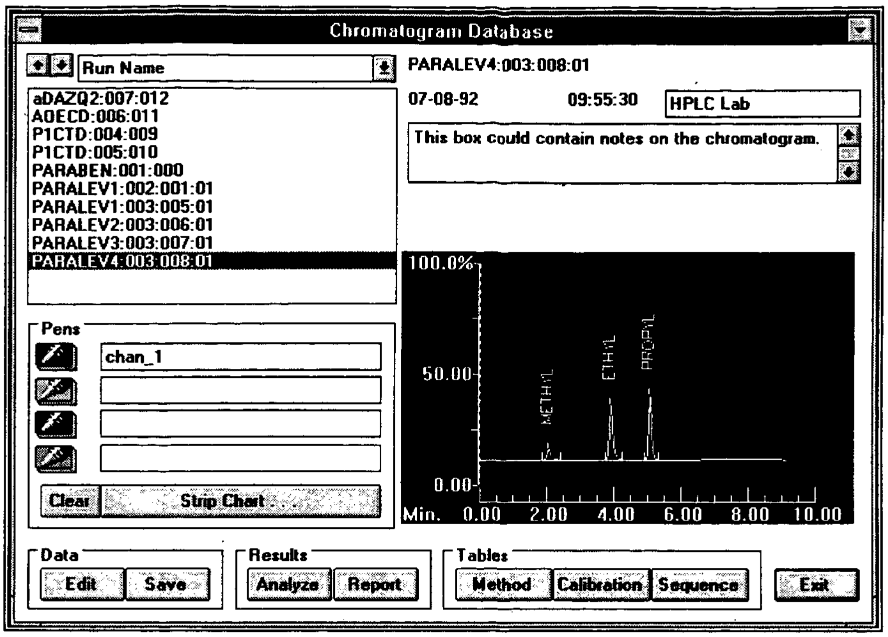


Select either **Single Method Run** or **Sequence Table Run** at the bottom of the screen. The methods or sequence tables available for reanalysis are listed in the box at the right of the screen. Click on the method or sequence table to be analyzed.

In some instances you will be asked to select the date or instance of the run. When the analysis is completed, the data will be available from the database screen.

Viewing Data with the Chromatogram Database

The Chromatogram Database allows you fast, easy access to all data files. To open the Chromatogram Database select the Database icon from the Easy Menu.



Sort Parameters & List Box

With the Chromatogram Database you have the option of sorting the data files with one of several different parameters. In the upper left corner of the screen are the sort parameters and a data file list box. The sort parameters are listed in a drop box that can be opened by clicking on the drop arrow to the right of the field. The data in the Chromatogram Database can be sorted by one of the following parameters:

Run Name - the Run Name is assigned in the database window that opens with each sample run. To differentiate between individual samples with the same run name ValueChrom adds .... and the Run ID number.

Method Name - this parameter will have the sequence table instance : injection number : and Run ID number after the method name to help identify the individual samples run with the method.

Sequence Table Name - this name will have the date and time of each injection in parenthesis to help identify the individual samples run with the same sequence table.

Date and Time - this is the date and time of the sample run

Run ID - This number, assigned by ValueChrom, is a sequential numbering of all injections on the system. Each data file has a unique Run ID.

Click once on any data file and the corresponding run information will be displayed in the run information boxes in the upper right corner of the screen, and all channels of data within that file will be displayed in the chromatogram field in the lower right corner of the screen. Use the up and down arrows, at the left of the sort parameters drop box to scroll through the data files.

Run Information

The boxes in the upper right corner of the screen contain information stored with the data file. In the event that one of these fields is left blank, the box is filled with the field title in parenthesis. The run information includes:

Run Name: When any run is initiated a Database window opens . In this window the user has the option of assigning a meaningful run name, up to 28 alphanumeric characters. If no name is assigned, ValueChrom will generate a name based on the method name, run instance, line number in the sequence table (1 for single method run) and the Run ID number. To differentiate data files with the same Run Name ValueChrom attaches the sequence table instance, the sample ID and the Run ID numbers to user defined Run Names.

Operator: At system start-up a log on window will open asking for identification of the current user. The name entered on this window will be used as a default in all Operator fields. The entry can be changed on the Chromatogram Information screen at the start of each method or sequence table, or edited post run.

Notes: This field allows unlimited text to be attached to a data file. When running a sequence table, the entry in this field is saved with each data file collected. This information is entered on the Chromatogram Information screen at the start of each method or sequence table, or can be edited post run.

Run Date: This field contains the date on which the sample was run and is filled by ValueChrom. This field is not editable at any time.

Run Time: This field contains the time at which the sample was run and is filled by ValueChrom. This field is not editable at any time.

Pens

The box labeled Pens contains a key for the colored pens used to display the channels within the data file. The data file listed in the pen field is displayed in the corresponding color on the chromatogram field.

Displaying a Data File:

To display a data file simply click on the file of choice and the associated run information will appear in boxes in the upper right of the screen and each channel of data will be displayed in the chromatogram window at the bottom right of the window.

Overlaying Two Data Files

To overlay two or more data files:

1. Depress one of the four pen buttons by clicking on it.
2. Click data file to be assigned to that pen. (If more than one channel of data is associated with the selected data file, a dialog box will open with the available choices displayed. Select the desired data channel. The name of the selected data file will appear in the field next to the pen and the displayed in the chromatogram field.)
3. Depress another of the four pen buttons.
4. Click the next data file to be displayed as described above.

When one or more of the pens are depressed, the newly selected data file will be assigned to the last pen depressed. If you release one of the depressed pen buttons (by clicking on it), its data channel assignment will be cleared with the next data selection. To clear all assigned pens, click on the **Clear** button.

Note: To return to the Chromatogram Database screen from the Strip Chart, double-click the right mouse button.

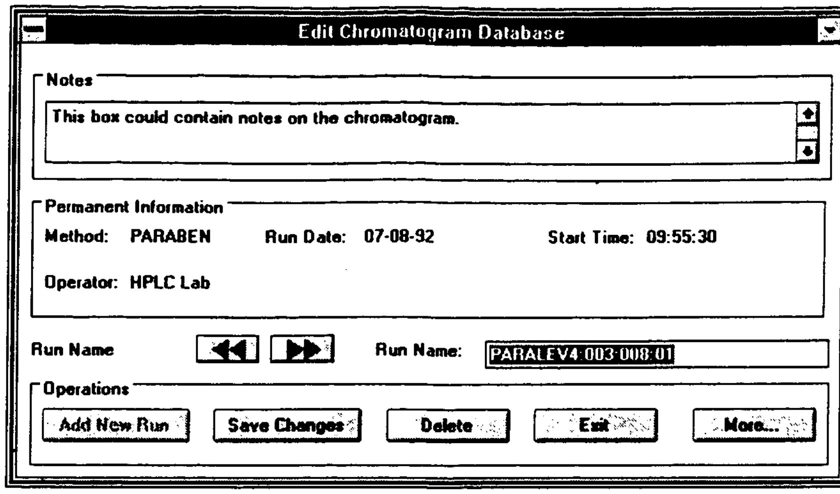
The **Strip Chart** button, or double clicking on the chromatogram, will expand the chromatogram portion of the database screen, with all currently selected data files, to the **Strip Chart** screen. For more information about the options available in the **Strip Chart** see "The Strip Chart" on page 65 for details.

Database Management

The buttons in the **Data** box allow the user to change some of the information saved with a data file.

Edit Button

From this screen you can edit the information stored with each data file.



Note: It is good practice to remove any unwanted data files as soon as you determine they are no longer needed.

This keeps the database smaller and operating faster.

It is possible to edit the **Notes** or the **Run Name**. All other fields are non-editable.

To save any changes click on the **Save Changes** button.

To scroll through the data files use the double arrow keys (<< and >>).

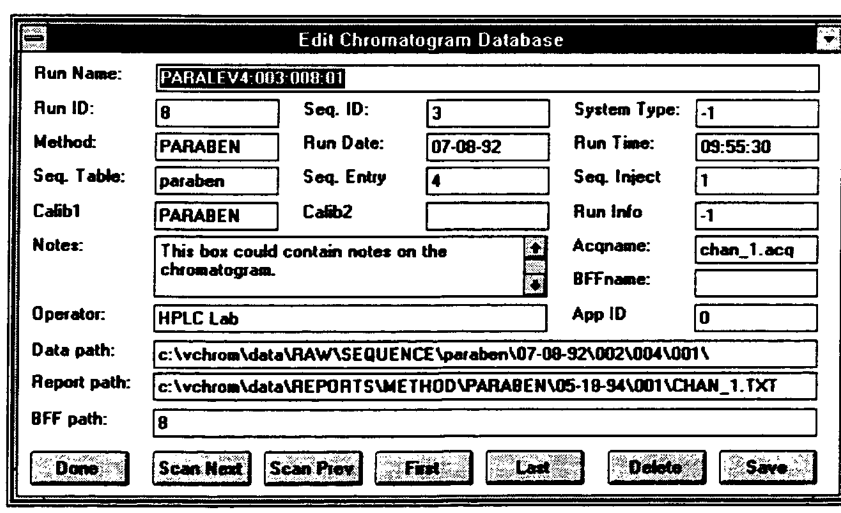
To delete a data file Click on the **Delete** key. To see the effects of the delete you must leave the database screen and return.

To return to the main database screen, click on **End**.

For more editing options, click on **More** and the **Edit Chromatogram Database** screen will appear.

Advanced Database Management

Under normal operation you will never need to use the Edit Database screen. This information is provided for your convenience.



This screen displays all of the information saved with the data file.

The **Run Name** field allows you to assign each sample a meaningful name with up to 28 alpha-numeric characters. If no name is specified by the user, **ValueChrom** will assign a name based on the method name and the time and date of injection. When running a sequence table the **Run Name** will be assigned to each sample included in the sequence table. ValueChrom attaches a number to the **Run Name** to help identify the individual samples. These number correspond to the sequence table line number (1 for a single method run) and injection number.

Run ID: The Run ID is a number generated by **ValueChrom** and incremented by one with each injection. This provides a search parameter that is unique for each data file, all other parameters can have more than one sample associated with any given entry.

Method: This field shows the method that was used to run this data file.

Run Date: This field contains the date on which the sample was run and is filled by ValueChrom. This field is not editable at any time.

Run Time: This field contains the time at which the sample was run and is filled by ValueChrom. This field is not editable at any time.

Seq. Table: This field shows the name of the sequence table used to run this data file.

Seq. Entry: This field shows the entry line number of this sample in the sequence table listed above.

Seq. Inj: In the event of replicate injections, this field shows the injection number of this data file.

Calib1: Each data channel can be linked to a calibration table to provide quantitative information about a sample. This field displays the calibration table linked to data channel 1.

Calib2: Each data channel can be linked to a calibration table to provide quantitative information about a sample. This field displays the calibration table linked to data channel 2.

Notes: This field allows unlimited text to be attached to a data file. When running a sequence table, the entry in this field is saved with each data file collected. This information is entered on the Chromatogram Information screen at the start of each method or sequence table, or can be edited post run.

Operator: At system start-up a log on window will open asking for identification of the current user. The name entered on this window will be used as a default in all Operator fields. The entry can be changed on the Chromatogram Information screen at the start of each method or sequence table, or edited post run.

Data Path: This field displays the entire path in which the raw data file is located. If it becomes necessary to move the data, this path must be changed to reflect the new location.

Report Path: This field displays the entire path in which the report file is located. If it becomes necessary to move the report, this path must be changed to reflect the new location.

BFF Path: This field displays the entire path in which the raw BFF data file is located. If it becomes necessary to move the BFF data, this path must be changed to reflect the new location.

Tables

The data file does not retain a copy of the method, calibration and sequence tables as they were when used to run each sample. These buttons will open the most recently saved versions of these tables.

The buttons contained in the Tables box, **Method**, **Calibration**, and **Sequence**, will open those tables that were used to collect the selected data file. For more information on Methods see "Creating a Method" on page 33, Calibration see "Creating a Calibration Table" on page 84, and sequence see "Creating a Sequence Table" on page 77.

Reports & Results

Area and Height Reports

The report heading gives information about the Run Name, method, file name, run time, calibration, and vial number.

AREA % REPORT
Run Name: PARALEV1 Date: 7/28/94 Time: 09:52:32
Method File: PARABEN Channel #: 1
Calibration Table: PARAB1 Calibration Type: None
Vial #: 51

RT	AREA	TYPE	AR/HT	AREA%	SYMMETRY	EFFICIENCY
2.003	449340	BB	0.165	11.398%	1.690	887
3.792	1851656	BB	0.241	46.971%	2.444	1750
5.810	1641136	BB	0.210	41.631%	2.466	5003

The table shows each peak retention time, area, type of peak, area to height ratio, area percentage, symmetry, and efficiency. For height reports, peak height and height percentage are given.

Peak types are letter codes where:

- B = baseline
- V = valley
- D = drop
- T = tangent.

A code of BB means the peak is integrated as, "baseline to baseline" or sometimes called baseline resolved. A code of BT would indicate the peak is baseline to tangent, or a skimmed peak.

At the bottom of the peak information is the total area.

Calibration Reports

Calibration reports show the amount of the sample and the name of the peak along with retention time, area, peak type, and Cal number.

RT	AREA	TYPE	CAL #	AMOUNT	PEAK LABEL
2.003	449340	BB	1R	1.267	METHYL
3.792	1851656	BB	2	3.517	ETHYL
5.810	1641136	BB	3	3.376	PROPYL

The information for this report is entered in the calibration table. CAL# refers to the number of the peak in the calibration table. A negative CAL# indicates the peak was not found in the calibration table. If a peak is not found in the calibration table the amount for that peak will be the peak's area and no name will be assigned.

The calibration report also gives the equation for each peak. The type of equation will depend on the type of curve fit. For example a quadratic curve fit will result in a quadratic equation.

PEAK LABEL	RT	TYPE	LEVELS	Equations
METHYL	2.00	LINEAR	4	Y=2.4899E+5 X + 1.33837E+5
ETHYL	3.79	LINEAR	4	Y=4.50893E+5 X + 2.65806E+5
PROPYL	5.81	LINEAR	4	Y=4.13062E+5 X + 2.46657E+5

The Strip Chart

The Strip Chart Icon

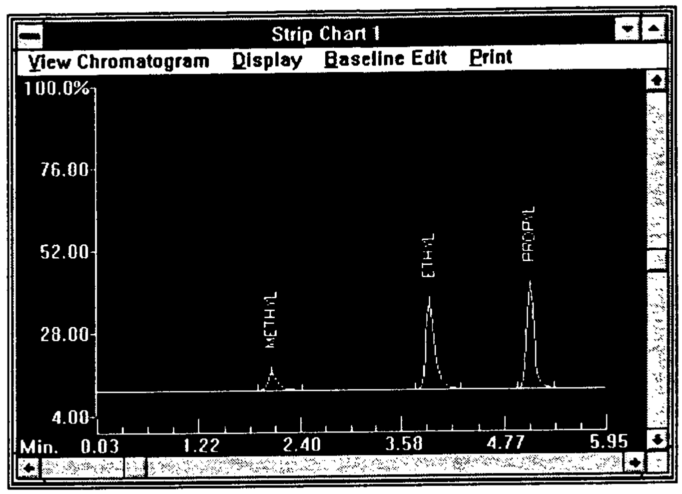
The video strip chart screens allow you to view up to 4 channels of data, edit the baseline then reanalyze, or print a chromatogram. While a method is running you can monitor up to 2 detector channels, 2 pump channels (gradient information) or an existing chromatogram (i.e., a standard to compare to an unknown).

The **Status Panel** screen and the **Strip Chart** screen open when a method starts. To access these screens at any time select **Status Panel A** (or **B**) from the **Control Panels** menu and select **View Chromatogram** from the **Easy Menu** or select **Strip Chart 1** (or **2**) from the **Chromatogram** menu.

Strip Chart Operation

Hint: Double clicking the right mouse button will restore the strip chart display to the database screen.

The **Strip Chart** screen is displayed whenever a method starts. To access this screen at any time select **View Chromatogram** from the **Easy Menu**, or select **Strip Chart 1** (or **2**) from the **Chromatogram** menu. While a method is running you can monitor up to 2 detector channels, 2 pump channels (gradient information) or an existing chromatogram (i.e., a standard to compare to an unknown). Select the data channels or data files displayed during a run in the **Assign Pens** screen in the method.



From the Strip Chart there are various ways to optimize the display for a given method.

Assign Axes

You may assign the pens to either of 2 axes:

Select **Assign Axes** from the **View Chromatogram** menu. The current pen assignments are shown in the **Axis pen assignments** screen.

To change an assignment, click on the desired pen and it will be assigned to the other axis.

Pen Colors

For a key to the pen colors or styles:

Select **Pen Colors** from the **View Chromatogram** menu

Scaling the Axes (Zooming)

When the method is started the time scale will be scaled to the time of the method automatically. The default for the vertical axis scale is 100% percent of full scale (the detector input is scaled with the Gain switches on the rear panel of the Instrument Interface, for instructions, see the Instrument Interface Users Manual). To adjust the scaling of this window there are two options:

Mouse Operation (scales both axes equivalently):

1. Move the mouse to one corner of the area you wish to expand.
2. Press the left mouse button and hold it down.
3. Drag the mouse to the opposite corner, diagonally, of the area to expand. An outline of the area you have selected will appear.
4. Release the mouse button and the strip chart will be re-drawn, showing only the area selected.
5. To reset the strip chart to the original settings, double click the mouse on the strip chart screen.

Scaling with the pull-down menu:

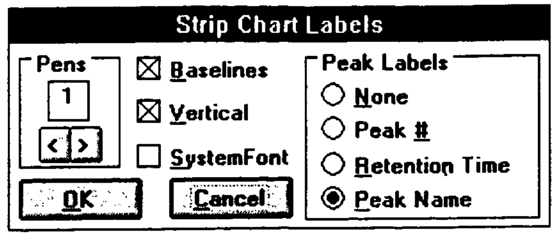
1. Select **Show Scaling** from the **Display** menu.
2. In the **Zoom Control** window you can scale each of the vertical axes individually in the **Vertical Scales** box.
3. The axes can be offset to avoid drawing pens from one axis on top of pens from the other. Enter a percentage offset in the **Axis Offset** box.
4. In the **Window Width** box the time scale can be set in minutes or seconds.
5. To see the effect of the changes, before closing the window, click on the **Show** button.
6. When the display is scaled properly, click **Done**.

To reset the strip chart to the original settings, double click the mouse on the strip chart screen.

Peak Labels

When the method is running, the incoming data is processed and peak labels are assigned in real time. To change the identifying information shown for each peak:

Select **Peak Labels** from the **Display** menu to bring up the **Strip Chart Labels** window.



If **Baselines** is selected, the baselines will be drawn in as each peak end is detected.

If **Vertical** is selected, the peak labels will be written on the screen vertically.

If **System Font** is selected, the peak labels will be written in the system font (the font used for the menu bars and window text).

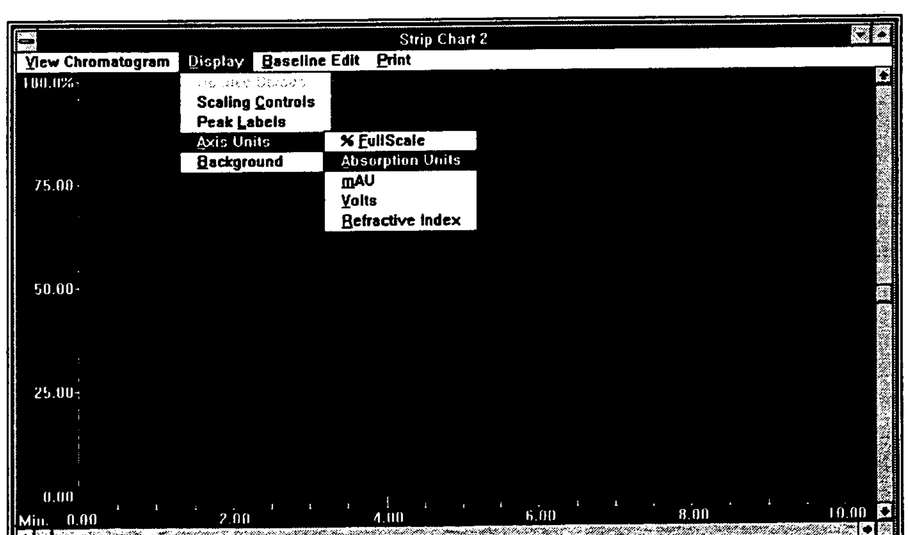
In the box labeled **Peak Labels** you can chose to have the peaks labeled with their **Peak #**, **Retention Time**, **Peak Name** (for a peak to be named the method must be calibrated, see "**Calibration**"), or not labeled (**None**).

Each pen must have its display characteristics set separately. To toggle from pen to pen click the **<** and **>** arrows in the **Pens** box.

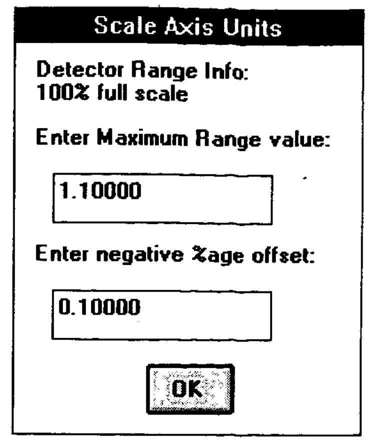
Axis Units

The units of the left vertical axis can be changed to reflect the type of data you are collecting. To change the units:

Select **Axis Units** from the **Display** menu. A list of available unit choices will be shown.



Select the appropriate unit choice for your method of detection. This brings up the **Scale Axis Units** window.



Enter the **Maximum Range Value** for your detector (i.e., for a UV detector this might be 2.000 Absorption Units).

Enter the **negative %age offset** (the percent of the maximum range that you wish to see below zero).

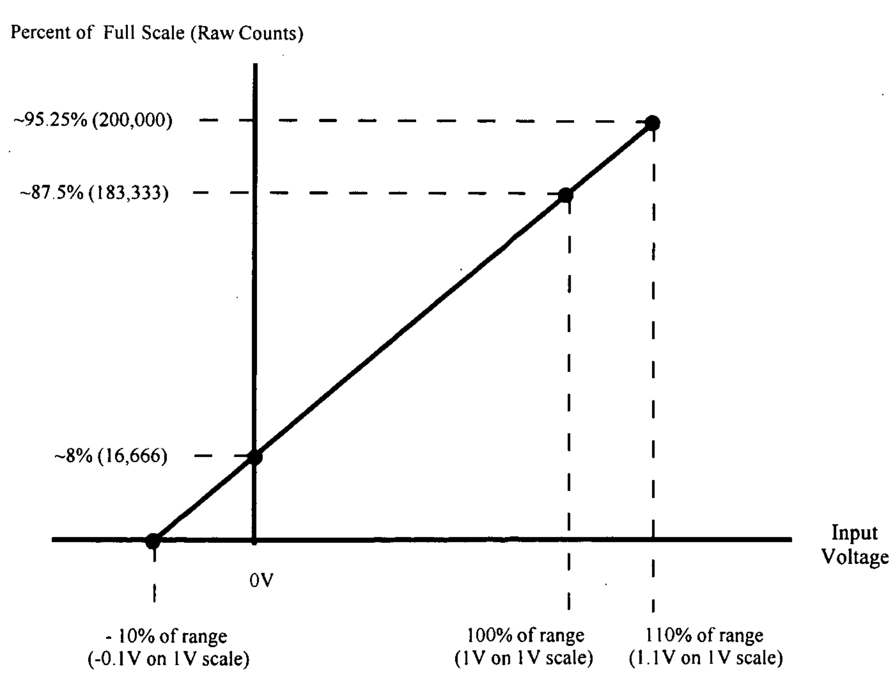
Click on **OK**.

Note: ValueChrom plots the baseline over a specified range as follows:

At an input of 0.0 volts the software will display the baseline at approximately 8% of full scale

At 100% of range (i.e., +1 V on the 0.0V to +1.0 V range) the software will display the baseline at approximately 87.5% of full scale

At 10% over range (+1.1 volts on the 0.0V to 1.0 V range) the software will display the baseline at approximately 95.25%.



Background

Selecting **Background** from the **Display** menu will change the **Strip Chart** to a black and white image. The pens will change to a variety of solid and dashed lines. This is helpful when cutting and pasting the strip chart into a black and white document.

Print

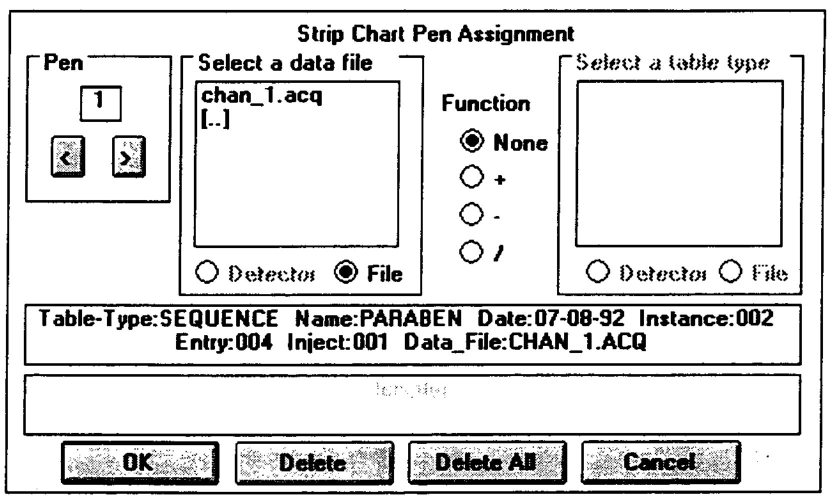
To print the chromatogram or data file at any time, click on **Print** in the menu bar. Any changes made to the display will be reflected in the printed data.

Viewing a File from the Strip Chart

To view an existing chromatogram from the strip chart:

Select **View a Chromatogram** from the **Easy Menu** or select **Strip Chart 1** (or **2**) from the **Chromatogram** menu.

Select **View data file** from the **View Chromatogram** menu. This brings up the **Strip Chart Pen Assignment** screen.



Up to 4 data channels can be viewed at one time. Each data file to be viewed must be assigned to a pen. The **Pen** box in the upper left corner of the window shows the pen currently selected. To scroll to the next or previous pen click on the < and > arrows.

Click on **File** in the **Select a data file** box.

Select the data file to be viewed. When a selection is made it appears in the box below the **Select a data file** box (when no data channel is assigned this box reads **[empty]**).

To delete a file selection click on the **Delete button** at the bottom of the screen.

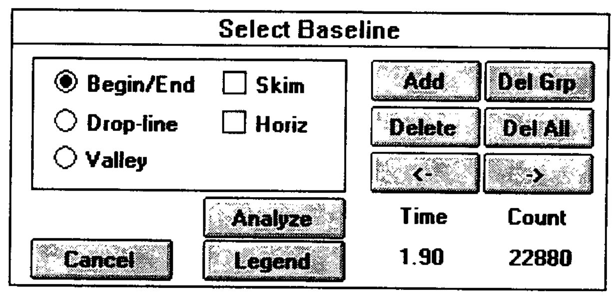
When viewing previously stored data you can choose to see the data file added, subtracted, or divided by another data file. After selecting data file above;

Select desired function (+, -, / buttons listed under **Function**).

Select data file to add to, subtract from, or divide into the original data file.

Baseline Edit

Baseline edit is used to alter the baseline and/or modify integration marker placement on existing data files. New integration reports that use the adjusted parameters can be generated. Changes made in **Baseline Edit** affect only **Pen #1**. Once a data file has been selected, you can access the **Baseline Edit** options by selecting **Baseline Edit** from the menu bar. The **Select Baseline** window will open and integration markers will appear on the chromatogram.



For a description of each type of marker, click on **Legend**.

Moving Integration Markers

Move the mouse to the marker you wish to move.

Clicking the left mouse button once will select the marker. The **Time** and **Sample** entries will reflect the location of the selected marker (**Time** in minutes, **Sample** in counts), the marker color will change to white.

The marker can be moved in 2 ways. Click on the **<-** or **->** keys to move the marker one data point at a time, or click on the marker and hold the left mouse button down while dragging to a new location. The **Time** and **Sample** entries will reflect these movements.

To see the integrated results of the new marker placement click on the **Analyze** button

The editor that displays the analysis is selected in the BRASD.INI file. Edit this file to change editors.

For this example we use Window's Notepad.

When analysis is complete, the editor program will open with the new report displayed. The changes made do not affect the integration parameters of the method with which the sample was run. The report is saved under the method **Composit** with each reanalysis assigned a unique date and instance file name. The report can be opened at any time by selecting **Open** from the Notepad **File** menu, select **Report** then select the appropriate file and click **Open**.

To print this report, select **Print** from the editor program you are using. The report is formatted for an 80 column dot-matrix printer. For other printers, you may have to adjust your printer margins to get a correctly formatted printout. To set the printer margins for the Windows Notepad program, select **Print Setup** from the Notepad **File** pull-down menu.

Adding or Deleting Integration Markers

Integration markers can be changed, added, or deleted. For a description of each type of marker, click on Legend.

To add a marker:

Click on the Add button

Select the type of marker:

- Begin/End to define a new peak,
- Drop-line to indicate a point at which a vertical integration boundary will be added,
- Valley to insert a valley point between two overlapping peaks,
- Skim for an integration boundary which skims the smallest of two overlapping peaks,
- Horiz for a horizontal extension of the baseline.

Place the mouse on the strip chart where you want the new marker then click the left mouse button. The Time display will indicate the exact placement of the marker. The Sample display shows the counts converted by the A/D converter.

To delete a marker:

Select the marker to be deleted by clicking on it with the mouse.

Click on Delete for Valley or Drop-line markers, or Del Grp for Begin/End, Skim, and Horiz markers.

When analysis is complete, the Windows Notepad program will open with the new report displayed. The changes made do not affect the integration parameters of the method with which the sample was run. The report is saved under the method **Composit** with each reanalysis assigned a unique date and instance file name.

The report can be opened at any time by selecting **Open** from the **File** menu, select **Report** then select the appropriate file and click **Open**.

To print this report, select **Print** from the **File** pull-down menu for the Notepad program. The report is formatted for an 80 column dot-matrix printer. For other printers, you may have to adjust your printer margins to get a correctly formatted printout. To set the printer margins for the Windows Notepad program, select **Print Setup** from the Notepad **File** pull-down menu.

Reanalysis and Integration

If the integration parameters were set inappropriately for a single chromatogram or an entire Sequence table you can adjust the parameters and reanalyze the data using the **Integrator 1** (or **2**) in the **Control Panels** menu or the **Reanalysis** command in the **Run** menu.

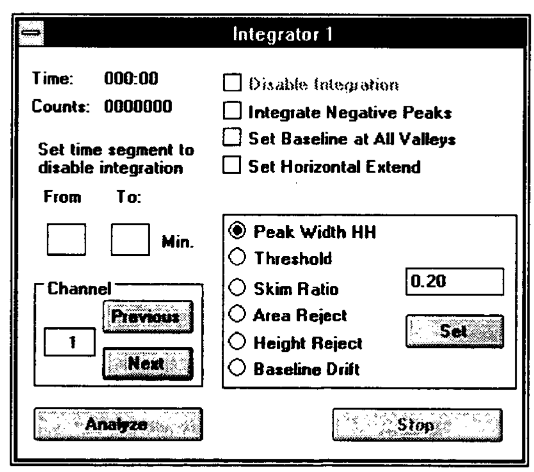
To re-integrate a single data file:

Select the data file you are interested using the database screen.

Click on the **Strip Chart** button to enlarge the strip chart.

From the ValueChrom main menu choose **Integrator** from the chromatogram pull-down menu.

The Integrator faceplate is the easy way to experiment with different integration values.



To disable integration, enter the time to stop and then restart integration in the fields labeled **From** and **To:**.

In the Integrator control panel you can select different files to integrate.

To **Integrate Negative Peaks**, **Set baseline at All Valleys**, or **Set Horizontal Extend** click in the appropriate box in the upper right portion of the screen.

You can also assign methods for reanalysis.

To edit the values for **Peak Width HH**, **Threshold**, **Skim Ratio**, **Area Reject**, **Height Reject**, or **Baseline Drift**, select the parameter to be changed, edit the value in the field above the **Set** button, click the **Set** button.

For more information regarding these parameters see "Integration Parameters" on page 41. When all parameters are adjusted appropriately, click on the **Analyze** button.

Once re-integration has started you may stop it at any time by clicking on the **Stop** radio button.

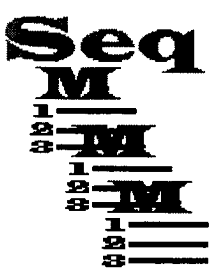
When analysis is complete, and you have selected an editor in the BRASD.INI file, the editor will open with the report. The changes made do not affect the integration parameters of the method with which the sample was run. The report is saved under the method **Composit** with each reanalysis assigned a unique date and instance file name.

The report can be opened at any time by selecting **Open** from the **File** menu, select **Report** then select the appropriate file and click **Open**.

To print this report, select **Print** from the **File** pull-down menu for the Notepad program. The report is formatted for an 80 column dot-matrix printer. For other printers, you may have to adjust your printer margins to get a correctly formatted printout. To set the printer margins for the Windows Notepad program, select **Print Setup** from the Notepad **File** pull-down menu.

Sequence Tables

Introduction

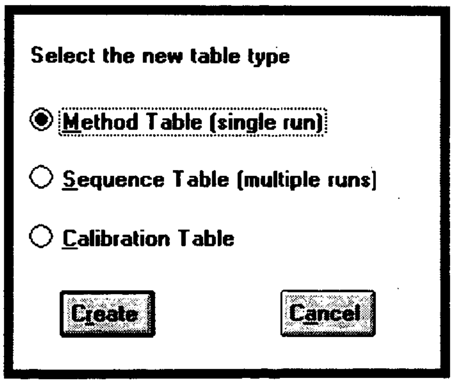


A sequence table is a set of list of methods to run. The Sequence Table allows you to program a series of samples, using the same or different methods, to be run automatically. When the system contains an autosampler the Sequence Table can be used for complete automation of your HPLC system.

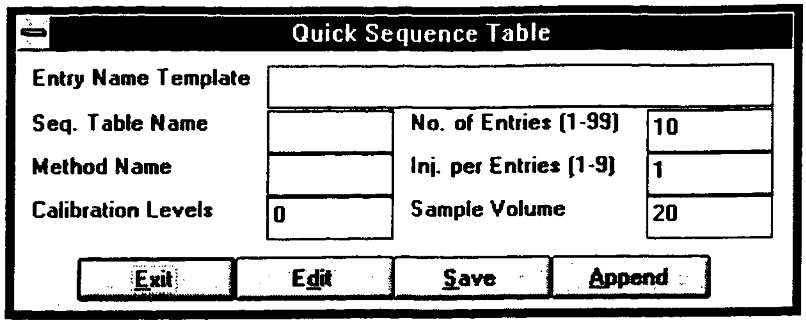
Creating a Sequence Table

Once all necessary methods have been created and saved you can create a **Sequence Table** to process a series of samples automatically. To create a **Sequence Table**:

Select **Create** from the **File** menu.



Select Sequence table, then click on **Create** and the Quick Sequence table will appear.



From the quick sequence table you can easily construct a sequence table.

1. Enter the entry name template, this will be the name used to identify the samples
2. Enter the name for the sequence table.
3. Enter the name for the method to run. (The method must be previously defined.)
4. Enter the number of entries (1 - 99).
5. Enter the number of injections per entry (1 - 9).

Select **edit** and the complete sequence table will appear.

Sequence Table Information

Sequence Table: test

Entry	Name	Vial No.	Volume	Method	No. Inj	Calibrate	Level
1	test:1	1	20	test	1	NONE	1
2	test:2	2	20	test	1	NONE	1
3	test:3	3	20	test	1	NONE	1
4	test:4	4	20	test	1	NONE	1
5	test:5	5	20	test	1	NONE	1
6	test:6	6	20	test	1	NONE	1
7	test:7	7	20	test	1	NONE	1
8	test:8	8	20	test	1	NONE	1
9	test:9	9	20	test	1	NONE	1
10	test:10	10	20	test	1	NONE	1

The sequence table will have the information you entered from the quick sequence screen.

Entry: This number is incremented automatically when a line is added to the Table.

Name: The sample name.

Vial Number: This field is for record keeping purposes only.

Volume: This field is for record keeping purposes only.

Method: Enter the name of the method to be used for this sample.

No. Inj: Enter the number of injections per sample. The allowable entries for this field are 1 - 99.

Calibrate: When running a calibration, enter **REPL** to replace any existing response factors and turn back the number of calibrations run to zero, **AVG** to average the new response factors with the existing values, an entry of **NONE** signifies no calibration.

Level: Each level of a calibration must be entered here. A calibration curve is normally run from the lowest level, or concentration, to the highest. This prevents contamination of the lower level samples with the higher level samples.

For more details see "Calibration" on page 83.

Inserting a Line in a Sequence Table

Note: To insert a line into the Sequence table, click on the **Entry** column with the left mouse button, then select **Insert** from the **Edit** menu. A line will be inserted above the line selected. To delete a line, click on the **Entry** column with the left mouse button, then press delete.

HINT: For Sequence tables with long columns of identical entries use the **Copy** and **Fill Down** options in the **Edit** menu.

Closing a Sequence Table

Select **Close** from the **Control-box** menu and the file will be saved with the name given on the quick sequence screen.

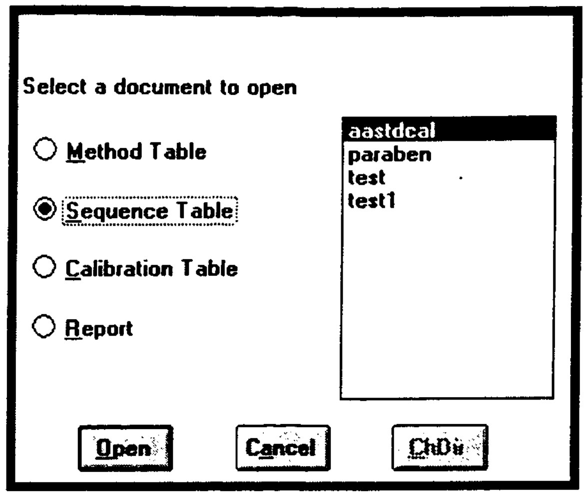
Editing an Existing Sequence Table

Once a sequence table has been created it can be used as a template for other sequence tables. Often large portions of the sequence table will stay the same from day to day, for example, you may only change sample names or the number of injections per sample. Editing an existing sequence table rather than creating a new sequence table can save valuable time.

To edit an existing sequence table:

Select the **Method** icon from the **Easy Menu** or **Open** from the **File** menu.

At the **Select a document to open** box select **Sequence table**. A list of available sequence tables will appear:



Select the sequence table to edit

Click on **Open**.

Edit the portion of the sequence table to be changed.

Saving an Edited Sequence Table

There are two options when saving an edited sequence table. The new sequence table can be saved under the same name as the old sequence table, overwriting it, or the new sequence table can be saved under a different name if you wish to keep the old sequence table.

To overwrite the old sequence table:

1. Click on the control menu box.
2. Click on Close.
3. In the box labeled Save changes to select the old sequence table name.
4. Click on Save. Only the new sequence table will be saved.

To save the new sequence table under a new name:

1. Click on the control menu box.
2. Click on Close.
3. In the box labeled Save changes to enter the new sequence table name.
4. Click on Save. Both sequence tables will remain on disk.

Running a Sequence Table

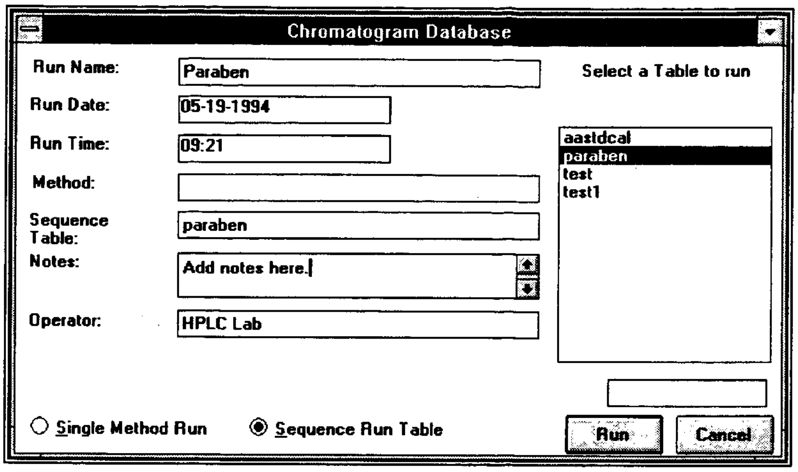
An HPLC method can be run individually or in a series as part of a sequence table. For more information about creating and running a single **Method** "Creating a Method" on page 33 and "Running a Method" on page 51.

To run a series of samples with a sequence table;

Select the **Run** icon from the **Easy Menu**, or select **Sequence table** from the **Run** menu.

This brings up the **Chromatogram Database** screen (below).

Select **Sequence Run Table** at bottom of the screen



Each time a sample is run ValueChrom automatically enters the data into the Chromatogram Database. The Chromatogram Database allows you to attach various information to the sample (with the screen shown above) so that it can be organized or easy access at a later date. For a thorough discussion on using the Chromatogram Database to access data see "Chromatogram Database" on page 55.

Run Information

The **Run Name** field allows you to assign each sample a meaningful name with up to 28 free form characters. If no name is specified by the user **ValueChrom** will assign a name based on the sequence table. The **Run Name** will be assigned to each sample included in the sequence table. ValueChrom attaches numbers to the **Run Name** to help identify the individual samples. These numbers correspond to the sequence table line number and injection number. After the injection you can rename these runs.

The **Run Date** and **Run Time** fields are filled by ValueChrom and are not editable. The data entered here corresponds to the date and time the sample is run.

The scroll box labeled **Select a Table to Run**, on the right of the screen, contains a list of all of the available sequence tables. Click on a sequence table name and it will appear in the **Sequence Table** field, or type the name of the sequence table to be used.

The **Notes** field allows you to enter additional information relevant to the samples in the table. All samples run with the sequence table will have this information stored with the data file. This field will accept unlimited free form text.

The **Operator** field will contain the **Log On** name provided when starting the software. This field may be edited if necessary.

Beneath the sequence table scroll box on the right of the screen is a field titled **Run ID**. The **Run ID** is a number generated by **ValueChrom** and incremented by one with each injection. This provides a search parameter that is unique for each data file, all other parameters can have more than one sample associated with any given value.

Select the **Cancel** button to leave the **Chromatogram Database** window without starting a sequence table.

To start the method click on **Run**. This will open the **Status Panel** and the **Strip Chart** screens, and begin the HPLC run.

When the run begins you will see the strip chart open and the status panel display.

Calibration

Introduction

A calibrated method provides quantitative information about the components in a sample. The concentration of each component is calculated by comparing its measured absorbance against a calibration curve.

To generate a calibration curve, a series of calibration mixtures containing known concentrations of pure standards of each component over an appropriate concentration range are injected. A curve is generated by plotting absorbance verses concentration. The concentration of each component in the unknown is calculated by comparing its absorbance to the calibration curve.

ValueChrom uses the sequence table to schedule each level of unknown mixture and calculate the response factors.

Step 1. Develop a Method

A method must be developed for your separation. For calibration, the peaks of interest must elute at or near the same time during every run.

ValueChrom performs Calibration through the interaction of the **Sequence Table**, and a **Method** linked to a **Calibration Table**. A method can only be calibrated after all the compounds of interest have been separated and identified.

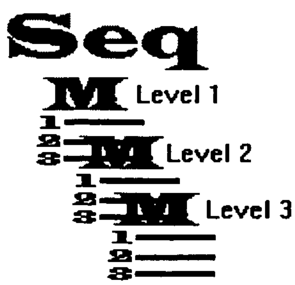
Step 2. Create a Calibration Table

Once a method has been developed and all the peaks of interest have been identified, that method can be calibrated. To calibrate a method it must be linked to a **Calibration Table**. See "Creating a Calibration Table" on page 84 for more information.

Step 3. Assign the Calibration Table to the Method

For ValueChrom to use a calibration table it must be linked to the method. Linking is done during method editing. See "Setting up the Method" on page 91 for more information.

Step 4. Run each Standard in a Sequence Table



The sequence table is used to designate each run and its corresponding level in the calibration table. Running each sample in a sequence table also allows multiple injections for each level, and an orderly process for ensuring quality quantification.

Each level is programmed as a different entry in the sequence table. It is good practice to run the lowest level first. This minimizes the effects of sample carry over.

Once a sequence table has been run, the response factors will be calculated and stored as part of the calibration table.

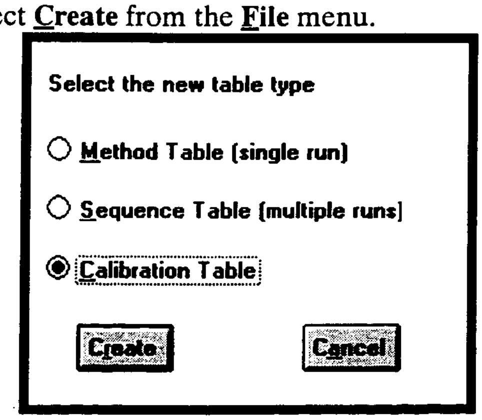
Step 5. Run Unknowns with the Calibration Table

If a calibration table has response factors, you can assign it to any method table. When the run is analyzed, calibration data will be produced, and peaks will be labeled with the name found on the calibration table.

Creating a Calibration Table

Once a method has been developed and all the peaks of interest have been identified, that method can be calibrated. To calibrate a method it must be linked to a **Calibration Table**. To create a **Calibration Table**:

Select **Create** from the **File** menu.

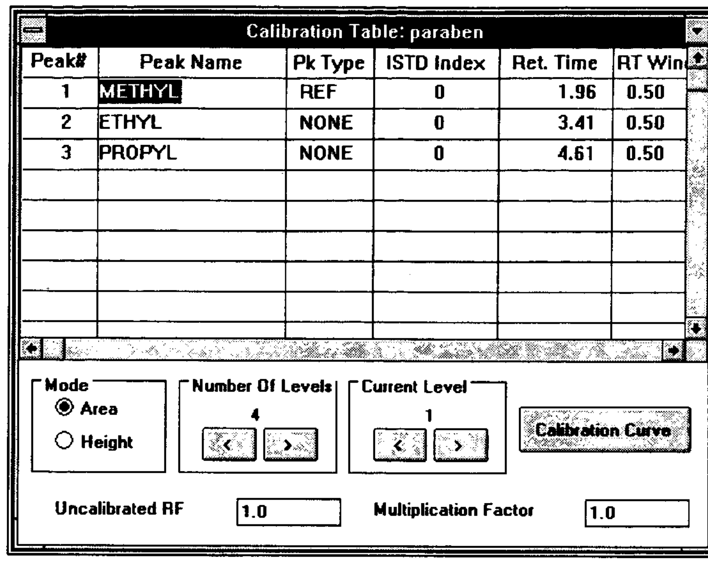


Select **Calibration Table**, then click on **Create**.

Calibration Table

Calibration table -- Example

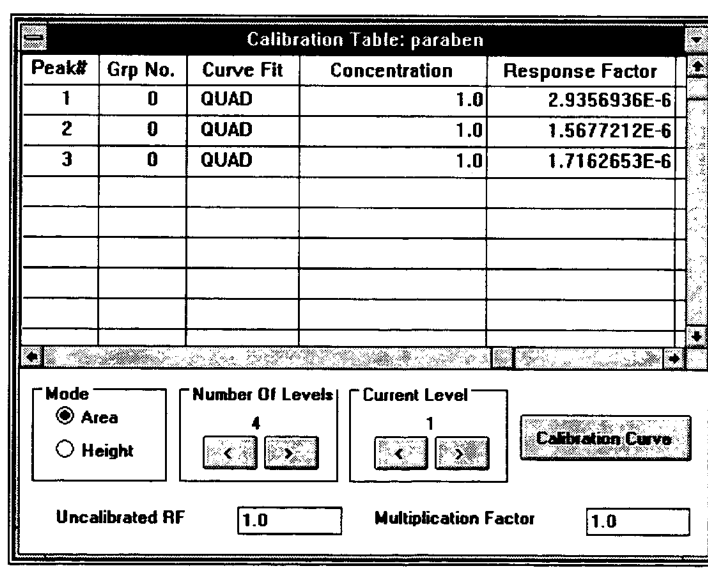
Below is an example of a **Calibration Table**. The screens represent the table as you scroll from left to right.



Calibration Table: paraben

Peak#	Peak Name	Pk Type	ISTD Index	Ret. Time	RT Win
1	METHYL	REF	0	1.96	0.50
2	ETHYL	NONE	0	3.41	0.50
3	PROPYL	NONE	0	4.61	0.50

Leftmost view of Calibration Table



Calibration Table: paraben

Peak#	Grp No.	Curve Fit	Concentration	Response Factor
1	0	QUAD	1.0	2.9356936E-6
2	0	QUAD	1.0	1.5677212E-6
3	0	QUAD	1.0	1.7162653E-6

Rightmost View of Calibration Table

At the bottom of the screen, in the box labeled **Mode**, select the parameter you wish to calibrate, peak area or peak height. **Peak Height** is the distance from the baseline to the apex of the peak. When the peak is Gaussian shaped this parameter is proportional to the peak area and may be used for calibration. **Peak Area** is the total area under the chromatographic peak.

Select the number of levels in the calibration in the **Number Of Levels** box. Each level of a calibration corresponds to a different standard concentration.

Enter a response factor for components that have no calibration standards (are not included in the table) in the **Uncalibrated RF** box. The default value for this parameter is 1.

Use the **Multiplication Factor** to adjust all calculated concentrations to compensate for sample dilution or to convert unit-concentration values to the original sample concentration.

Entering Information on the Calibration Table

For each level in the calibration table you must enter the following information (all information except **Amount** is usually the same throughout all levels and is carried over as you scroll from level to level):

Peak# is the line on the calibration table (filled in automatically).

Peak Name: Enter the name of the peak (up to 15 characters).

Pk Type: Enter **REF** for a reference peak, **ISTD** for an internal standard peak, **RISTD** for a peak that is both a reference and an internal standard peak, or **NONE** for all other compounds.

Reference Peak (REF) -- Use a reference peak to correct for retention time shifts that can occur with any method.

Internal Standard (ISTD) -- Use an internal standard to compensate for variations in sample handling techniques, dilutions, or instrument performance.

ISTD Index permits the use of more than one internal standard in the same analysis.

Ret.Time: Enter the retention time for each peak.

RT Window: Enter either a time window (in minutes) or a percentage of the retention time (enter a number followed by %) within which the peak may be found. For example if the retention time entered is 5 minutes and the retention window is 1 minute or 10%, the software will look for the peak between 4.5 minutes and 5.5 minutes.

Units refers to the concentration of the sample component (i.e., picomoles/injection volume).

Grp No. allows you to add the peak areas of a group of peaks. To assign a group, enter the same number for all peaks included. An entry of zero indicates no grouping.

Curve Fit: There are three possible Curve Fits, **(POINT)**, **(LINEAR)**, and **(QUAD)**. See "Point-to-Point Calibration", "Linear Regression", or "Quadratic Calibration".

Concentration: Enter the concentration for each component in the units previously provided **(Units)**. Each calibration level should have different concentration information (the lowest level should correspond to the lowest concentration).

Note: An individual sample component may have a different response factor when run under different conditions, for example; different flow rates, detectors, mobile phases or gradients. Although you can enter values in this field manually, it is not recommended.

Response Factor is the amount per unit response for this component. This value is calculated during the calibration run and is entered on the table automatically.

Calib Runs is the number of calibration runs performed. This value is non-editable and is incremented by 1 each time a calibration is done and all internal standard and reference peaks are found.

When all Calibration Table pages have been completed for all calibration levels, click on the **Control-menu box**, then on **Close**. When the **Select a Table to Close** box appears enter the table name in the **Save changes to** box, then click on the **Save** button.

HINT: For Calibration tables with long columns of identical entries use the **Copy** and **Fill Down** options in the **Edit** menu.

Calibration Levels

A **Single Level (Point) Calibration** assumes that a plot of detector response versus concentration will be a straight line passing through the origin, therefore the data point from a single standard is used to create a straight line through the origin.

A **Multi-level Calibration** has up to 5 concentration entries per peak. When calibrating a method, it is always more accurate to use multiple levels rather than a single level to calculate a curve.

The box labeled **Current Level** displays the level currently visible on the table. With the **<** and **>** keys you can scroll through the concentration levels of the calibration table. Each calibration level requires different concentration information. Up to five levels are allowed.

Reference and Standard Peaks

Reference Peak (REF) -- Use a reference peak to correct for retention time shifts that can occur with any method. When there is a shift in the retention time of the reference peak the calibration table will update the retention times of the peaks eluting after it so that proper peak identification will be made. The software will weight the different retention times as follows:

New RT Value = [0.75 x observed RT value] + [0.25 x current RT value]

Internal Standard (ISTD) -- Use an internal standard to compensate for variations in sample handling techniques, dilutions, or instrument performance. To use an internal standard, add a constant, known amount of the internal standard (a compound similar in chemical properties to the analytes) to the calibration standards and to each sample. Any change that affects the internal standard and the unknown equally will not affect the results. The calculation used to correct sample concentration for internal standard response is:

C(sample) = [RF(sample) x A(sample) x C(int. std.) x multi. factor] divided by [RF(int. std.) x A(int. std.)]

Where **C** equals concentration, **RF** equals response factor, **A** equals absorbance, **int. std.** equals internal standard and **multi. factor** equals multiplication factor.

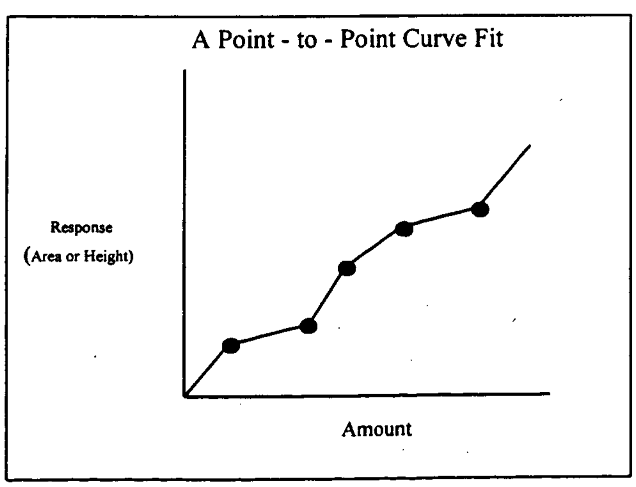
Note: Use an **External Standard** when the response factor of each sample component is independent of any other sample component. When this is the case the calibration standards containing the compounds of interest are the external standards. The concentration calculation is as follows:

C(sample) = A(sample) x RF(sample) x multi. factor

ISTD Index permits the use of more than one internal standard in the same analysis. Enter a unique number in this field for each internal standard and enter the same number in the ISTD field for each sample peak to be calibrated with that internal standard.

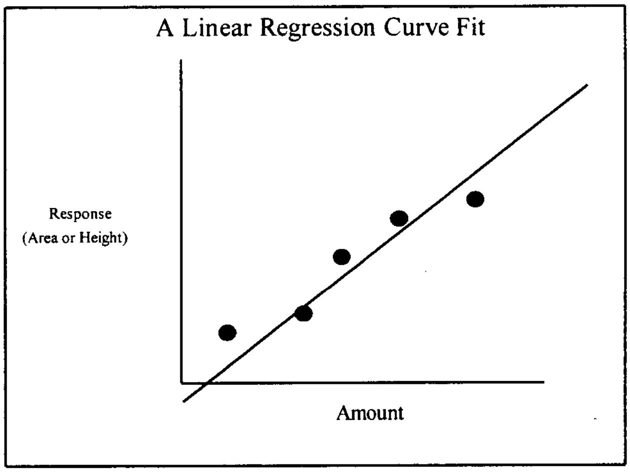
Point-to-Point Calibration

Point - to - Point is a series of straight line segments connecting the calibration points. The software will assume that the segment of response below the lowest level of calibration extends through the origin and that the segment between the second highest and the highest calibration levels extends linearly.



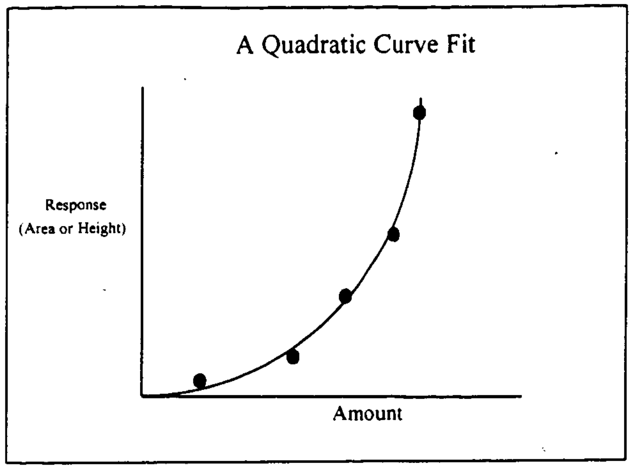
Linear Regression

Linear Regression will draw the best fitting straight line through the data points. For a linear curve fit there must be at least 2 levels in the calibration table. The curve is not forced through the origin.



Quadratic Calibration

Quadratic curve fit will construct a second order polynomial response curve. For a quadratic curve fit there must be at least 3 levels in the calibration table. Use the quadratic curve fit to model detector responses that are second order.



Running a Calibration

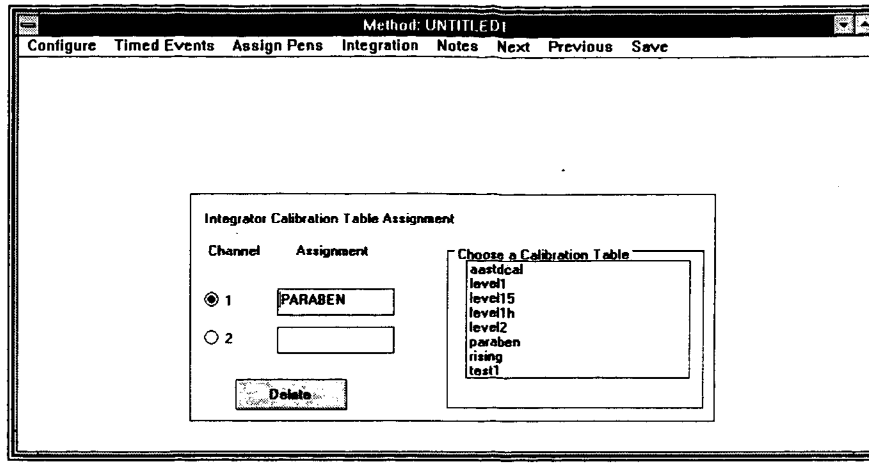
To run a calibration curve, the method and calibration table must be linked, then a series of calibration standards must be run using the **Sequence Table.**

Setting up the Method for Calibration

To receive quantitative information from a method it must be linked to a calibration table, then a calibration curve must be generated through the **Sequence Table.**

To link a calibration table and a method:

1. Select the **Edit** icon from the **Easy Menu** or select **Open** from the **File** menu.
2. Select **Method Table** Click on the method to be calibrated, then click on **Open.**
3. Select **Calibration** from the **Integration** menu.



4. In the **Integrator Calibration Table Assignment** window a list of available calibration tables will be displayed in the **Choose a Calibration Table** box.
5. Select the integrator channel to be linked to a calibration table. Each integrator channel can be assigned a separate calibration table.
6. Click on the appropriate calibration table displayed in the **Choose a Calibration Table** box. The linked table will appear in the **Assignment** box.
7. To delete an assignment, click on the **Delete** button at the bottom of the screen.
8. Select **Save** from the menu bar.

Setting up the Sequence Table

Once the method has been linked to the calibration table a calibration curve can be generated through the **Sequence Table**. To set up a Sequence Table to run a calibration:

Select **Create** from the **File** menu.

Select **Sequence Table**, then click on **Create**.

The sequence table screen will open.

If the calibrate column is "NONE" then response factors are not changed.

*If Calibrate is "**REPL**" then new response factors replace old ones.*

If Calibrate is "AVG" then new response factors are averaged in with existing ones.

Sequence Table: paraben

Entry	Name	Vial No.	Volume	Method	No. Inj	Calibrate	Level
1	PARALEV1	91	17	PARABEN	1	NONE	1
2	PARALEV2	91	3	PARABEN	1	NONE	2
3	PARALEV3	91	8	PARABEN	1	NONE	3
4	PARALEV4	91	20	PARABEN	1	NONE	4

Entry: This number is incremented automatically when a line is added to the Table. There must be one line on the Sequence Table for every level in the calibration table.

Name: Enter the sample name; it may be helpful for all calibration levels to have the same sample name.

Vial Number: This field is for record keeping purposes only.

*HINT: For Sequence Tables with long columns of identical entries use the **Copy and Fill Down** options in the Edit menu.*

Volume: This field is for record keeping purposes only.

Method: Enter the name of the method to be calibrated.

No. Inj: Enter the number of injections per sample. The allowable entries for this field are 1 - 99.

Calibrate: Enter either **REPL** to replace any existing response factors and turn back the number of calibrations run to zero, or **AVG** to average the new response factors with the existing values using the following formula:

Avg. RF = [New RF + Old RF]/2

An entry of **NONE** signifies no change in response factors.

Level: Each level of the calibration must be entered here. A calibration curve is run from the lowest level, or concentration, to the highest.

Note: The calibration standards must appear in the Sequence Table in the same order as the levels are arranged in the calibration table. For example; if level 1 in the calibration table is 0.5 mg/ml, level 2 is 1.0 mg/ml and level 3 is 2.0 mg/ml then the first sample injected must be 0.5 mg/ml, the second must be 1.0 mg/ml and then the third must be 2.0 mg/ml.

Select **Close** from the **Control-box** menu.

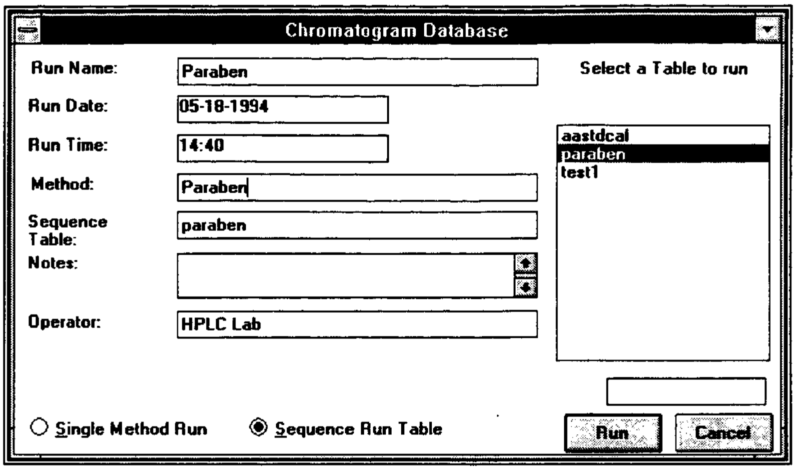
Enter the Sequence Table name in the **Save changes to** box, click on **Save**.

Generating a Calibration Curve

To generate the Calibration Curve (run the Sequence Table):

Select the **Run** icon from the **Easy Menu**, or Select **Sequence Table** from the **Run** menu.

This opens the Chromatogram Database screen:



Each time a sample is run ValueChrom automatically enters the data into the Chromatogram Database. The Chromatogram Database allows you to attach various information to the sample (with the screen shown above) so that it can be organized for easy access at a later date.

The **Run Name** field allows you to assign each sample a meaningful name with up to 28 free form characters. If no name is specified by the user **ValueChrom** will assign a name based on the sequence table. The **Run Name** will be assigned to each sample included in the sequence table. ValueChrom attaches a number to the **Run Name** to help identify the individual samples. These number correspond to the sequence table line number and injection number, for example:

The **Run Date** and **Run Time** fields are filled by ValueChrom and are not editable. The data entered here corresponds to the date and time the sample is run.

The scroll box labeled **Select a Table to Run**, on the right of the screen, contains a list of all of the available sequence tables. Click on a sequence table name and it will appear in the **Sequence Table** field, or type the name of the sequence table to be used.

The **Notes** field allows you to enter additional information relevant to the samples in the table. All samples run with the sequence table will have this information stored with the data file. This field will accept unlimited free form text.

The **Operator** field will contain the **Log On** name provided when starting the software. This field may be edited if necessary.

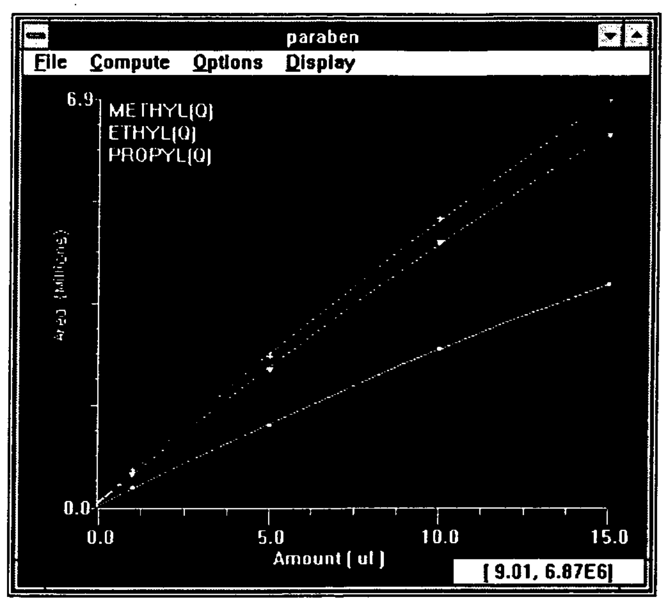
Viewing a Calibration Curve

When the Sequence Table is completed the software will generate a calibration curve for each sample component by plotting absorbance over concentration.

To view the calibration curve:

1. From the Database screen choose the method icon
2. Select Calibration table and a list of calibration tables will appear
3. Click on the one you are interested in and choose **Open**.

The calibration table will open.



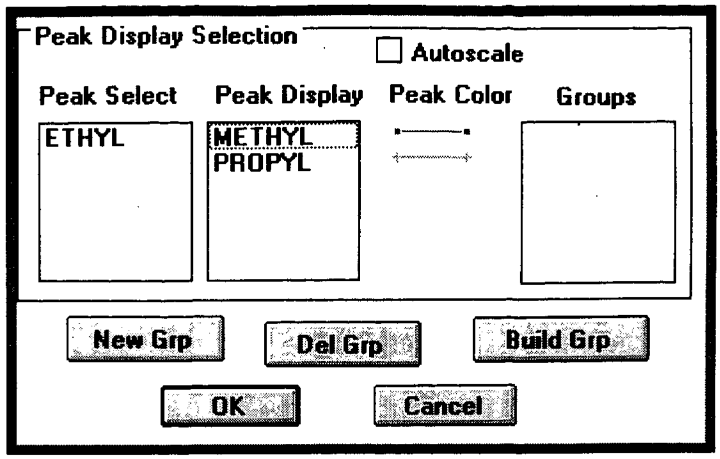
To display a different calibration curve, select **Open** form the **File** menu.

The screen displays all available calibration tables. To open a calibration file click on the file name, then click on **OK**.

Once a file is open the calibration curves for the first 4 peaks listed in the Calibration table are displayed.

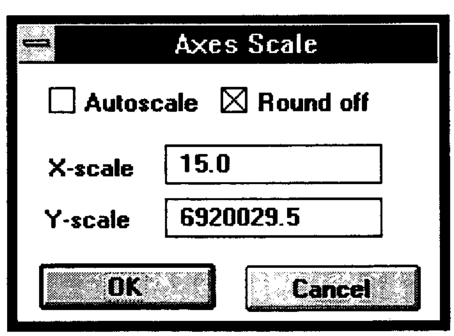
In the **Compute** menu select the appropriate curve fit, **Point** (straight line segments connecting the generated point), **Linear** (a best fit line drawn through the points), or **Quadratic** (a second order polynomial curve fit).

To view the calibration curves remaining peaks select **Peak Select** form the **Options** menu. The peaks not currently displayed are listed in the **Peak Select** box. Those currently displayed are listed in the **Peak Display** box.



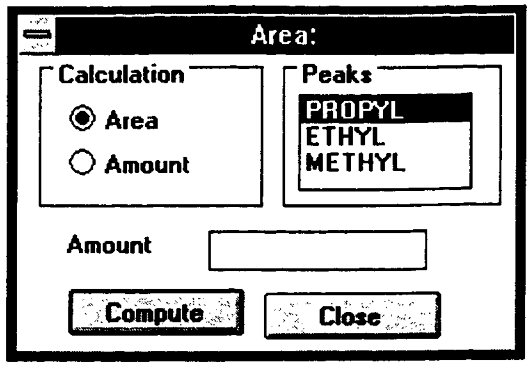
From this window it is possible to group the results from several peaks into a single curve. To form a peak group click on the **New Grp** button. Click on the peaks to be included in the group. The new group will appear in the **Groups** box. To add to an existing group click on the **Build Grp** button, click on the group to add to, then click on the peak to add. To delete a group click on the **Delete Grp** button then click on the group to be deleted.

Select **Axis Scale** in the **Options** menu to scale the curve axes. Click on the **Autoscale** box and when you change one axis value, the other axis will scale automatically.



When **Show Data** in the **Options** menu is selected the data point from which the curve was calculated will appear on the curve. If one of these point is the result of an error (sample handling, injection volume) it can be discarded and the curve recalculated ignoring the outlying point. Simply click on the point to be discarded and the curve will be re-drawn. The data point will remain on the screen, however it will appear white, to indicate that it was not used in the curve calculation.

Evaluate Formula in the **Options** menu opens the window shown below. If you know the area of a peak you can use this screen to calculate the amount or if you know the amount of sample then you can predict the area of its peak.



With **Cross Hairs** is selected in the **Display** menu, pressing the right mouse button causes cross hairs to appear with the axes values shown in the lower right corner.

When **Values** is selected in the **Display** menu, the position of the cursor on the window will be reflected with respect to the curve axes.

With the **Peak Labels** option listed in the **Display** menu you can chose to have the peak names displayed in the upper left of the window. When seen in color, the color of the peak name corresponds to the color of its displayed curve.

Removing Outliers

Note: You cannot remove a point from a curve if it is the only point for that level.

Often several samples are run for each calibration level. Each of these samples will be plotted on the calibration screen. If one of the points is an outlier, and you do not want it included in the calculations, you can remove it by clicking on the point with the mouse. When a point is removed the color changes to yellow on the screen, and the line will change shape. Your curve will now be based on the equation of the line without the point you removed. To restore the point click on it again. Before reanalyzing your chromatograms you must update your response factors.

Updating Response Factors

After removing outliers or changing the type of curve you may want to reanalyze your data. To do this you must update the response factors in the calibration table. Choose **Update Table** from the **File** pull-down menu. When you choose Update Table, the settings you have selected for the calibration will be transferred to your calibration table.

To see the results, reanalyze your chromatography using the reanalyze button on the Database screen.

System Menus

Introduction

This chapter covers every item available through the system menus of ValueChrom. ValueChrom brings most of the common operation into easy to use screens like the database screen and the easy menu. Less often, but more powerful commands are available through the system menus. The underlined letter indicates the short cut key. Pressing ALT and the short cut key will activate that selection.

The File Menu

Create...

Select Create to set up a new method , Sequence Table, or calibration table.

Open...

Select Open to view or edit an existing method), Sequence Table), calibration table, or report.

Save:

Select Save to store methods, Sequence Tables and calibration tables on the hard disk.

Save As...

Select Save AS to Name or rename a method, Sequence Table, or calibration table then store it on the hard disk.

Close:

Select Close to close (remove from the screen) methods, Sequence Tables, calibration tables, reports, or data files.

Delete..

Delete permanently removes methods, Sequence Tables, calibration tables, reports or data files from the hard disk.

View Reports:

Select View Reports to view the last report generated or the last batch of reports.

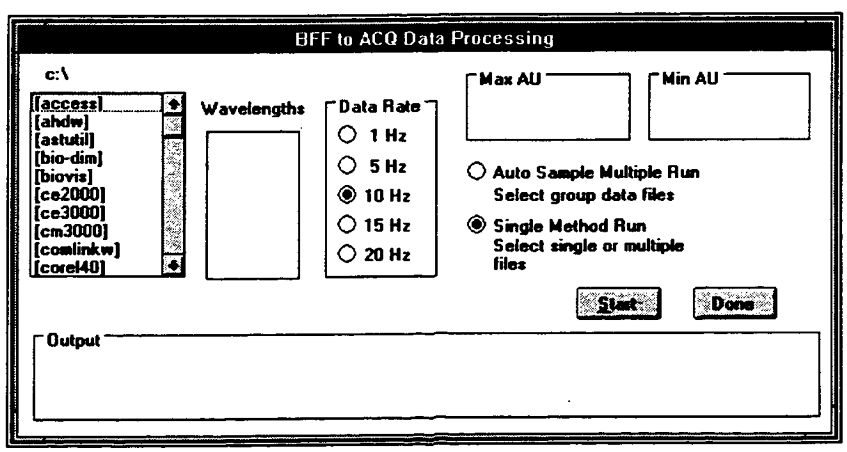
DataFile Utilities:

Data File Utilities contains three options, Bio-Dimension for manipulation of data from the Bio-Dimension detector, ASCII Data Conversion and File Backup for translation of data files and data backup, and Notepad for direct access to the Windows Accessory, Notepad, for viewing reports.

Bio-Dimension:

Select Bio-Dimension to accesses the Bio-Dimension conversion window. From this window you can convert Bio-Dimension detector files into Integrator files to be used with ValueChrom, or to ASCII files for use with third party programs.

BFF to Integrator:



To convert Bio-Dimension detector data (files with the extension **BFF**) to a format that ValueChrom can read (files with an **ACQ** file extension):

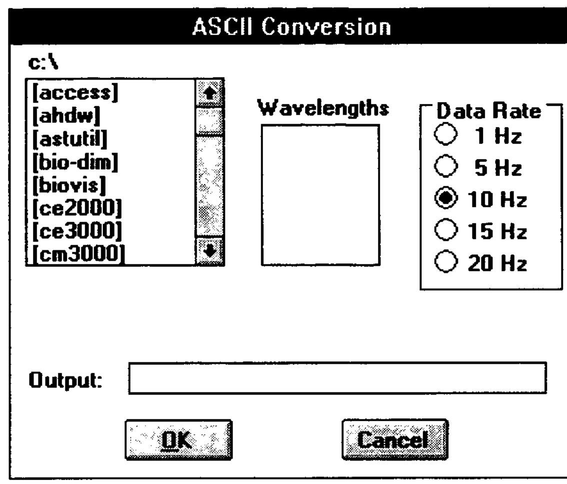
Select the **BFF** file to convert in the window labeled **c:\HPLC**

A list of available wavelengths will appear in the box labeled **Wavelengths**. Select the wavelength to be converted. In the box labeled **Output** the file name and directory path for the converted data will appear.

Select the data rate and click on **START**. The data will be converted and added to the database.

BFF to ASCII

Note: The Bio-Dimension ASCII conversion program converts the entire BFF file into a large ASCII text file.

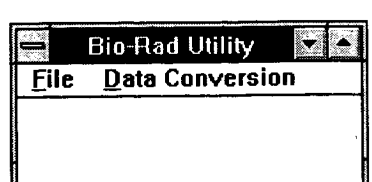


To convert Bio-Dimension detector data to an ASCII format:

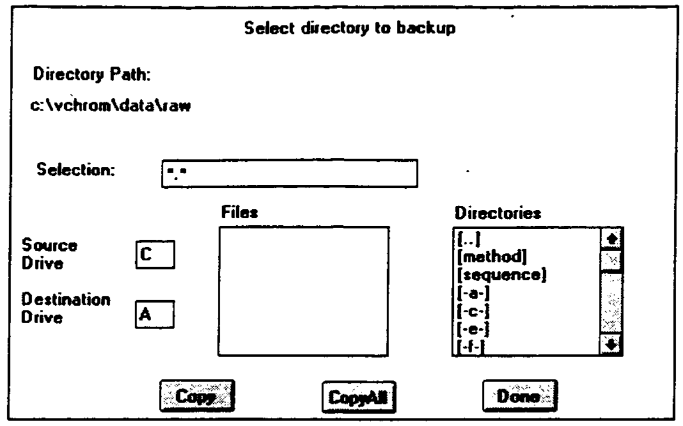
1. Select the **BFF** file to convert in the window labeled c:\HPLC
2. Select to wavelengths to be converted
3. Select a data rate
4. In the box labeled **Output** enter a file name for the converted file.
5. Click on **OK**

ASCII Data Conversion and File Backup:

Brings up the **Bio-Rad Utility** window. From this window you can backup data files or convert data files into an ASCII format for use with other software programs.



Data Backup (in **File** menu): Allows you to backup, copy or move data files. HPLC data files can be very large and a tape backup or other mass storage device is strongly recommended.



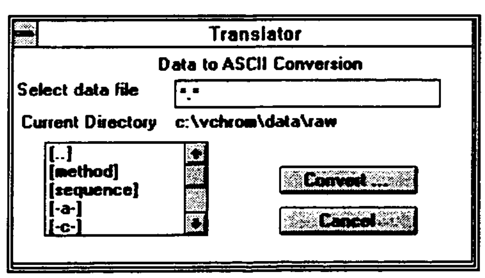
1. Select the data file(s) to be copied.
2. Set the destination drive.
3. Click on **Copy** to copy a single file, or **Copy All** to copy multiple files
4. When you have finished backing up your data, click on **Done**.

Restore Data

To restore data to your hard disk, follow the instructions for backing up data except choose the source drive as your floppy disk, and the destination drive as your hard disk. In some cases you may need to run batch reanalysis to get the data properly into the database.

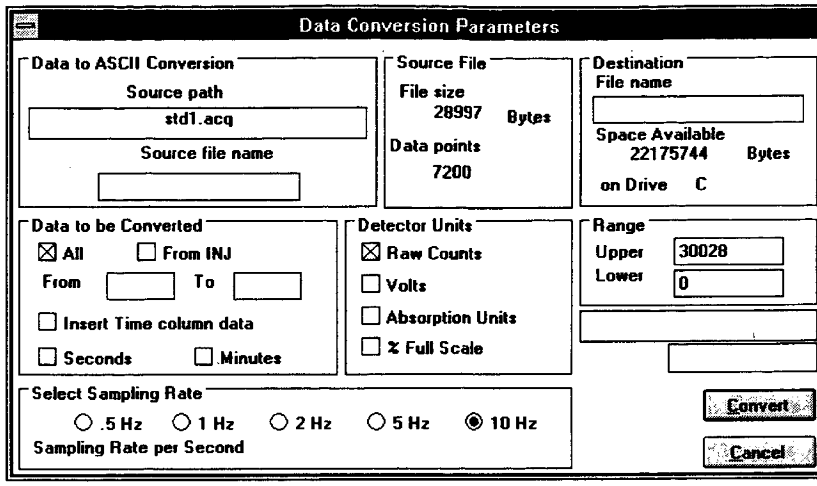
Data Conversion:

Data Conversion brings up the **Translator** window for converting data files to an ASCII format.



Select the data file to be translated

Click on **Convert.** This brings up the **Data Conversion Parameters** window:



The box titled **Data to ASCII Conversion** shows the currently selected data file.

The box titled **Source File** gives information (file size, in bytes and data points) about the selected data file.

The box labeled **Destination** shows the computer assigned name for the new ASCII file. You may enter another name in this field. This box also shows the amount of disk space on the current drive.

In the box labeled **Data to be Converted** you can choose to convert **All** data (data from time zero through the end of the method), data **From INJ** (all data from the time of injection to the end of the method), or in the boxes labeled **From** and **To** you can enter specific regions of the data to be converted.

In the box labeled **Detector Units** select the appropriate detector units. The box labeled **Range** will display the Upper and Lower limits for the detector.

In the box labeled **Select Sampling Rate** select the sampling rate per second.

When all information is entered, click on **Convert**. When conversion is complete the status box in the lower right corner will read "processing complete".

Page Setup...

Brings up the setup screen for the printer installed on your system.

Print:

Select Print to print a method, Sequence Table, calibration table, or report.

Exit:

Select Exit to quit the Series 5000 HPLC Software.

DiskInfo:

DiskInfo provides information about the available disk space and data acquisition space.

About:

About provides title, version number, and copyright information about **ValueChrom**.

The Edit Menu

Cut:

Cut allows you to delete a selected area of text and place it into the Clipboard, replacing the old Clipboard contents. That text can then be placed in another location using the Paste command.

Copy:

Copy allows you to copy a selected area of text, leaving the original in tact, and place it into the Clipboard, replacing the old Clipboard contents. The copied text can then be placed in another location using the Paste command.

Paste:

Paste copies information from the Clipboard into the file.

Delete:

Delete removes the current selection from the screen

Fill Down:

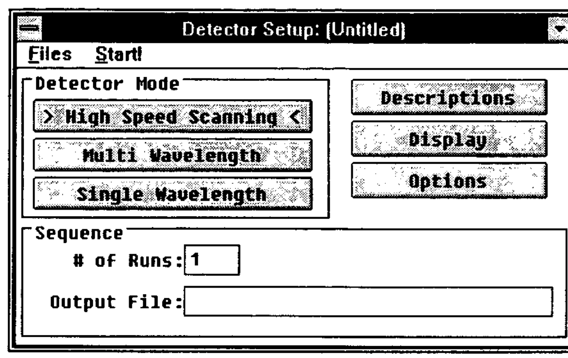
Fill Down will fill a column with information stored in the Clipboard.

The Run Menu

Bio-Dimension Detector:

Bio-Dimension Detector brings up the Bio-Dimension **Detector Setup** window. From this screen you can create methods for the Bio-Dimension Detector. Refer to the Bio-Dimension instruction manual for further information.

Note: This command is only available if you have purchased the Bio-dimension software.



Method:

Select Method to start a method. For a thorough discussion of **Running a Method** see "Running a Method" on page 51.

Sequence Table:

Select Sequence Table to start a Sequence Table. For a thorough discussion of **Running a Sequence Table** see "Running a Sequence Table" on page 80.

Reanalysis:

Select Reanalysis to re analyze data from a Sequence Table. For a more thorough discussion on **Reanalysis** see "Reanalysis and Integration" on page 74.

The Chromatogram Menu

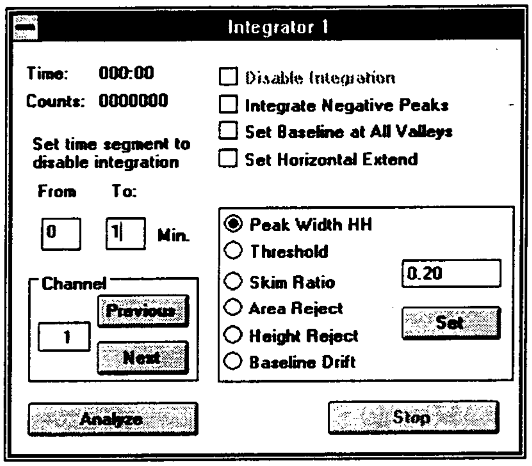
Strip Chart 1 (2):

Selecting Strip Chart 1 or 2 opens the Strip Chart window for viewing a chromatogram.

Integrator 1 (2):

Select Integrator to reintegrate a single data file. The integrator is an easy way quickly try different parameters for you chromatogram. To use the integrator:

1. Select the run of interest on the database screen.
2. Select Integrator from the Chromatogram menu



The Integrator Faceplate

3. Enter a time to not integrate data. (This is useful for skipping over void peaks, or other peaks that are of no interest.)
4. Select the parameters to change and enter a new value. Click on the set button to record each change.
5. Click on the Analyze button.

When the analysis is done you will see the new baseline on the strip chart. If you have selected an editor in the initialization file (BRASD.INI), the editor will open with the report file.

The Control Panels Menu

Status Panel A (or B):

The system Status Panel shows the current running state of the system. This window is automatically opened when a method begins.

Pumps A (or B):

Selecting Pumps A (or B) opens the Solvent Delivery System control panel for manual operation of the pumps.

Relays A (or B):

Selecting Relays A (or B) opens the **I/O Control Panel**. A window shows the input and output relay status.

Shutdown:

This exits the **ValueChrom** program.

Configure:

Before the ValueChrom can be used, information regarding the devices in your system must be entered into the system configuration.

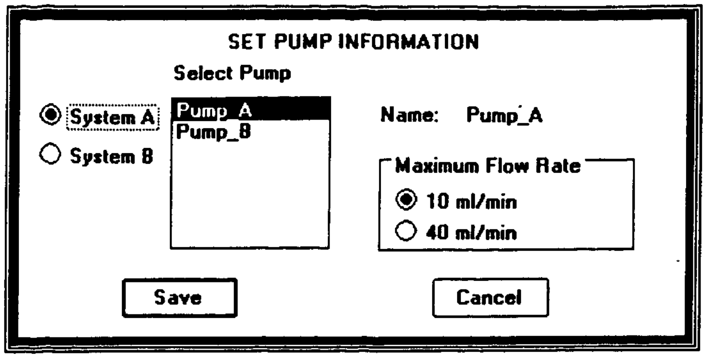
Set System:

Use the Set System Configuration window to assign available modules to your HPLC system.

Set Autosampler:

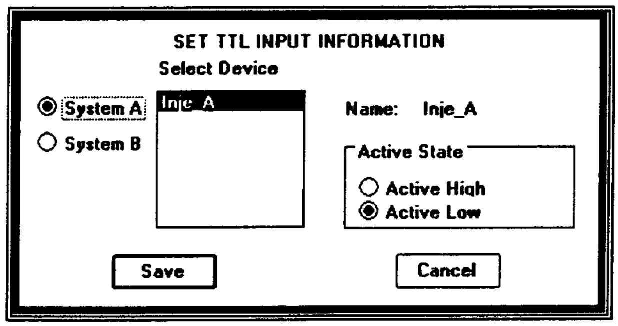
The autosampler configuration window is used to supply the software with information about your autosampler.

Set Pumps



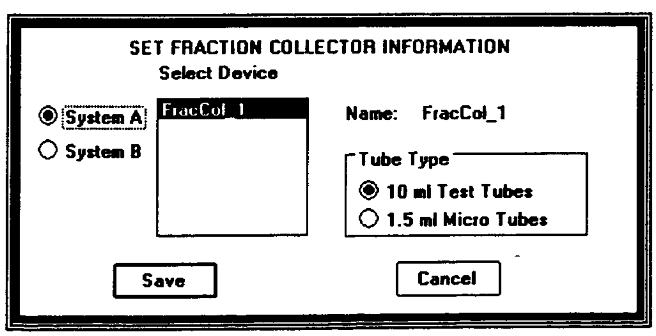
From this screen you can configure your pumps for analytical or preparative flow rate capabilities.

Set Inputs



From this screen you can determine how the software will interpret input information.

Set Fraction Collector



From this screen you can configure the tube volume for a fraction collector.

Reboot II

Selecting this menu item resets the Instrument Interface.

Column Information

Affinity Columns

Affinity chromatography (AFC) is a liquid chromatographic technique which separates molecules based on some inherent specific activity. Affinity separations occur by the specific binding of a particular sample solute to the functional group of the support matrix.

Bio-Rad has two distinct AFC column lines: Affi-Prep 10 and Bio-Gel Protein A. The Affi-Prep 10 columns contain activated N-hydroxysuccinimide (NHS) moieties bound to a polymer support matrix. This functional group is able to react with primary amines of various biomolecules. The Bio-Gel Protein A columns contain protein A covalently bound to the polymer support matrix. These columns are especially designed to bind immunoglobulins with an affinity towards protein A. This binding can be enhanced by Bio-Rad's proprietary buffering system.

Affinity Operating Conditions

Affi-Prep 10 AFC Columns

The operating conditions for the "reacted" Affi-Prep 10 AFC columns are those usually used for the affinity purification of proteins and peptides. The mobile phases are generally 10 - 100 mM aqueous buffer in a pH range of 2 - 12 as the starting buffer, and 10 - 100 mM aqueous buffer containing various additives as the eluting buffer. These additives include 1 - 2 M NaCl, urea, and guanidine HCl. Affi-Prep 10 columns are predominantly run under step gradient conditions at flow rates of 0.5 - 1.0 ml/min.

Bio-Gel Protein A Columns

The operating conditions for the Bio-Gel Protein A columns are those commonly used for the purification and analysis of antibodies. The mobile phases are generally 10 - 100 mM aqueous buffer in a pH range of 2 - 12 as the starting buffer, and 10 - 100 mM aqueous buffer containing various additives as the eluting buffer.

These additives include 1 - 2 M NaCl, urea, and guanidine HCl. Also, Bio-Rad has developed unique and proprietary buffers that enhance the binding of certain antibodies to protein A. Bio-Gel Protein A columns are predominantly run under step gradient conditions at flow rates of 0.5 - 1.0 ml/min.

Affinity Cleaning Conditions

Before performing any cleaning procedures, all guard columns and precolumn filters should be removed.

Affi-Prep 10 Columns

The Affi-Prep 10 HPLC columns can be cleaned (after reaction) by running 0.1 N sodium hydroxide (NaOH) in 1.0 M NaCl and injecting 1 M urea. The injection volume should be 2 - 5 full injector loops (i.e., approximately 40 - 200 ul). The flow rate during cleaning should be 0.2 - 0.5 ml/min. The storage solvent should be 0.02% azide.

Bio-Gel Protein A Columns

The Bio-Gel Protein A columns can be cleaned by running 50% methanol and injecting 50% methanol. The injection volume should be 2 - 5 full injector loops (i.e., approximately 40 - 200 ul). The flow rate during cleaning should be 0.2 - 0.5 ml/min. The storage solvent should be 50 mM sodium phosphate, pH 7.5, containing 0.02% azide.

For more complete operating, cleaning, and application information, please refer to Bio-Rad's *HPLC Columns, Methods, and Applications* guide (bulletin #1833) or call your local Bio-Rad sales representative.

Hydrophobic-Interaction Columns

Hydrophobic-interaction chromatography (HIC) is a liquid chromatographic technique which separates molecules based on their hydrophobicity. Hydrophobic interaction separations occur by differential hydrophobic interactions of sample solutes with the functional groups of the support matrix.

The difference between HIC and reversed phase chromatography (RPC) is based on the density (or carbon load) of the functional groups on the support matrix. HIC has a relatively low carbon load; while RPC has a higher carbon load.

Bio-Rad offers two distinct HIC columns: the Bio-Gel MP7 and the Bio-Gel Phenyl-5-PW. The Bio-Gel MP7 HIC column is a polymer-based column containing a weakly hydrophobic moiety for very hydrophobic biomolecules such as membrane proteins and immunoglobulins. The Bio-Gel Phenyl-5-PW. HIC column is a polymer-based column containing a highly hydrophobic moiety for weakly to moderately hydrophobic molecules, including proteins, peptides, and nucleic acids.

Hydrophobic-Interaction Operating Conditions

Bio-Gel MP7 HIC Columns

The operating conditions for the Bio-Gel MP7 HIC columns are those usually used for the purification and analysis of proteins, peptides, and nucleic acids. The mobile phases are generally 50 - 100 mM aqueous buffer containing 1.0 - 2.0 M salt (e.g., ammonium sulfate) in a pH range of 2 - 12 as the starting buffer, and 50 - 100 mM low salt buffer in a pH range of 2 - 12 as the eluting buffer. Bio-Gel MP7 HIC columns are predominantly run under linear gradient conditions at flow rates of 0.5 - 1.0 ml/min for analytical columns.

Bio-Gel Phenyl-5-PW HIC Columns

The operating conditions for the Bio-Gel Phenyl-5-PW HIC columns are those usually used for the purification and analysis of proteins, peptides, and nucleic acids. The mobile phases are generally 50 - 100 mM low salt buffer containing 1.0 - 2.0 M salt (e.g., ammonium sulfate) in a pH range of 2 - 12 as the starting buffer, and 50 - 100 mM aqueous buffer in a pH range of 2 - 12 as the eluting buffer. Bio-Gel Phenyl HIC columns are predominantly run under linear gradient conditions at flow rates of 0.5 - 1.0 ml/min for analytical columns.

Hydrophobic-Interaction Cleaning Conditions

Before performing any cleaning procedures, all guard columns and precolumn filters should be removed.

Bio-Gel MP7 and Bio-Gel Phenyl-5-PW HIC Columns

The Bio-Gel MP7 and Bio-Gel Phenyl-5-PW HIC columns can be cleaned by running 0.1 N sodium hydroxide (NaOH), 5% methanol, or 6 M urea as the mobile phase and injecting 0.2 N sodium hydroxide (NaOH). The injection volume should be 2 - 5 full injector loops (i.e., approximately 40 - 200 ul). The flow rate during cleaning should be 0.2 - 0.5 ml/min. The storage solvent should be 0.05% azide or 5% methanol.

For more complete operating, cleaning, and application information, please refer to Bio-Rad's *HPLC Columns, Methods, and Applications* guide (bulletin #1833) or call your local Bio-Rad sales representative.

Ion-Exchange Columns

Ion-exchange chromatography (IEX) is a liquid chromatographic technique which separates molecules based on their ionic charge or isoelectric point (pI) at a given pH. Ion exchange separations occur by differential ionic interactions of the sample solutes (molecules or ions) with the charged functional groups of the support matrix.

Bio-Rad offers two distinct IEX column lines: the Bio-Gel MA7 and the Bio-Gel PW. The Bio-Gel MA7 IEX columns are nonporous ion-exchange columns that provide rapid and high-resolution separations. The Bio-Gel PW columns are macroporous ion exchangers that provide high binding capacities for proteins and nucleic acids.

Ion-Exchange Operating Conditions

Bio-Gel MA7 IEX Columns

The operating conditions for the Bio-Gel MA7 IEX columns are those usually used for purification and analysis of various biomolecules. The mobile phases are usually 10 - 50 mM aqueous buffer in a pH range of 5 - 9. The starting buffer usually has a very low ionic strength while the eluting buffer has a much higher ionic concentration, approximately 0.5 - 1.5 M (i.e., higher salt concentration). Bio-Gel MA7 columns are usually run under linear gradient conditions at flow rates of 5 - 10 ml/min for analytical columns.

Bio-Gel PW IEX Columns

The operating conditions for the Bio-Gel PW IEX columns are those usually used for purification and analysis of various biomolecules. The mobile phases are usually 10 - 50 mM aqueous buffer in a pH range of 5 - 9. The starting buffer usually has a very low ionic strength, while the eluting buffer has a much higher ionic concentration, approximately 0.5 - 1.5 M (i.e., higher salt concentration). Bio-Gel PW columns are usually run under linear gradient conditions at flow rates of 0.5 - 1.0 ml/min for analytical columns.

Ion-Exchange Cleaning Procedures

Before performing any cleaning procedures, all guard columns and any precolumn filters should be removed.

Bio-Gel MA7 IEX Columns

The Bio-Gel MA7 IEX columns can be cleaned by running 1 M buffered NaCl as the mobile phase and injecting 0.1 N sodium hydroxide (NaOH). The injection volume should be 2 - 5 full injector loops (i.e., approximately 40 - 200 ul). The flow rate during cleaning should be 0.5 - 2.5 ml/min. The storage solvent should be 0.02% sodium azide or 10% ethanol.

Bio-Gel PW IEX Columns

The Bio-Gel PW IEX columns can be cleaned by running 1 M NaCl as the mobile phase and injecting 0.2 N sodium hydroxide (NaOH). The injection volume should be 2 - 5 full injector loops (i.e., approximately 40 - 200 ul). The flow rate during cleaning should be 0.2 - 0.5 ml/min. The storage solvent should be 0.02% sodium azide or 5% ethanol.

For more complete operating, cleaning, and application information, please refer to Bio-Rad's *HPLC Columns, Methods, and Applications* guide (bulletin #1833) or call your local Bio-Rad sales representative.

Multimodal Columns

Multimodal chromatography (MM) is a liquid chromatographic technique which separates molecules based on their hydrophobic or polar characteristics. Multimodal separations occur by the differential hydrophobic or polar interaction of sample solutes with the functional groups of the support matrices. These interactions are greatly influenced by the mobile phases being used.

Bio-Rad has three distinct MM columns: Bio-Sil Disaccharide, Bio-Sil NH_2, and Bio-Sil CN. The Bio-Sil Disaccharide column is a silica-based, high-load aminopropyl column especially designed for high-resolution separations of various disaccharides. The Bio-Sil NH_2 and the Bio-Sil CN are silica-based, bonded columns that provide very selective and high resolution separations of drugs, steroids, and other small organic molecules.

Multimodal Operating Conditions

Bio-Sil Disaccharide Columns

The operating conditions for the Bio-Sil Disaccharide columns are those generally used for the analysis of disaccharides. The mobile phase is usually 70% acetonitrile. Bio-Sil Disaccharide columns are predominantly run under isocratic conditions at flow rates of 0.5 - 1.0 ml/min for analytical columns.

Bio-Sil NH_2 and CN Columns

The operating conditions for the Bio-Sil NH_2 and CN columns can be those used for normal phase or reversed phase separations. Bio-Sil NH_2 and CN columns are predominantly run under linear gradient conditions at flow rates of 0.5 - 1.0 ml/min for analytical HPLC columns.

Multimodal Cleaning Conditions

Before performing any cleaning procedures, all guard columns and precolumn filters should be removed.

Bio-Sil Disaccharide, NH_2, and CN Columns

The Bio-Sil Disaccharide, NH2, and CN columns can be cleaned by running 100% acetonitrile as the mobile phase and injecting 20% dimethylsulfoxide (DMSO). The injection volumes should be 2 - 5 full injector loops (i.e., approximately 40 - 200 ul). The flow rate during cleaning should be 0.2 - 0.5 ml/min. The storage solvent should be 5% acetonitrile, 0.05% sodium azide, or 5% methanol.

For more complete operating, cleaning, and application information, please refer to Bio-Rad's *HPLC Columns, Methods, and Applications* guide (bulletin #1833) or call your local Bio-Rad sales representative.

Normal-Phase Columns

Normal-phase chromatography (NPC) is a liquid chromatographic technique which separates molecules based on their polar characteristics. Normal phase separations occur by differential polar interactions of the sample solutes with the functional groups of the support matrices.

Bio-Rad offers a slightly modified NPC column: Bio-Sil Polyol 90-10. The Bio-Sil Polyol 90-10 column is a polyhydroxyl, silica-based column that provides efficient separations of steroids, lipids, fatty acids, petroleum products, and other small organic molecules.

Normal-Phase Operating Conditions

Bio-Sil Polyol 90-10 NPC Columns

The operating conditions for the Bio-Sil Polyol 90-10 column are those generally used for normal phase chromatography. The starting solvent is a 70 - 100% nonpolar organic solvent (e.g., hexane) containing 0 - 30% polar organic solvent (e.g., tetrahydrofuran or isopropanol, and 100% polar organic solvent as the eluting buffer. The Bio-Sil Polyol 90-10 column is usually run under linear gradient conditions at flow rates of 0.5 - 1.0 ml/min for analytical columns.

Normal-Phase Cleaning Conditions

Before performing any cleaning procedures, all guard columns and precolumn filters should be removed.

Bio-Sil Polyol 90-10 NPC Columns

The Bio-Sil Polyol 90-10 columns can be cleaned by running 100% acetonitrile as the mobile phase and injecting 20% dimethylsulfoxide (DMSO) in acetonitrile. The injection volume should be 2 - 5 full injector loops (i.e., approximately 40 - 200 ul). The flow rate during cleaning should be 0.2 - 0.5 ml/min. The storage solvent should be 10% acetonitrile or 10% methanol.

For more complete operating, cleaning, and application information, please refer to Bio-Rad's *HPLC Columns, Methods, and Applications* guide (bulletin #1833) or call your local Bio-Rad sales representative.

Reversed-Phase Columns

Reversed-phase chromatography (RPC) is a liquid chromatographic technique which separates molecules based on their hydrophobicity. Reversed-phase separations occur by differential hydrophobic interactions of sample components with the hydrophobic functional groups attached to the support matrix.

Bio-Rad offers two distinct RPC column lines: Bio-Sil and Hi-Pore columns. The Bio-Sil RPC columns are silica-based, reversed phase columns for high- resolution separations of small biomolecules (e.g., amino acids, nucleotides, or nucleosides) and other small organic compounds, including drugs. Hi-Pore columns are large-pore, silica-based RPC columns for extremely efficient separations of peptides, proteins, and oligonucleotides.

Reversed-Phase Operating Conditions

Bio-Sil RPC Columns

The operating conditions for the Bio-Sil RPC columns are those generally used for reversed-phase applications for drugs, amino acids, nucleotides, and other small organic molecules. The mobile phases that are commonly used are HPLC grade water or < 5 mM aqueous buffer as the starting buffer, and 100% water miscible organic solvent (e.g., acetonitrile) as the eluting buffer (solvent). These mobile phases may contain various additives (e.g., triethylamine) to help improve separation. Bio-Sil RPC columns are usually run under linear gradient conditions at flow rates of 0.5 - 1.0 ml/min for analytical HPLC columns.

Hi-Pore RPC Columns

The operating conditions for the Hi-Pore RPC columns are those generally used for reversed-phase applications for drugs, derivatized amino acids, larger peptides, proteins, and oligonucleotides. The mobile phases that are commonly used are HPLC grade water or < 5 mM aqueous buffer as the starting buffer, and 100% water miscible organic solvent (e.g., acetonitrile) as the eluting buffer (solvent). These mobile phases may contain various additives (e.g., triethylamine) to help improve separation. Hi-Pore columns are usually run under linear gradient conditions at flow rates between 0.5 and 1.0 ml/min for analytical HPLC columns.

Reversed-Phase Cleaning Procedures

Before performing any cleaning procedures, all guard columns and precolumn filters should be removed.

Bio-Sil RPC Columns

The Bio-Sil RPC columns can be cleaned by running 100% acetonitrile as the mobile phase and injecting 20% dimethylsulfoxide (DMSO). The injection volume should be 2 - 5 full injector loops (i.e., approximately 40 - 200ul). The flow rate during cleaning for analytical RPC columns should be 0.2 - 0.5 ml/min. The storage solvent should be 50% - 100% methanol or acetonitrile.

Hi-Pore RPC Columns

The Hi-Pore RPC columns can be cleaned by running 100% acetonitrile as the mobile phase and injecting 20% dimethylsulfoxide (DMSO). The injection volume should be 2 - 5 full injector loops (i.e., approximately 40 - 200ul). The flow rate during cleaning for analytical RPC columns should be 0.2 - 0.5 ml/min. The storage solvent should be 50% - 100% methanol or acetonitrile.

For more complete operating, cleaning, and application information, please refer to Bio-Rad's *HPLC Columns, Methods, and Applications* guide (bulletin #1833) or call your local Bio-Rad sales representative.

Size-Exclusion Columns

Size-exclusion chromatography (SEC) is a liquid chromatographic technique which separates molecules according to their size (Stokes Radius) and molecular weight. Size exclusion separations occur by repeated diffusions of sample solutes (molecules) into and out of SEC support matrices' pores.

Bio-Rad offers three distinct SEC column lines: Bio-Sil SEC, Bio-Silect SEC, and Bio-Gel SEC XL columns. The Bio-Sil SEC columns are silica-based columns for high-resolution separations of proteins and nucleic acids in a molecular-weight range of 5,000 - 1 million daltons. The Bio-Silect SEC columns offer biocompatible, non-metallic columns for the separation of proteins using Bio-Rad's unique silica SEC matrices. The Bio-Gel SEC XL columns are polymer-based columns that offer wide pH (2 - 12) and separation ranges (<1,000 - 200 million daltons) for larger peptides, proteins, and nucleic acids.

Size-Exclusion Operating Conditions

Bio-Sil and Bio-Silect SEC Columns

The operating conditions for the Bio-Sil SEC columns are those generally used for the purification and analysis of proteins and other biomolecules. The mobile phase is usually a 10 - 50 mM aqueous buffer (e.g. phosphate, TRIS, acetate, etc.) containing 0.15 - 0.5 M salt in a pH range of 6 - 8. Bio-Sil SEC columns are predominantly run under isocratic conditions at flow rates of 0.5 - 1.0 ml/min for analytical HPLC columns.

Bio-Gel SEC XL Columns

The operating conditions for the Bio-Gel XL SEC columns are those generally used for the purification and analysis of biomolecules of any size. The mobile phase is usually a 10 - 100 mM aqueous buffer (e.g. phosphate, TRIS, acetate, etc.) containing 0.15 - 0.5 M salt in a pH range of 2 - 12. Bio-Gel XL SEC columns are predominantly run under isocratic conditions at flow rates of 0.5 - 1.0 ml/min for analytical HPLC columns.

Size-Exclusion Cleaning Procedures

Before performing any cleaning procedures, all guard columns and precolumn filters should be removed.

Bio-Sil SEC and Bio-Silect SEC Columns

The Bio-Sil SEC and Bio-Silect SEC HPLC columns can be cleaned by following a two-step process. First, run five (5) column bed volumes of 0.3 M sodium phosphate (NaH_2PO_4), pH 3.5. Next, run five (5) column bed volumes of 0.1% acetonitrile (ACN), or run a 0% - 100% methanol gradient. The flow rate for analytical SEC columns during each of the two cleaning steps should be 0.2 - 0.5 ml/min. The storage solvent should be 0.05% azide or 5% methanol.

Bio-Gel SEC XL Columns

The Bio-Gel SEC XL columns can be cleaned by running 0.1 N sodium hydroxide (NaOH) or 10% methanol as the mobile phase and injecting 0.2N sodium hydroxide (NaOH). The injection volume should be 2 - 5 full injector loops (i.e., approximately 40 - 200 ul). The flow rate during cleaning for analytical SEC columns should be 0.2 - 0.5 ml/min. The storage solvent should be 0.05% sodium azide.

For more complete operating, cleaning, and application information on Bio-Rad's SEC columns, please refer to Bio-Rad's *HPLC Columns, Methods, and Applications* Guide (bulletin #1833) or call your local Bio-Rad sales representatives.

Appendix

Pump Pressure Conversion

To convert Kilograms per square centimeters to pounds per square inch (PSI) multiply by 14.223

To convert Kilograms per square centimeter to bar multiply by 0.9806.

Optimizing Computer Performance

There are several things you can do to maintain optimal system performance for your ValueChrom system.

1. Delete any unwanted files from the database when you no longer need them.
2. Run the DOS program DEFRAG.EXE or some other disk optimization software often. In some cases weekly if you do a lot of data collection.
3. Experiment with different printer driver options. Many printer drivers allow fast printing or high priority printing. In some cases this can interfere with real time data collection.
4. Run the DOS program MEMMAKER.EXE or some other suitable memory management program to ensure you are using your memory optimally.
5. Change your Windows to use a permanent swap file. (See your Windows manual for instructions.)
6. Use SMARTDRV.EXE
7. Do <u>not</u> use Double-space or stacking programs.
8. Do not use Screen savers or other terminate and stay resident (TSR) programs.

Getting Help from Bio-Rad

Bio-Rad maintains well trained support staff around the world to help with your HPLC problems. If you need assistance contact your local Bio-Rad office.

In the US Bio-Rad can be contacted through the Technical Resources Department by calling (800) 4-BIORAD. Alternatively you can reach Bio-Rad by fax at (510) 741-1047. The Technical Resources Department is open 6 AM to 5 PM Pacific time.

Before you call have the following information handy.

1. A description of the problem -- particularly when it occurs. Printing the chromatograms showing the problem is helpful since faxing chromatograms expedites problem resolution.
2. Details of your chemistry, i.e. type of column, running conditions, type of sample... etc.
3. The details of your computer system. The easiest way to do this is to run the program MSD.EXE from DOS, and print the report. For computer problems you may need to fax this report to Bio-Rad. (The program MSD.EXE is supplied as part of Windows and is in the Windows directory on your disk.)
4. A copy of your purchase order if your system is under warranty.
5. A service contract number if you purchased a service contract from Bio-Rad.

Removing ValueChrom from your System

To remove ValueChrom from your hard disk delete the following files

1. All ValueChrom files located in the main ValueChrom directory (usually C:\VCHROM.)
2. All data stored in subdirectories under the main ValueChrom directory.
3. The file BRASD.INI to your Windows directory

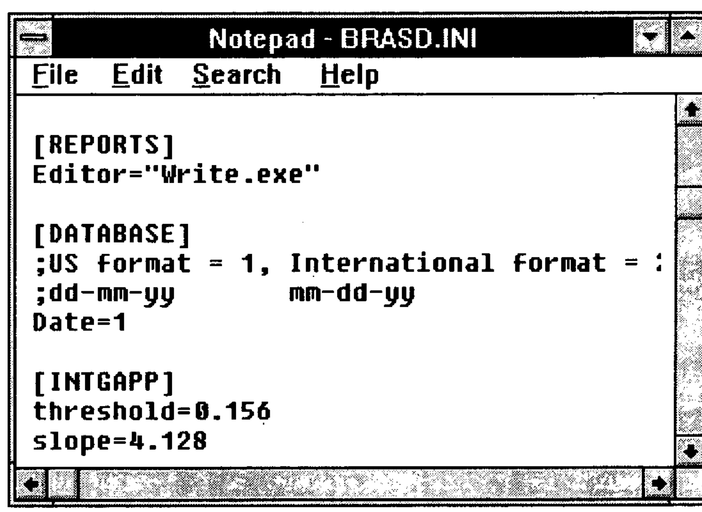
ValueChrom also modifies the WIN.INI file with a section labeled {VCHROM]. you may want erase this section of the WIN.INI file. DO NOT DELETE THE WIN.INI FILE.

Customizing the BRASD.INI File

Selecting an Editor

The BRASD.INI file allow you to select which editor you want to use when performing reanalysis. To change the editor

1. Open the BRASD.INI file by selecting **Datafiles Utilities** from the **File** pull-down menu.
2. Choose **Edit Initialization File**. and the Notepad editor will open
3. Scroll down to the section labeled [REPORTS].
4. For editor enter the exact file name of the editor you want to use.
5. Choose SAVE from the FILE pull-down menu.
6. Close the editor.



Changing the Editor and Date format for BRASD.INI

Changing the Date Format

International date format can be selected in the BRASD.INI file by editing the section labeled [DATABASE]

Default Parameters

*Note: The **BRASD.INI** file contains a section labeled [INTGAPP]. The values in this section are calculated by the software when baseline monitoring is performed. Do not alter to this section of the **BRASD.INI** file.*

The default parameters for integration can be customized and the **DEFAULT** names changed by editing the **BRASD.INI** text file. Below is an example of one set of default parameters from the **BRASD.INI** file.

```
[1INTGAPP]
width=0.100
threshold=0.010
skim=10.0
arearej=100.0
htrej=0.100
drift=0.100
name=Narrow
```

Getting Your Results Ready to Publish

By taking advantage of Windows it is easy to get chromatograms, integration results, and calibration plots into word processing documents.

All ValueChrom integration reports are stored as ASCII text files. They can easily be read by all the major word processing programs on the market today.

After each analysis a copy of the integration report is stored in the TEMP directory. Each time a reanalysis is performed, this file is copied over with the new report data. For system A, the report is called REPORTA.TXT, and for system B the file is called REPORTB.TXT. When a sequence table is analyzed all of the reports are stored in a single file called REPORT1.TXT. These files are very convenient for manipulating data with spreadsheet programs such as MicroSoft Excel.

Raw data can be exported into spreadsheets and other data analysis packages. See "ASCII Data Conversion and File Backup:" on page 101 for instructions.

Chromatograms can be cut and pasted into Windows based word processing documents by following these steps.

1. Start your word processor and open a document.
2. Start ValueChrom and from the East Menu choose Database.
3. Select the chromatogram file by clicking on it with the mouse.
4. Click on the **Strip Chart** button to enlarge the display.

5. Zoom in to the desired size, or accordingly adjust the scaling controls.
6. Change the background to white by selecting **Background** from the **Display** pull-down menu
7. Press and hold the **ALT** key and then press the **Print Screen** key.
8. Using the mouse switch focus back to your word processor.
9. Place the mouse at the desired insertion point and select **Paste** from clipboard.
10. The strip chart will be pasted into your document.

Pressing print screen button without holding the ALT button in step 6 will cause the entire screen to be copied into the clipboard.

HPLC Standards

HPLC standards are very important for evaluating column performance. Most Bio-Rad columns are shipped with the appropriate standard. The standard should be run using the test condition specified in the instruction manual, or printed on the test chromatogram shipped with your column. The test chromatogram should be stored. When you suspect there is a problem with your column, re-run the standard with the exact conditions. This allows direct comparisons of the columns performance from when it was new.

125-0561 Protein Standard for Anion Exchange Chromatography

Equine myoglobin	1.0mg
Conalbumin	2.5mg
Chicken ovalbumin	3.0mg
Soybean trypsin inhibitor	2.5mg

125-0562 Protein Standard for Cation Exchange

Equine myoglobin	1.0 mg
Ribonuclease A	4.0 mg
Cytochrome c	1.0 mg

151-1901 Gel Filtration Standard

Thyroglobulin	5.0 mg
Bovine gamma globulin	5.0 mg
Chicken ovalbumin	5.0 mg
Equine myoglobin	2.5 mg
Vitamin B-12	0.5 mg

125-0559 HIC Protein Standard

Cytochrome c	2.0 mg
Bovine serum albumin	4.0 mg
Soybean trypsin inhibitor	4.0 mg

151-1905 Protein Standard for Hydrophobic Interaction

Cytochrome c	1.4 mg
Equine myoglobin	0.8 mg
Lysozyme	0.8 mg

125-0083 Paraben Standard for Reversed Phase Chromatography

Methyl paraben	0.44 mg/ml
Ethyl paraben	0.5 mg/ml
Propyl paraben	0.5 mg/ml

Accessories and Replacement Parts

Catalog #	Product Description
Systems	
125-1531	**Model 5000T System**, titanium with IBM compatible computer, includes: Model 2800 Solvent Delivery System with built in high pressure mixing Model 7125T injector Prime/purge valve System interface provides control over system and accessories, plus communication to an IBM compatible computer MicroSoft Windows software, version 3.1 ValueChrom, with VGA color graphics 100% IBM compatible computer
125-1528	**Model 5000T System**, titanium without IBM compatible computer
125-1529	**Model 5000S System**, stainless steel with IBM compatible computer, includes: Model 2800 Solvent Delivery System with built in high pressure mixing Model 7125T injector Prime/purge valve System interface provides control over system and accessories, plus communication to an IBM compatible computer MicroSoft Windows software, version 3.1 ValueChrom, with VGA color graphics 100% IBM compatible computer
125-1527	**Model 5000S System**, stainless steel without IBM compatible computer
Series 1350 Soft-Start Pumps	
125-1350	**Model 1350 Soft Start Pump** 100 / 120 / 220 / 240V, 316 stainless steel construction, 9.9ml / min. maximum flow rate
125-1352	**Model 1350T Soft Start Pump**

	100 / 120 / 220 / 240V, titanium construction, 9.9ml / min. maximum flow rate
125-1355	**Model 1355T Soft Start Pump**
	100 / 120 / 220 / 240V, titanium construction, 40ml / min. maximum flow rate
127-0300	**Lo-Pulse Pulse Damper**
125-1348	**Solvent Draw-Off Valve**, for Model 1350 pump
125-1361	**High-Pressure Dynamic Mixer**, titanium
125-0302	**Piston Seals**, 2 ea., Model 1330 or 1350 pump
125-0303	**Piston Seal Insertion Tool**
125-0304	**Sapphire Piston**, Model 1330 or 1350 pump
125-0305	**Piston Insertion Tool**
125-0306	**Inlet Check Valve**, Model 1330 or 1350 pumps
125-0307	**Outlet Check Valve**, Model 1330 or 1350 pumps
125-0308	**Spare Parts Kit**, Model 1350 pump, includes one each of 125-0302, 125-0304, 125-0305, 125-0306, 125-0307
125-0317	**Solvent Inlet Filter**, Model 1350 pumps
125-1345	**Static Mixer**, 250µL stainless steel
125-1353	**Piston Seals**, Model 1350T pump
125-1354	**Spare Parts Kit**, Model 1350T pump, includes one each of 125-1359, 125-1353, 125-0305, 125-1356, 125-1357
125-1356	**Inlet Check Valve**, titanium
125-1357	**Outlet Check Valve**, titanium
125-1358	**Piston Seals**, Model 1355T Pump
125-1359	**Sapphire Pistons**, Model 1355T pump
125-1360	**Spare Parts Kit**, Model 1355T pump, includes one each of 125-1358, 125-1359, 125-0305, 125-1356, 125-1357

Model AS-100 Automatic Sampling System

167-1100	**Model AS-100 Automatic Sampling System**, 100 / 120V
167-1101	**Model AS-100 Automatic Sampling System**, 200 / 240V
167-1102	**Model AS-100T Automatic Sampling System**, 100 / 120V, titanium
167-1103	**Model AS-100T Automatic Sampling System**, 200 / 240V, titanium
**125-1654	**Model AS-100 Digital I/O Cable**
223-9500	**Micro Test Tubes**, 1.5ml polypropylene, 500/box, without caps
167-1160	**Glass Micro Test Tubes**, 1.5ml, 100/box, without caps
167-0645	**Pierceable Caps**, 500/box
167-1115	**Standard Needle**, for Model AS-100
167-1116	**Side Point Needle**, for Model AS-100
167-1110	**Standard Vial Sample Holder Set**, 100 x 1.5ml
167-1111	**Small Vial Sample Holder Set**, 100 x 0.5ml
167-1112	**Microplate Sample Holder Set** 96 well
167-1117	**Titanium Needle**, for Model AS-100T only

Bio-Dimension Scanning UV-Vis Detector

167-0940	**Bio-Dimension Programmable Detector**, with stainless steel flow cell
167-0941	**Bio-Dimension Programmable Detector**, with Kel-F flow cell
167-0944	**Bio-Dimension Software**, complete with RS-232 cable
167-0912	**Variable Pathlength Preparative Flow Cell**

Model UV-1806 HPLC Detector

167-0950	**Model UV-1806 UV-Vis HPLC Detector**
167-0951	**Analytical Flow Cell**, for Model UV-1806 detector
167-0952	**Deuterium Lamp**, for Model UV-1806 detector

Model 1740 Fixed Wavelength UV-Visible Monitor

167-1000	**Model 1740 Fixed Wavelength UV-Visible Monitor**, with Analytical Flow Cell, Mercury Lamp, and 280nm Filter, 110 / 120 / 220 / 240V
167-1005	**Model 1740 Fixed Wavelength UV-Visible Monitor**, with Preparative Flow Cell, Mercury Lamp, and 280nm Filter, 100 / 120 / 220 / 240V
167-1010	**Zinc Lamp**, for Model 1740, 214 and 308nm
167-1011	**Cadmium Lamp**, for Model 1740, 229 and 326nm
167-1012	**Mercury Lamp**, for Model 1740, 254 and 280nm
167-1013	**Tungsten Lamp**, for Model 1740, 350 to 800nm
167-1015	**214nm Filter**, for Model 1740
167-1016	**229nm Filter**, for Model 1740
167-1017	**254nm Filter**, for Model 1740
167-1018	**280nm Filter**, for Model 1740
167-1019	**308nm Filter**, for Model 1740
167-1020	**326nm Filter**, for Model 1740
167-1023	**436nm Filter**, for Model 1740
167-1025	**570nm Filter**, for Model 1740
167-1026	**660nm Filter**, for Model 1740
167-1030	**Sapphire Windows**, for Model 1740 analytical flow cell, 1 pair
167-1002	**Analytical Flow Cell**, for Model 1740
167-1007	**Preparative Flow Cell**, for Model 1740

HPLC Column Heater

125-0425	**Column Heater**, 115V, inserts not supplied
125-0426	**Column Heater**, 220V, inserts not supplied

Inserts with Allen wrench and screws, one required

125-0428	**1/4" x 5cm**, fits 50 x 4.0 or 4.6 mm columns
125-0429	**1/4" x 10cm**, fits 100 x 4.0 or 4.6mm columns
125-0430	**1/4" x 25cm**, fits 250 x 4.0 or 4.6mm columns
125-0434	**1/4" x 15cm**,fits 150 x 4.0 or 4.6mm columns
125-0431	**3/8" x 10cm**, fits 100 x 7.8mm columns
125-0432	**3/8" x 30cm**, fits 300 x 7.8mm columns
125-0433	**1/2" x 30cm**, fits 200 x 10mm column
125-0438	**Injector Brackets**, with screws, right
125-0439	**Injector Brackets**, with screws, left
125-0441	**Mounting Brackets**, with clamps

Injector Valve Assembly

125-0324	**Injector Valve Assembly**, includes Model 7125 injector, Prime / Purge Valve, and valve organizer bracket, assembled for mounting on Model 1350 Soft-Start Pumps
125-0325	**Injector Valve Assembly**, titanium
125-0322	**Purge / Prime Valve**, high pressure, 3-way taper seal, 1/16" tube, stainless steel tube (1/16"OD x 0.030"ID x 6")
125-1349	**Prime / Purge Valve**, titanium

Injectors and Sample Loops

125-0200	**Model 7125 Syringe Loading Sample Injector**, with 20µl sample loop
125-0208	**Model 7125T Injector**, titanium with 20µl titanium sample loop
125-0201	**Rotor Seal**, for Model 7125 Injector
125-0202	**Rotor Seal**, for Model 7010 Injector
125-0203	**Model 7010 Sample Injector**, with 20µl sample loop

125-0206	**Model 7012 Loop Filler Port**

Sample Loops, SS 316

125-0218	**Sample Loop**, 5μl
125-0210	**Sample Loop**, 10μl
125-0211	**Sample Loop**, 20μl
125-0212	**Sample Loop**, 50μl
125-0213	**Sample Loop**, 100μl
125-0214	**Sample Loop**, 200μl
125-0215	**Sample Loop**, 500μl
125-0216	**Sample Loop**, 1.0ml
125-0217	**Sample Loop**, 2.0ml
125-0219	**Sample Loop**, 10.0 ml

Sample Loops, Titanium

125-0250	**Sample Loop**, 10μl
125-0251	**Sample Loop**, 20μl
125-0252	**Sample Loop**, 50μl
125-0253	**Sample Loop**, 100μl
125-0254	**Sample Loop**, 200μl
125-0255	**Sample Loop**, 500μl

Syringes

125-0220	**HPLC Sample Syringe**, with needle, 10μl
125-0221	**HPLC Sample Syringe**, with needle, 25μl
125-0222	**HPLC Sample Syringe**, with needle, 50μl
125-0223	**HPLC Sample Syringe**, with needle, 100μl
125-0224	**22 Gauge Needle**, with CTFE luer hub

HPLC Protein System Fittings Kit

125-0018	**Protein System Fittings Kit**, includes:	
	ZDV Union with 0.010" through-hole	QTY 2
	Fingertight III plastic male nut with ferule	4
	Male Nut, titanium, short	2
	Male Nut, titanium, long	2
	Ferrules, titanium	4
	Tubing, titanium, 4" x 0.010"	2

Stainless Steel Fittings Kit

125-0010	**Stainless Steel Fittings Kit**, includes:	
	Union with 0.050" through-hole and external threads	2
	ZDV Union with 0.020" through-hole	2
	Male Nut	5
	Female Nut	5
	Ferrules	10
	Tefzel Plugs	2
	Assorted Rheodyne Nuts	8
	Rheodyne Ferrules	10
	SS Tubing 2" x 0.010" and 2" x 0.020"	8
	SS Tubing 4" x 0.010" and 4" x 0.020"	8

SS Tubing 8" x 0.010" and 8" x 0.020"	8
Fingertight I nut-ferrule combination	2
Tefzel Cap	2

Glossary of Terms

Batch Reanalysis

Batch reanalysis is used to import existing data into the chromatogram database.

Calibration

At a uniform flow, peak area is directly related to the concentration of the eluting compound. A **Calibration Curve** is a plot of the absorbencies of standards over a known concentration range. The absorbence of an unknown can be compared to this plot to determine its concentration. **Calibration curves** are generated through a **Sequence table**, and linked to individual methods. When the method is run, a **Calibration report** may be printed along with the integration report.

Click

A click consists of pressing and releasing the left mouse button without moving the mouse pointer. Click is the method of selecting objects and actions.

Configure

The term configure is used both to designate what type of modules you have in your HPLC system and how you want them to operate. Before using **ValueChrom** you must establish which peripheral equipment you have connected to each system. Use the **Configure** screen in the method table to select sampler type or mode of fraction collection.

Data Channel

Each detector is associated with one of the available data channels on the Instrument Interface. Data channels are assigned to strip chart pens in the method table using the **Assign Pens** screen. To integrate data from a detector, you must also assign it to an integration channel.

Data File

Data files are the DOS files that contain the data from your HPLC detector. Raw data files are created by the analog to digital converter and are stored on your hard disk during an HPLC run. While data files cannot be edited or changed once they are collected, the parameters for analyzing the data can.

Direct Manipulation

Direct manipulation is similar to drag select; the mouse is positioned over a particular item on the screen, the left mouse button is pressed and held down while moving the mouse to a new location, and the item beneath the mouse pointer is changed in size or position.

Direct manipulation allows users to control the appearance of the computer desk top using the mouse. For example, users can size a window by selecting the window's border segment with the mouse and moving the window border until the window is the desired size.

Double-click

A double-click consists of two rapid, consecutive clicks without moving the mouse pointer. **When you double-click on an object that has a default action, that default action is performed**. For example, the default action of the control menu box is closing the window. When users double-click on the control menu box for any window, that window is closed.

Drag Select

Drag select consists of pressing the left mouse button and holding it down while moving the mouse so that the pointer travels to a different location on the screen. Dragging ends when the mouse button is released. All items between the button down and button up points are selected.

Gradient

A gradient involves a change in solvent composition over time. Gradients are programmed on the **Timed Events Screen** of the method.

Integration

Integration is the analysis of a raw data file. Integration parameters determine the boundaries that the computer will use to define chromatographic peaks.

Main Menu Bar

The main menu bar is the collection of pull-down menus that appears across the top of the **ValueChrom** window.

Method

A table with all the necessary conditions to run a separation. It includes flow rate and running time, along with integration parameters.

Response Factor

A number relating the peak area to the absorbance measurement.

Sample

A sample is the substance to be analyzed.

Sequence Table

A sequence table is a series of methods and samples programmed to run, automatically.

Slope Sensitivity Parameters

Slope sensitivity measures the noise and drift of a baseline and uses that measurement to determine peak beginnings and endings.

Soft-Start

Soft-Start is a process that slowly increases the pump flow rate to eliminate column damage from shocking the column.

Strip Chart

A strip chart is used for graphical representation of data files (both raw and analyzed). The strip chart allows you to set the display parameters, alter the integration parameters, generate new integration reports and print chromatograms.

Index

P

Q

R

S

T

U

V

W

Z

BIO-RAD

Bio-Rad Laboratories

Life Science Group

Bio-Rad Laboratories Main Office, *2000 Alfred Nobel Dnve, Hercules, California 94547, Ph. (510) 741-1000, Fx. (510) 741-1060*
Eastern Regional Office, *85A Marcus Dr., Melville, New York 11747, Ph. (516) 756-2575, Fx. (516) 756-2594*
Also In: **North Ryde, Austrailia**, Ph. 02-805-5000, Fx. 02-805-1920 **Wein, Austsria**, Ph. (1) 877 89 01, Fx. (1) 876 56 29 **Nazareth, Belgium**, Ph. 09-385 55 11, Fx. 09-385 65 54 **Mississauga, Canada**, Ph. (905) 712-2771, Fx. (905) 712-2990 **Beijing, China**, Ph. (01) 2046622, Fx. (01) 2051876 **Copenhagen, Denmark**, Ph. 45-39 17 99 47, Fx. 45-39 27 16 96 **Ivry sur Seine Cedex, France**, Ph. (1) 49 60 68 34, Fx. (1) 46 71 24 67 **Munchen, Germany**, Ph. 089 31884-0. Fx. 089 31884- 100 **New Delhi, India**, Ph. 91-11-461-0103, Fx. 91-11-461-0765 **Milano, Italy**, Ph. 02-21609.1, Fx. 0221609.399 **Tokyo, Japan**, Ph. 03-3534-7665 Fx. 03-3534-8497 **Veenandaal, The Netherlands**, Ph. 0318-540666, Fx. 0318-542216 **Auckland, New Zealand**, Ph. 09-443 3099, Fx. 09-443 3097 **Kowloon, Hong Kong**, Ph. 7893300, Fx. 7891257 **Flamingo Valley, Singapore**, Ph. (65) 443 2529, Fx. (65) 442 1667 **Solna, Sweden**, Ph. 46 (0) 8 735 83 00, Fx 46 (0) 735 54 60 **Madrid, Spain**, Ph. (91) 661 70 85, Fx. (91) 661 96 98 **Glattbrugg, Switzerland**, Ph. 01/809 55 55, Fx. 01/809 55 00 **Hemel Hempstead, United Kingdom**, Ph. 0800 181134, Fx. 01442 259118

SIG 093094 Printed in USA